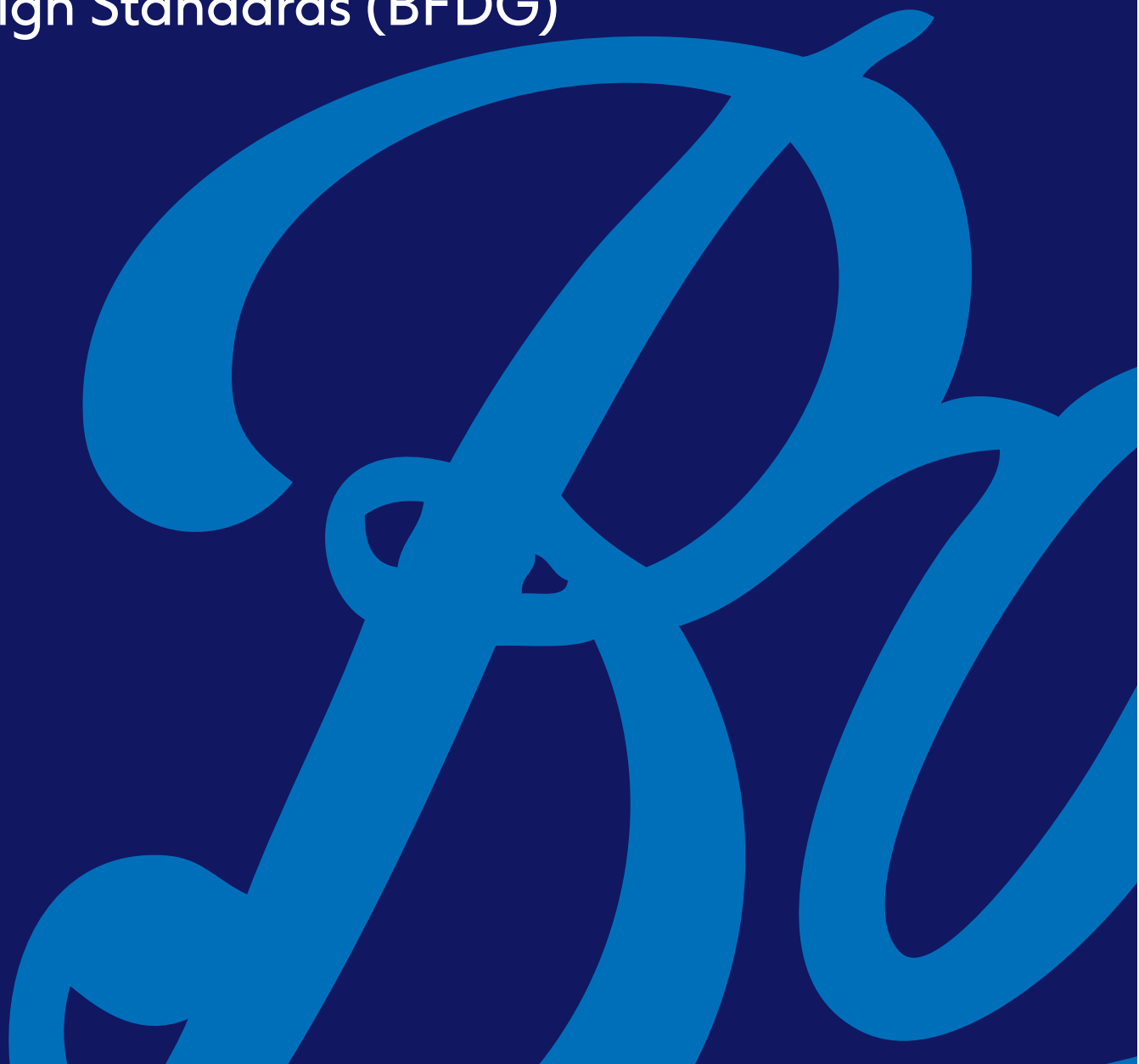




With you. For life

Boots In-Store Equipment Design Standards (BFDG)



V6 June 2026
Specification and Supply Chain Team
Boots Property



The Boots Group

Revisions shown in red

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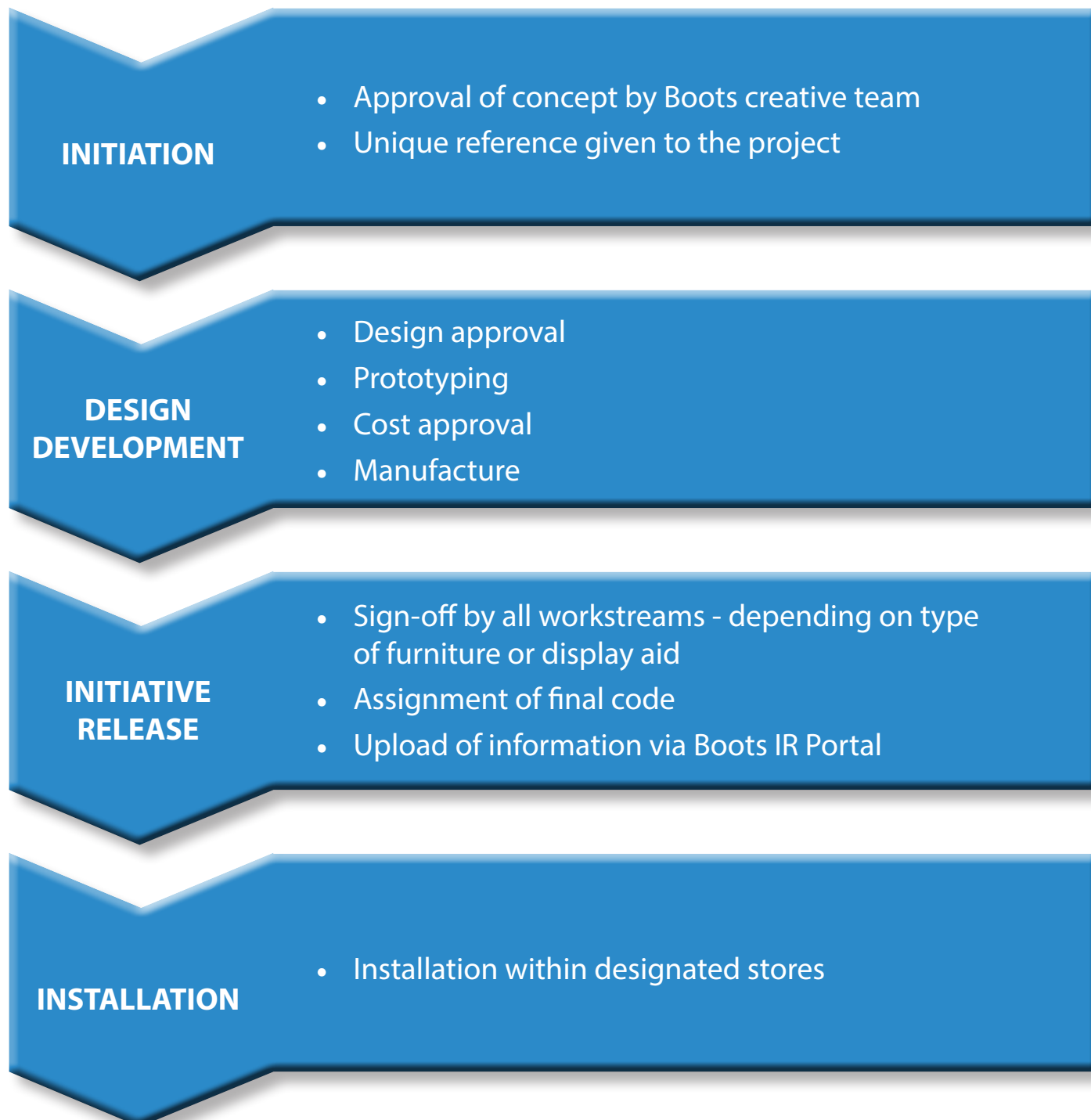
Change Log

Version No.	Date	Revision details	Changed by
v1	20/12/20	First draft issue	CS
v2	21/04/21	Live document	CS
v3	10/06/21	Updated the 5.0.2 Boots In-store Equipment - Measured Force Topple Test Certification BTC001	DH
v4	10/06/21	Internal changes - Version not published	RH
v5	7/12/22	Definition of furniture. Introduction of ESC & IR process - Revised sections shown in red	RH
v6	01/06/26	Amends to electrical, sustainability and safety standards & other minor definitions. Appendix C, D & E added.	RH



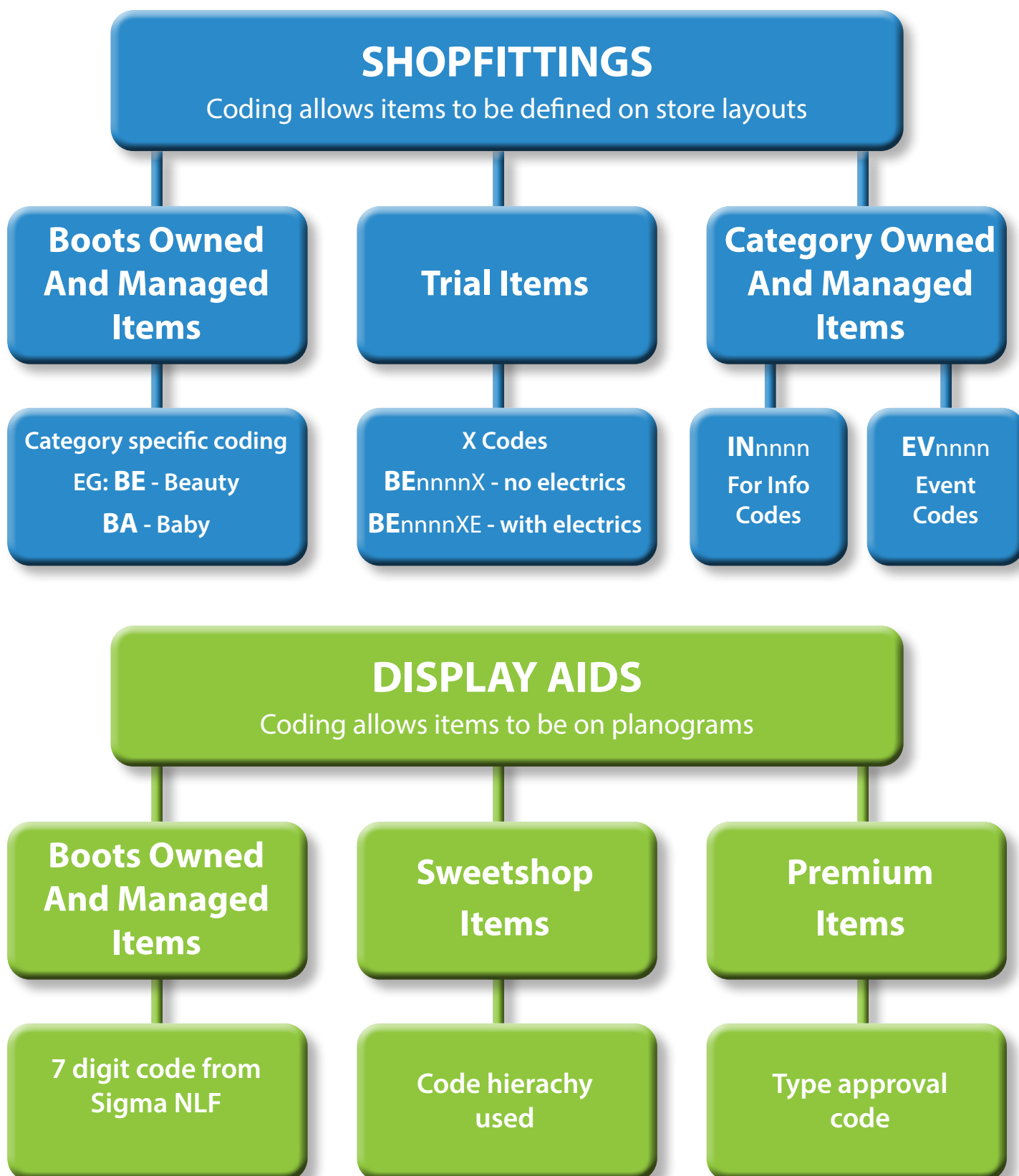
Foreword - Boots Standard Project Process

Delivery of Boots new retail equipment and GNFR (goods not for resale) projects follow a standard format and the illustration below shows the process that should be used for each project.



Foreword - Boots Codes Explained

Boots uses standard codes for all equipment and display aids. The type of code issued defines the component category and use, and the following table shows the code structure that Boots use.



1

About this guidance



Section 1 About our standards

1.1 Introduction

Boots understand the benefits that creative, safe and fit for purpose shopfittings, display aids and signage play in the success of a retailer. At the same time understanding the potential impact that poor design and quality can have on our brand.

The aim of this guidance is to provide a clear outline of Boot's needs when designing equipment for our stores, and to set out a simple, robust process that allows a close working relationship between designers and manufacturers.

Boots require constant support from designers, engineers and manufacturers to design, develop and produce our retail equipment. It is our objective that with your continued support, we can accomplish this in an ever more simple, efficient and robust way. This document will assist us in achieving these goals.

1.2 Objectives

The primary objectives of the Boots ISE Code Of Practice are as follows:

1. This document will form the foundation for managing the implementation and installation of in-store equipment.
2. Where applicable to our furniture and display aids, we aim to comply with the latest:
 - BSI standards,
 - European standards,
 - North American Standards where a situation arises where a British or European standard is available.
3. Ensuring hazard identification and risk analysis are at the heart our design process.
4. Actively promoting sound environmental planning and design to reduce the impact that any furniture or equipment has on natural resources.
5. Inclusion of compliance and audit documentation.
6. Implementation of Boots ISE testing and certification standards (Boots Test Certification - BTC).

1.3 Risk assessment

The main thrust of our risk analysis process will be centered around the concept of 'Four C's'. This involves identifying hazards, risks and controls in relation to the persons who will interact with in-store furniture. These persons are:

Contractors – These are the designers and manufacturers of in-store equipment. This is inclusive of risk control during manufacture and installation into our stores.

Colleagues – These are Boots in-store colleagues who will merchandise, clean, maintain and interact on a daily basis with the installed furniture.

Customers – They interact with the in-store furniture and the designers should account for this activity during their design processes.

Children - Children view the world differently and in much simpler terms than adults and their perception of risk and danger is much less than an adults view. Furniture that a designer views as being a display, may be viewed by a child as a play ground. Therefore the designer is required to consider how children may view the furniture they create and identify control measures that, so far as is reasonably practicable, minimise or eliminate the possibility of injury

1.4 Vulnerable Persons

Any designs should consider the breadth and range of persons who may interact with an item of furniture once it is in store. Boots caters for customers who may have functional diversities, who may fall within the scope of the Equality Act (UK) and The Equal Status Acts 2000 – 2018 (Republic of Ireland), or are well beyond retirement age or are young children. All of these amazing people regularly visit our stores and it needs to be part of your design process to consider how people will interact with the realisations of your design visions.

1.5 Definitions

Design - A documented creative process undertaken by parties who are proposing and developing ideas that will subsequently be realised as items that will be manufactured as in-store furniture and display aids.

Designer – A person or party that has a design or creative input into any plans, designs, schematics, drawings, or technical documentation in relation to or in support of an idea or item intended for manufacture or display.

Manufacturer - A party who manufactures, creates, or builds the physical representation of any supplied design, drawing or plan from a designer.

Supplier – A party who, via a procurement process, sources, manufactures and / or subsequently provides or delivers any item that is incorporated into any prototype, trial, or final finished product as an output from the design process.

Boots Owned – Any item, product, software, process, image, document, plans and designs that have been determined via agreement as being the sole property of Boots. This also includes Intellectual Property and rights to said items.

Third-Party Owned – Any items, furniture, displays, objects, equipment, etc. in place within a Boots building, store or controlled area that belong exclusively to a third-party company or business.

Brand – The mark or recognised image of any product purveyor, including Boots owned / labelled or associated goods, who presents or intends to sell their wares within a Boots building, store, or on-line retail platform.

Event Space - An area within a Boots operational building or retail environment that has been allocated for product promotional activities by external parties or Boots. The space must be of a suitable size to effectively permit the promotional event and all associated displays and furniture to achieve its goals while not interfering with the day-to-day operational functionality of the building or environment within which it sits.

Free Standing Display Unit (FSDU) – A temporary structure, display or platform intended for presenting, showcasing, or advertising products or items, that stands alone and on the floor in the open retail environment of a Boots store or building and is not physically or mechanically connected in any way to another display, unit, or structure that may be considered as being a permanent fixture.

Premium – Any brand, line, product, or item that would be considered as being a 'premium selling item' above and beyond the expected recognition of items that may be considered as 'essential' or 'day-to-day' from a customer preference point of view.

Display Aid – Any item, stand, shelf, tray, frame, table, prop, or back-drop whose purpose is to present and 'showcase' any product for sale.

Shop Fitting – If a brand-new Boots store was to be opened from new, all of the fixtures and fittings placed within the store with the intention of displaying goods and items for sale, e.g. gondola, shelving, cabinets, wall cladding, back and end panels, display cases, till units, counters, chairs, stock room racking and shelving, etc. would be considered as being shop fittings.

Safety Critical Fixing – where component parts are to be joined together, where heavy or bulky items are secured to or within a display, or a display or item of furniture is to be secured to the fabric of a building, any fixings utilised to achieve this aspect are classed as being 'Safety Critical Fixings'. The fixings MUST BE CLEARLY IDENTIFIED on any design, schematic or installation plan. Where a safety critical fixing is to be used in relation to the display item itself, the intended location of that fixing MUST be indicated on the equipment to ensure that the installer knows exactly where a safety critical fixing is to be used. Once in position, the installer is required to make a photographic record of the fixing in place and this is to be kept as part of the records of any installation.

2

Boots standards overview



Section 2 Boots standards overview

Boots require furniture and equipment to comply with all relevant specification standards in each of its geographical locations. It is a requirement of working with Boots that our suppliers understand and apply all geographical standards and statutory regulations to products supplied and fitted within the Boots estate.

2.1 Standards relating to furniture (UK and Republic of Ireland):

BS EN 16122:2012 Non domestic storage furniture – test methods for the determination of strength, durability and stability.

BS EN 16139:2013 Test Level 2 Furniture. Strength, durability and safety. Requirements for non-domestic seating.

BS EN 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites. Performance rating of any material must meet the 'Medium Hazard' category.

TBGE1 - Shopfittings, Signage and Displays –Electrical guidance.

BS7449 Specification for inclusion of glass in the construction of furniture, other than tables or trolleys, including cabinets, shelving systems and wall hung or free standing mirrors.

BS EN 15372:2023 Furniture — Strength, durability and safety — Requirements for non-domestic tables.

BS EN 1730:2012 Furniture — Tables — Test methods for the determination of stability, strength and durability.

UL 962 – Standard for Safety – Household and Commercial Furnishings

In order to provide concise test criteria for our retail estate, Boots have quantified the type and method of test required for retail fixtures and provided details of the BTC (Boots Test Certification) codes later in this document. It should be noted that the relevant BTC tests stipulated by Boots are based on the above standards and it remains the vendors responsibility to ensure compliance to all standards, at the same time as satisfying Boots own test criteria. All tests should be carried out by a qualified person or specialist, documented and recorded with both video footage and photography to prove test validity.

2.2 Environmental policy and sustainability objectives:

- Boots expect environmental consideration to be part of each retail project. An understanding of the life of the furniture is required from creative through manufacture, use and finally disposal, and this needs to be considered to minimise environmental impact.
- We expect our suppliers to have robust environmental policies which are adhered to when working with Boots.
- Material should be used sparingly but sufficiently to allow the appropriate lifespan and not create maintenance issues.
- Materials should be selected from sustainable sources, with recycled content and should be reusable or recyclable where possible
- Waste to landfill MUST be reduced to a minimum.
- Flexible Magnetic cladding or displays MUST be accompanied by a statement advising how to recycle these items.
- Packaging should be considered for recycling or reuse whenever possible
- The furniture should be designed to be easily disassembled to allow for efficient recycling or reuse.
- Shipping needs to be minimised where possible, and flat pack should be considered where appropriate.

2.3 Requirements for design approval:

- Approval must be given at every stage of the design process as detailed in 4.1 design development process. Documents must be shared with Boots Property Operations team using the IR Upload Portal. Each individual design of shopfitting must have a fully completed ISE Compliance Certificate before it can be installed within a Boots property. Listing the BTC certification test standards detailed later in this document.
- Any unit requiring variation from a standard design, such as those required in specific stores due to layout or installation issues must have a ISE Compliance Certificate. Reference can be made to the "parent" design so that only the variation to design are certified. Units without compliance certificates will not be installed.

3 Safety



The Boots Group

Section 3 Safety

It is imperative that all shop fittings, displays and signage used in-store are designed for a life cycle that prevents the risk of injury to anyone who may come into contact with it. Furniture must comply to any relevant standards, be tested and approved and comply with any specific requirements of Boots, with particular attention given to the newly implemented Boots BTC certification testing.

Consideration **MUST** also be given to the age range of our customers. We cater for all age groups, from toddlers to customers well beyond retirement age. Therefore consideration of all design solutions must take into account the specific needs and considerations of our customers age range - for example: children may view our displays as a play-thing and try to climb or move elements. Therefore designs should pay special attention to the perspectives of our entire customer age range.

3.1 Processes and design considerations to minimise risk:

- Safety requirements are based upon sound risk assessment principles that identify foreseeable hazards and risks associated with the intended item and the use of control measures to eliminate or reduce to manageable levels those identified risks.
- Items of furniture or display items must by their very nature be stable and resistant to toppling or over turning. Your designs must take account of the stability of any piece you manufacture and ensure that it will not fall while in place if subject to reasonably foreseeable forces. BTC testing criteria is listed in section 5 of this guide and all equipment must pass any applicable test to achieve certification.
- All components must be rounded, chamfered or deburred to prevent sharp edges, particularly at corners.
- Tubular components need to be capped or closed.
- Adjustable and movable parts must be designed to prevent accidental operation or injury during use.
- Load bearing sections of furniture must not come loose unintentionally.
- Lubricants must be contained to prevent leaks and safety issues.
- Vertical roll fronts and sliding doors will not move independently from any position to 200mm from their closed position.
- Moving units shall have at least 100mm clearance from the floor to avoid any crush or pinch points.
- **Identification and management of risks relative to vulnerable persons (see 1.4) should be an integral part of your design processes.**

3.2 Risk assessments:

Whenever any item is designed that is to be utilised in a retail environment, a full and comprehensive risk assessment must be conducted by the designer and / or manufacturer.

The assessment process must begin at conceptual stage and risks must be identified relative to a designs function within its intended operational environment. Including how it is to function as an item, how persons are to interact with it and considerations for potential misuse, risk or injury.

The risk assessment should consider as a minimum requirement:

- **It is recommended that risk assessments are carried out by suitably qualified personnel and if in doubt suppliers should engage the services of an independent safety specialist to carry out risk assessments on their behalf.**
- Risks associated with design, e.g. stability, sharp edges, height, materials used, electrical issues, etc.
- Risks associated with transportation from manufacturing to the retail environment, e.g. weight, manual handling, etc.
- Risks associated with the in-store installation process by shop fitting personnel.
- **Risks from interactions with the "4 C's" (Ref:1.3).**
- **Risks to vulnerable peresons (Ref:1.4).**
- Specific hazards, e.g. mechanical, electrical, shelving, anticipated loadings from merchandising, Plasma or L.E.D. screens.

Risk assessments, safety and due diligence are the responsibility of the supplier and must be carried out, documented and supplied to Boots as part of final documentation and information issue. All risk assessments should be completed and based on approved, finished equipment designs and updated should substantive change (that affects risk profiles) occur.

It is the vendors responsibility to ensure that change management processes include updated risk assessments where required. Should an amended risk assessment not be carried out it is Boots assumption that the vendor has considered all original risk assessments to be adequate for any change or update.

Section 3 Design safety considerations

3.3 Design Safety Considerations:

Stability	Is the footprint of any base unit stable enough to fully support the weight and height of the unit without any risk to tipping, movement or falling? Has the design been considered and tested against relevant British, European or N American standards.
Crushing and Trapping	Are there any gaps, moving parts or risks where fingers, toes, feet or limbs could become crushed or trapped within or by any equipment? Has specific evaluation of children and the elderly been considered?
Weight and Lifting	Have design considerations taken into account the weight of a unit in comparison to its use and users? Has a single person lift weight limit been taken into account.
Corners and Edges	Has consideration been given to the finishing and management of sharp corners, edges or points of equipment?
Structural Fixing Criteria	Is there a requirement for specialist fixing types or methods required by equipment designs, and if so have specific instructions, risks and method statements been prepared?
Fittings and Installation	Has the security and fixing of equipment within Boots stores been taken into account as part of the design process? Ensuring that any equipment hanging, clipping or in any way fitting to Boots equipment is safe?
Specific Mirror and Glass Considerations	Has special attention been paid to the type, location and fitting of glass and mirror components? Designs should take into account Boots policy to only use mechanical fixings for glass and mirrors.
Mobile Equipment	Has special attention been paid to the type of castor, wheel or slide used in any design? Ensuring that all mobile components can be locked in place to avoid any unexpected movement in-store.
Tapes, glues and adhesives	It is Boots standards to avoid where possible the use of any kind of adhesive as a primary fixing method. Should adhesive be proposed it is the suppliers responsibility to seek specific Boots approval of material and type.
Ceiling Fixed or Overhead Eqpt And Signage	Any hanging signage or equipment that is fixed at high level requires special attention and due diligence to the type, location and method of fixing. Ensuring there is no risk of failure of either fixings or suspension fittings.
Safety Critical Fixings	Critical fixings used to ensure the security and integrity of any furniture or display (see 1.5 definitions).

Note: The above steps are not exhaustive and illustrative only, designed to bring suppliers attention to specific risk areas when providing proposals for equipment to be used in the Boots estate. It is the suppliers responsibility to carry out full due diligence, testing and risk assessments of their designs.

Section 3 Safety

3.3.1 Safety considerations:

Designs should consider the full life of any furniture from design, manufacture, implementation, use and maintenance through to removal and disposal. Safety of the relevant user must be considered at every stage. Below are a list of safety considerations. This is not exhaustive and you should react accordingly to any further issues you discover as you design the furniture for its full life cycle.

3.3.2 Designing for Customers Striking Against/ Bumping Into:

- Retail equipment you are proposing must be designed to eliminate the risk of injury should someone bump into or fall against it. There must be no sharp corners or edges. If necessary, they should be rounded off or bevelled. These considerations must be clearly identified in your risk assessment. No parts of the unit are to protrude into the aisle, or have features out of eye line which may be missed by a customer walking past.
- Avoid features of the design which may become dislodged or damaged if a customer bumps into them. Counter tops must be secured, glass shelving must be effectively held in place and laminate edges protected against impacts.
- Avoid using materials likely to chip or damage easily if bumped into or if they receive a knock from a trolley or push chair.
- Consider including corner protection in the overall design, as Boots stores are very busy environments and retail equipment is heavily used.

3.3.3 Stability.

- Any display item especially if it is free-standing, **MUST** be stable by design and **MUST** have a degree of resistance to toppling if struck by a person, roll-cage, shopping trolley, or mobility scooter, etc.
- Tall items with narrow base-widths are intrinsically unstable. To improve stability, aspects such as larger base-plates, joining items together to increase footprint size, use of stage weights or the unit having a heavy base should be investigated.
- Any base plate **MUST NOT EXCEED 5mm IN THICKNESS / GAUGE**. This minimises it becoming a trip hazard in itself.
- The same is evident where an item is 'top heavy'. **THESE ARRANGEMENTS MUST BE AVOIDED AT ALL TIMES.** (Top Shop In-queue merchandising unit is sobering reminder).
- Any item of equipment **MUST** be reviewed against the testing requirements as described in Section 5. Each test that relates to the design item specifically, must be undertaken.
- Once testing is completed, Appendix 1 – BTC Compliance Certificate must be completed in full (including the N/A boxes). This is your attestation that applicable tests have been successfully undertaken.

3.3.4 Slips, Trips and Falls:

- If liquids are to be used with the fixture; e.g. if hand washing, or testing with customers; then consideration should be given to what happens if liquids fall on the floor. The risk of spills should be designed out of the fixture.
- Avoid any aspects of the design which may cause a trip hazard at low level. This is especially important when considering the use of storage drawers in units, how can these be safely used to ensure that customers and staff do not fall over them if they are left unattended.
- No cables should be allowed to be trailed across the floor unless fitted beneath appropriate conduits or trunking Ref WBA 1 (E).
- All cables within a cabinet or shop fitting should be installed correctly and retained within the unit. No cables should be hanging within the units in areas which may cause tripping hazards or entanglement Ref WBA 1 (E).
- Litter can also be a trip hazard in store, so any fixture that encourages interaction with a display, such as product testing, should also include a bin for the disposal of wipes or tissues if they are used with the fixture.
- Fixtures should not leak liquids or powders onto the floor around the fixture as this can also cause slips to occur. Cosmetics powders are especially dangerous on hard floor surfaces.
- Projections and Tripping Hazards - Designs should avoid the use of features that are out of eye line, in particular those low to the ground, and which may be missed by customers or staff when manoeuvring near fixtures, thus creating tripping hazards.

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- Designs should also limit or avoid the use of cantilevered elements that protrude into aisles or paths of circulation, in particular any that may occur at the head height of children. All drawers and doors should naturally close where possible and be held in the closed position until used.
- Stability - Fixtures and displays are to be solid and stable and shall not fall or topple over if leaned upon or bumped into under normal use. Equipment should be designed to withstand predictable impact from shoppers, shopping carts or standard cleaning equipment. Furniture should be shown not to topple over at less than 15 degrees and the actual angle of topple should be shown at the approval stage. IE: equipment should return to its original vertical position when pushed to an angle of 15 degrees.
- Fixtures must have an adequate footprint relative to the height of the unit. In general, a 1:3 ratio (base to height) may be applied as a starting point, but this is not a hard and fast requirement, and should be evaluated in combination with other design features.
- Designs should also establish a low centre of gravity, located within the overall foot print of the unit.
- Levelers should be used whenever possible to help level and stabilize the fixture in cases when the floor is not level. Levelers should leave 6mm reveal when retracted. Levelers should have a top adjustment; either slotted or hex with an access hole above if possible.
- Any temporary flooring should have a minimum slip resistance of 30 wet PTV and should not be between 7mm and 22mm high.
- All edging must present a minimum 30 points LRV difference to flooring.

3.3.5 Stacking/ Storage:

- The retail equipment must remain stable at all times, and stock should be able to be displayed securely without it falling off the display and causing injury.
- No display element should be able to be easily pulled away from the unit, everything should be securely located in place, with the exception of display trays.
- The use of retractor cords must be carefully considered as these can easily destabilise displays if pulled hard.
- Do not Locate heavy items at height or where they could cause injury if they fell onto a person.
- All parts of the display unit should be securely fixed together, so that they will not come apart in-situ.
- You must consider that low level podiums or staging may be attractive to children and could entice them to climb upon it. Your designs should take this factor into consideration.

3.3.6 Health:

- Toxic, corrosive, harmful or irritating substances must not be used in the unit.
- A COSHH assessment is required for any substance associated with the manufacturing process or any substance associated with the use of the final product.

3.3.7 Mechanisms:

- It is essential that any moving parts included in the design do not cause injury by trapping, crushing, drawing in or shearing injuries.
- Doors, drawers, access panels or other mechanisms should be so designed to prevent any injury through use or 'predictable' misuse.
- Access panels which are opened for access to powered, moving components should have isolation switches fitted to minimise the risk of injury once the panel has been removed.
- Any panel that should only be opened by trained staff or specialists should be clearly labelled and be secured in place using a mechanical fixing.

3.3.8 Thermal / Fire:

- Adequate ventilation of any item likely to generate heat must be incorporated into the design of the unit, to ensure that there is no unnecessary build up of heat which may cause fire or product degradation.

Section 3 Safety

- Any item which is likely to get hot which can be exposed to the touch must be tested to ensure that the item doesn't get hot enough to burn or shock, or cause product degradation.
- Any apparatus whose purpose is to heat solutions for application to skin MUST HAVE A THERMOSTATIC CUT-OFF SWITCH AS PART OF THE OPERATIONAL CONTROL OF THAT APPARATUS.

3.3.9 Water:

- Where any display requires the use of water, e.g. for washing or cleaning, the aspect must be included in your risk assessment. You must consider how you will transport clean water to the unit, how it will be stored, and how dirty or used water will be captured, stored and removed from the shop floor.
- You must also consider how spills will be managed and have a ready supply of disposable cloths to mop up any spills immediately should they occur.
- A manual handling assessment may be required for the activity of moving and handling volumes of water.

3.3.10 Pressure:

- There should be no risk of a build up of pressure in the unit, which may cause a risk of explosion, implosion or injection.

3.3.11 Chairs and stools:

- Any chairs, stools or tables to be used in Boots stores must comply as a minimum to the relevant British and European Standards. Documentation to confirm compliance must be provided before approval will be given.
- Where stools, chairs or other seating is provided for customers or colleagues to sit upon, Boots require that you include in your risk assessment the European / British Standard reference number for the seating used.
- The furniture must be stable by design and be capable of supporting a minimum weight of 110Kg / 18 stones.
- Your risk assessment must consider the stability of these items in terms of persons climbing on and off them.
- BS EN 16139:2013 Test Level 2 Furniture. Strength, durability and safety. Requirements for non-domestic seating.
- BS 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites.
- BS 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites.
 - Performance rating of any material must meet the 'Medium Hazard' category.
- Castors are not permitted on any chairs or stools used in Boots retail areas.

3.3.12 Electrical:

- All electrical components should comply with TBGE1 - Shopfittings, Signage and Displays –Electrical guidance' as specified on the Boots Property Central website (www.bootspropertycentral.co.uk).
- This specification is updated regularly and it is essential that upon commencing the design of a new fixture or fitting the current specification is located and used for the design of that fixture.
- Any fixture that doesn't comply with the electrical specification will not be approved.

3.3.13 Glass:

- All glass used in store must comply to the requirements identified in section 4.7 of this guide.

4 Design guidance



Section 4 Design guidance

Boots design requirements are constantly evolving in line with our retail needs. Relying heavily on both internal and external resources for creative, development and production solutions.

4.1 Design Guidance:

When generating new design solutions for Boots, safety and risk management must be of prime importance. The safety of our customers and staff is paramount, therefore the following guidance should be applied during the design of any Boots equipment:

- Furniture must comply to relevant standards or statutory requirements and to any specific requirements of Boots.
- The age range of our customers should be taken into account - from babies and toddlers through to persons past retirement age..
- Size, weight and stability of our equipment is a key consideration.
- Design safety is based upon the knowledge that furniture and its parts are likely to cause injury when they are heavy and fall. This is possible if floor standing units overbalance, wall or screen mounted units become detached, heavy parts become detached from units or units collapse due to failure.
- **Safety, stability, material selection and usage, transportation and handling from point of manufacture to store implementation should form part of any new design strategy.**
- **Designs must consider any potential risks related to the life cycle of any furniture or equipment component.**
- The scale, volume and life-span of any equipment designed by Boots should form a key part of any new design strategy.
- CSR, material selection, recycling and end of life reuse or disposal strategies should be incorporated into any new design strategy.
- Any new designs should be subject to a design risk assessment prior to any approvals or prototype. This should then develop into a full risk management and assessment of finished products.
- The objective of the design development process is to establish clear expectations regarding design guidelines, technical specifications, development and approval processes, compliance testing methods and other requirements relating to the development of fixtures and displays for Walgreens Boots Alliance.

4.1.1 IP, Data and Documentation:

It is a requirement that any new design generated for Boots is owned by Boots. Therefore any designer, consultant, vendor or other, generating or providing designs to Boots agrees unconditionally that all original intellectual property, copyright and design right is owned by Boots. Unless specifically agreed and approved in contractual form by Boots and the originating party.

Upon completion of the design process it is a requirement that all originating parties provide the following original documentation and data files to Boots for inclusion in Boots design and equipment records:

- General arrangements, exploded views and drawings in PDF format.
- Risk assessments in PDF format.
- Operations manual in PDF format.
- Installation manual in PDF format, if assembly is required.
- Final, production ready 3D CAD files in either Solidworks or Inventor format.
- Production drawings in either DXF or DWG format.
- Include any associated artwork or cutter guide details in high resolution PDF or original source format

All documentation and data should be the latest version and complete. All Solidworks files should include parts and assemblies in their original, linked file structure and be editable and production ready.

4.1.2 Change Control:

Any changes, modifications or updates carried out by any supplier or consultant should be reflected in a design change control record. With fully updated drawing and data-sets, risk assessments, operations manuals and CAD data updated and supplied to Boots immediately upon completion.

It is the vendors responsibility to ensure Boots receive all updates.

Section 4 Design guidance

4.1.3 Design Process: It is Boots intention to establish a standard approach to the design process, with the ability to modify or add further stages should specific projects require this. Should this be the case, your Boots project team will advise and agree any deviation of approach at the briefing stage so that all parties are clear and aligned with any specific process.

The standard Boots design stage process is illustrated below:



bootspropertycentral.co.uk - 'Boots Property Central', 'BPC' or 'ICON'.

Boots use an online database for store specifications and for managing design projects. All design projects undertaken for Boots should be recorded on BPC as a record of the design process. This ensures that Boots can prove 'due diligence' processes have been followed in the case of an audit or litigation. For access and training on how to use BPC, please request from your Boots contact.

The responsibilities of the supplier when using the ICON system are summarised in this document

Section 4 Design guidance

4.1.4 Brief and response to brief:

- The Brief contains all the essential information that a supplier will need to understand and quote for the design development project.
- The Supplier is expected to be proactive in responding to the brief. Boots expect the Supplier to provide a formal, detailed 'Response to brief', including a budget set against each of the design development stages outlined in the brief, plus critical timeline.
- Following is a basic list of information that should be included in a brief. Understanding in detail each of these points will allow the designer or supplier to make informed decisions, and for Boots to get best results and ROI from design solutions.

Design intent:

- A full understanding of the design intent is required covering the full customer experience, expectations including its use, who will interact with it, how they will interact with it, for how long, what are the priorities in terms of use, what are the aesthetic expectations and restrictions.

Design and trial requirements:

- An explanation of the specific design and trial requirements, location, date, period of time, expected outcomes against KPI's.

Budgets:

- What is the budget for the equipment? This should include design IP, manufacture, delivery, installation, removal, disposal and an understanding of ongoing maintenance and running costs. Is there a specific ROI target?

Quantity:

- What quantity of units are required? Are they to be manufactured in a single order or in batches? What is the BAU quantity required after an initial manufacture run? Spares should be considered during main production to leverage of economies of scale. Volumes that directly affect the most efficient production methods and tooling investment should also be considered.

Delivery dates:

- What is the target delivery date? A critical path should then be generated to include all design lead times, approvals, testing, manufacture and delivery. The critical path should be approved by the Boots project team and help guide on-time project delivery.

Expected life-span and durability:

- What is the target life of the equipment in a store environment? Is there a requirement for guarantees or maintenance agreements? If so is it possible to design solutions that are available with guarantees?

Where to be used:

- Different countries have different standards to comply with. If geographical information is included in the brief a design solution should be generated that can adhere to local standards and be adapted to for market specifically.

Manufacture/Assembly:

- Where is the furniture to be manufacture? Is this UK manufacture or overseas? Who is assembling the units? What is their level of skill? Do they have access to tools? Where is the furniture to be assembled? Place of manufacture, before delivery, on-site? What is the QC process? **NOTE: It is Boots policy for TOOL-FREE assembly of equipment by store colleagues. IE: Any equipment that is specified to be built in-store by Boots store staff should be designed for assembly with no requirement for any tools or equipment unless expressly agreed as part of the project design.**

Freight and transport:

- What is the most commercially viable location to produce the equipment? If it is in the Far East has consideration been given to materials, construction, shipping volumes and flat pack or knock-down shipping methods?

Section 4 Design guidance

Delivery:

- What is the intended delivery strategy? Straight to store? 3rd party logistics? Installation team?
- What is the delivery date, is this date to store or to manufacturers for assemble? Are there any long lead time items which need to be considered which may affect this delivery?
- Who is delivering the units?
- Will the units be delivered by post, through Boots own logistics, courier or installation partners?
- What are the restrictions in weight and size, what packaging will be required?

Other delivery considerations:

- Size restrictions, no back shop space so unit has to go through the front door, store is upstairs with no lift or a small lift.
- Weight restrictions, no fork lift and no one in the store to help delivery.
- Time restrictions, on a high street or mall, night time delivery only additional costs for out of hours staff and security.

Installation:

- Who is installing the units, store staff, installation specialists, shopfitters, electricians?
- What is their level of qualification, competence and do they have professional indemnity insurance?
- It is Boots ambition that installation can be carried out by a non-specialist team.
- Does the work have to be done out of hours?
- What are the restrictions?
- Out of hours restrictions?
- Working at height restrictions?
- Noise restrictions?
- Access equipment restrictions?

Storage:

- Is the furniture to go straight into all stores or go into warehousing?
- Are there storage restrictions on weight, pallets, size etc?
- What packaging is required?
- Is the furniture seasonal and will it require storage in store and assembly/disassembly annually?

Spares/maintenance:

- Is there a maintenance program? What restrictions do they have, working at height etc?
- What is the level of competence?
- Who will hold spares, the number of specific spares, where will these be stored, how will they be implemented?

Removal:

- Who will be responsible for removal and is there a value to the equipment after its life in store? Has a process been set up for removal?

Disposal:

- What statutory requirements are there? Is there a value to the equipment after its life in store? Is it recyclable?

Environmental:

- Are there project specific environmental requirements for the equipment or does it fall within Boots environmental policy?

Section 4 Design guidance

4.1.5 Project Deliverables:

This section should detail the physical design output at each stage of the design development process detailed in the brief, including:

- Insight / research / site visits & surveys.
- Presentation visuals - Hand-drawn or 3D visuals.
- 3D walkthroughs.
- CAD layout / production / details drawings.
- Reports / Presentation documents.
- Models / Prototypes.
- Risks Assessments / Method Statements.
- Specifications / guidelines.
- Operating & Maintenance manuals and A working Risk Register, which is updated throughout the process.
- Change control logs and decision logs to be kept.

4.1.6 Project Timeline:

- The Supplier must provide a full and detailed programme of works, showing the start, finish and approval periods for all design stages.
- The Supplier should be realistic with regards to approvals periods. Discussion must be held with the Boots Design Manager as to the appropriate time to allow for approvals. 'Drop dead' dates and critical deadlines should be included.
- The Supplier should include all stages which they have been asked to quote for as clarified during the briefing process.
- Critical approval dates and deadlines will be included in the timeline in agreement with the Boots Design Manager.

4.1.7 Fee Proposal:

This section should detail the full fee proposal for the project, broken down into the design stages outlined in the brief, including:

- Design costs are expected to be capped at the number of hours work required, by the standard rate per hour as agreed with Boots. Boots will then choose a supplier with which to progress the brief into the remainder of the process under the agreed rates. Provision should be made for the creation of a risk register during the Concept Design stage. The Supplier and Boots Design Manager will monitor the risk register throughout the project.
- Provision for a reasonable number of design changes to occur at the Concept Design stage in order to collaboratively explore all relevant avenues. This can be discussed with the Boots project team when creating the Response to Brief.
- The Supplier must be clear about what exclusions apply to the fees. If no exclusions are detailed, it will be Boots understanding that any reasonable design works required will thus be included within the fee proposal, and available to be carried out if requested.
- An all-inclusive fee proposal is preferred but if expenses are to be included in the proposal all expenses must be fully detailed, and once approved no other expenses will be allowed without approval from the Creative Manager.
- Boots standard payment terms will apply in all cases.

Section 4 Design guidance

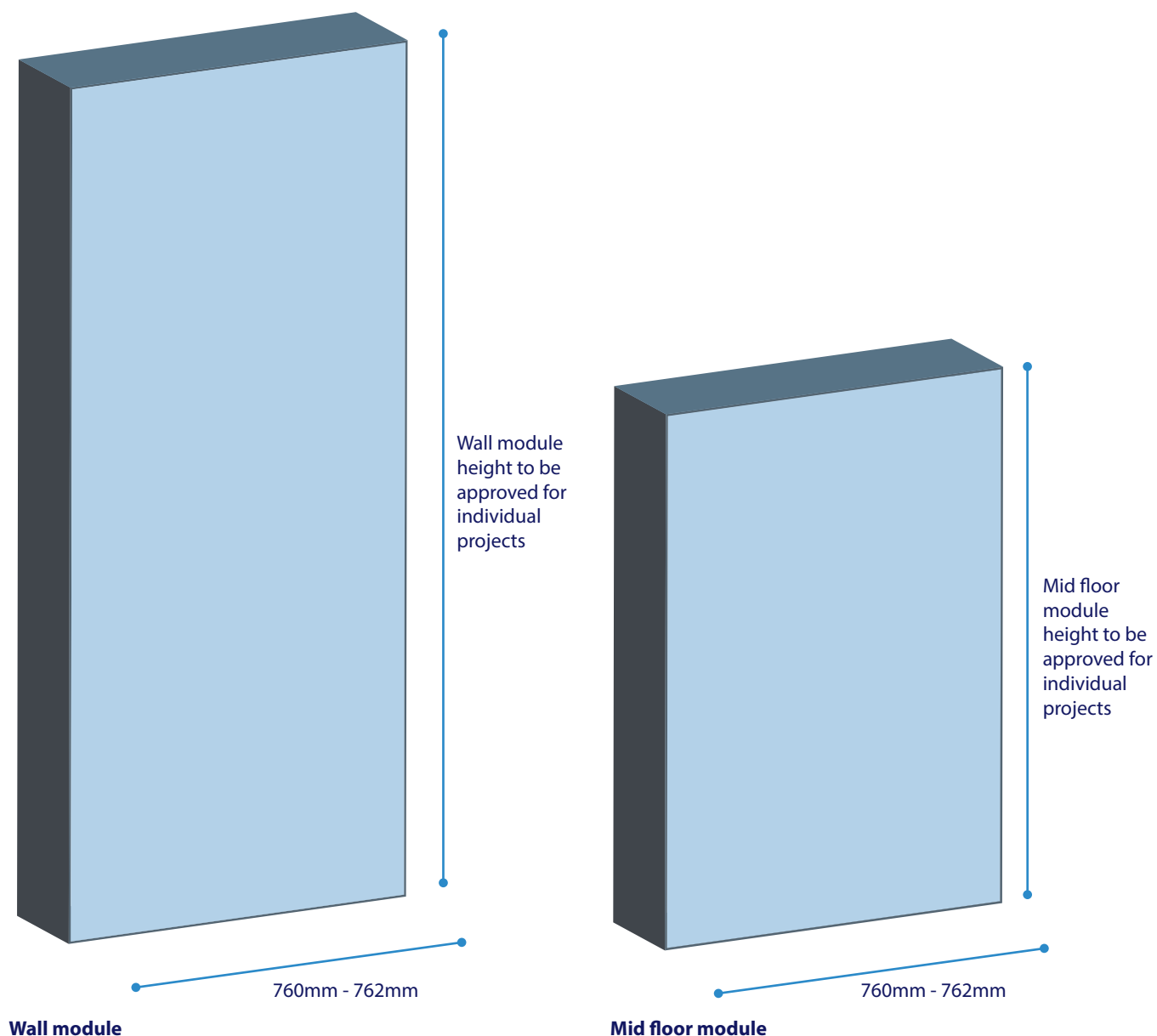
4.1.8 Supplier Appointment:

- The Boots Team will review the Response to Brief's and will select one or more Suppliers to continue to the development design stage of the process.
- The Boots Project Team will initiate a purchase order to be raised to confirm selection of the preferred Supplier.
- No work will be undertaken by the Supplier until a purchase order is raised, or a letter of intent is received.
- Any work undertaken by the Supplier prior to receiving the purchase order for the project is undertaken entirely at their own risk.
- It is expected that the Suppliers that work for Boots should have the ability to carry out all relevant design development work in-house, negating the risks associated with subcontracting to third party suppliers. However, we understand that in some cases Suppliers need to sub-contract out specialist services. If this is the case the Supplier must confirm full details of what services these are, and the details of the organisation and persons involved, to allow the Boots Design Manager to understand the implications for the project.
- It is expected that Suppliers will have suitable premises that can host design presentations, demonstrations and other client meetings in a suitable, professional environment.
- It is the responsibility of the Supplier, if appointed, to deliver a design solution which meets the requirements of the brief. Failure to do so may result in withholding of payment.
- No design work is to be undertaken by the Supplier unless instructed by the Boots Design Manager. If the Supplier receives instruction from other areas of the Walgreens Boots Alliance group of companies, the supplier must seek confirmation from the Boots project Team that this work is approved and should be undertaken. Failure to do so will result in the Supplier being excluded from consideration for all future design development project work.
- A Supplier will not be paid more than what was proposed in the 'Response to the Brief' unless they have gone through the agreed change control process and received the appropriate approvals from the Boots Design Manager.
- Boots withholds the right to cancel any and all services with the Supplier at any stage of the process.

Section 4 Design guidance - module height and width

4.1.9 Module Height and Width Guidance - Project Specific:

Boots have module height and width restrictions for design of equipment used within stores. The illustration below shows the standard width and height for Boots merchandising shelving and any equipment that is required to fit to the standard shelving should work within these module sizes.



When designing any equipment to fit into the slotted uprights of Boots equipment (Boots code 563686 gondola T frame and 563589 perimeter post), it is the vendors responsibility to ensure that the bracket and fitment is correct and matches Boots standard brackets.

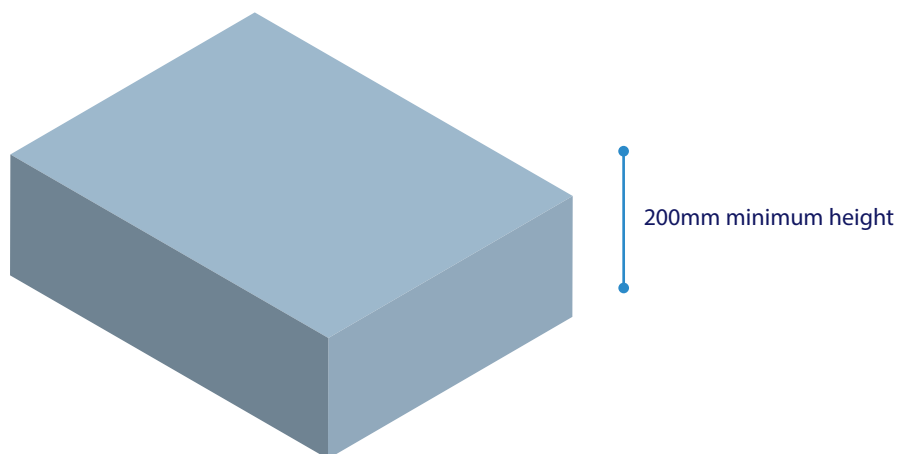
Suppliers and designers should also be aware that Boots have had three system variants over many years across their estate. In some variants there is a +2mm difference to the centre line of the module size. This should be taken into account and fully tested on all variations of equipment as suitable to task because oversize modules can cause “creep” along a run, with subsequent fitting issues.

Any equipment designed for Boots should be checked against 3D models of Boots GLF, GL and GF modular systems to ensure compatibility across all three system types (with drawings available for reference on ICON).

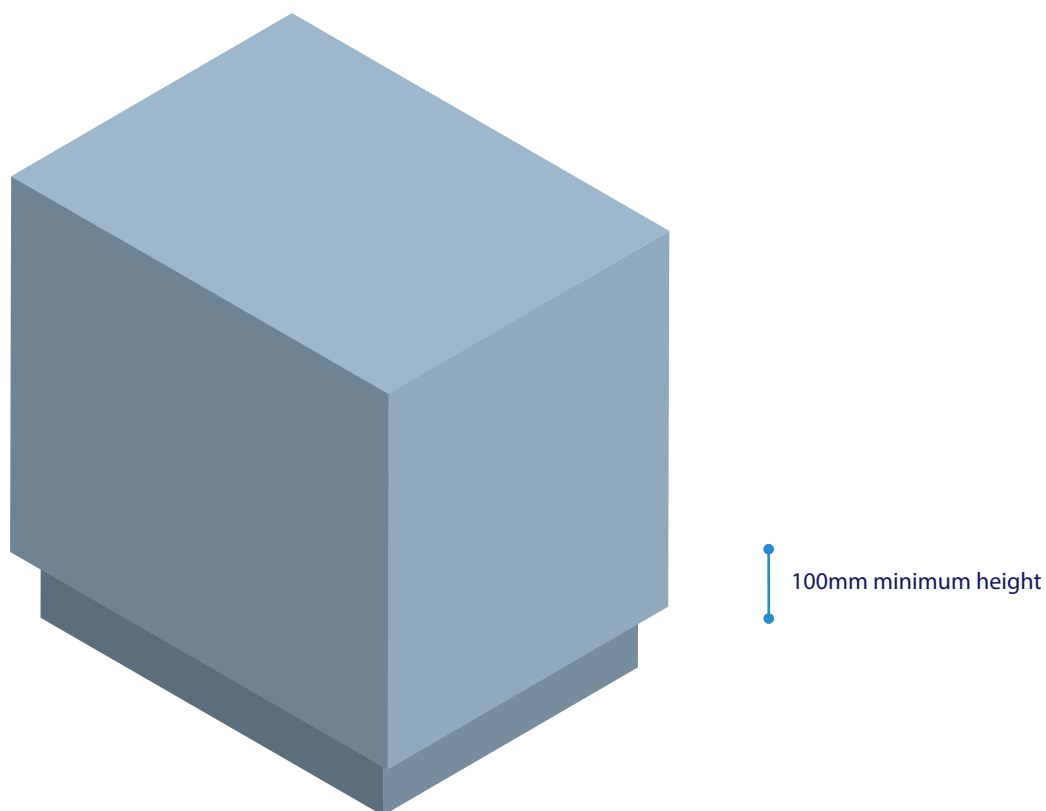
Section 4 Design guidance - minimum heights

4.1.10 Display Plinth and Recessed Plinth Height Guidance:

To alleviate specific safety concerns, Boots have minimum requirements for the type and size of display and recessed plinths. The following guidance should be followed when designing equipment for use in the Boots estate.



Display Plinth



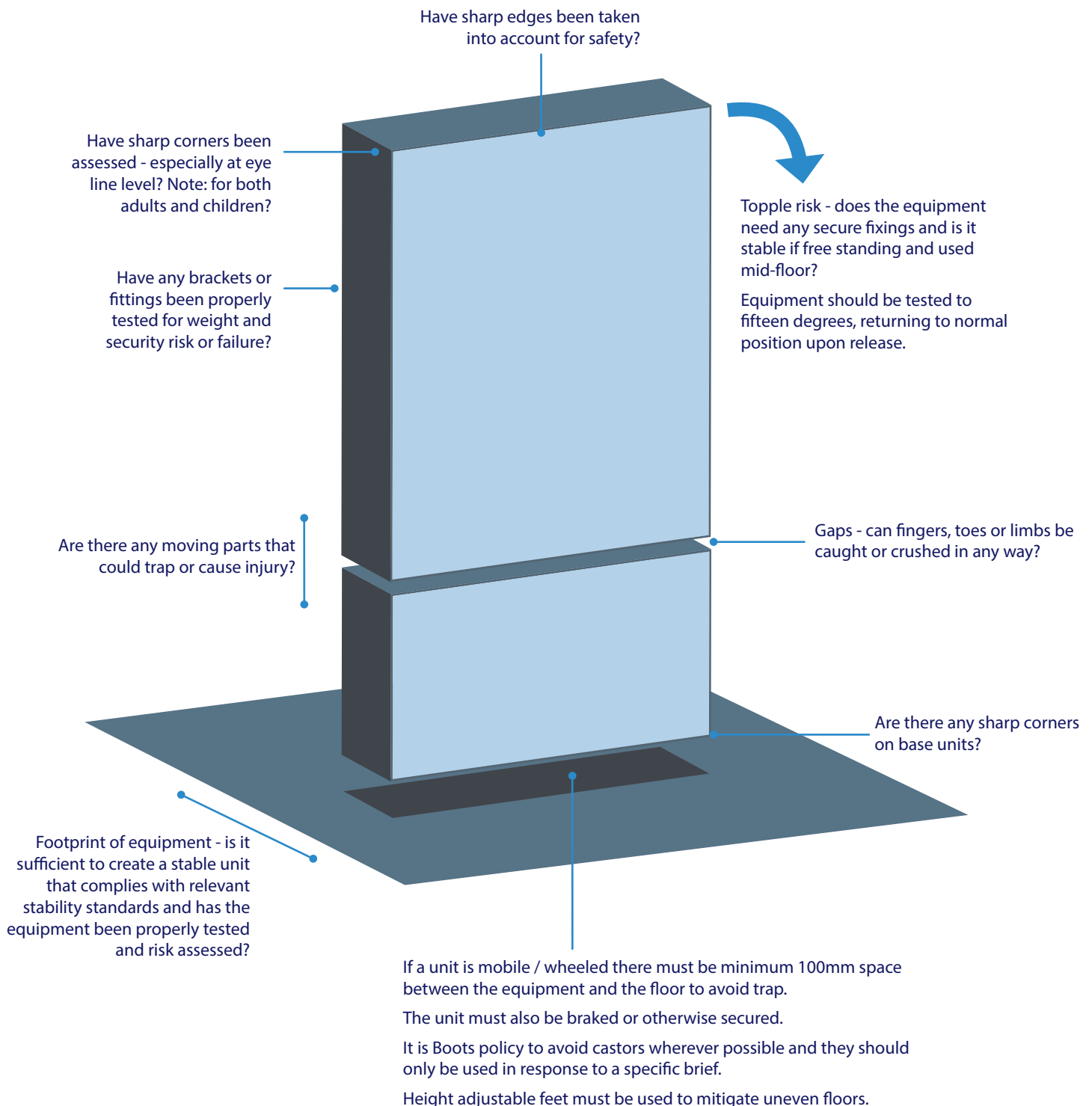
Recessed Plinth



Section 4 Design guidance - design risk illustration

4.1.11 Design Risk Illustration

When designing equipment for use in Boots stores, special attention should be paid to the safety and risk management of any equipment that could cause damage or injury to our customers or store personnel. The following illustration shows points and features to be considered, noting that it is not exhaustive and that it is the designers responsibility to carry out a complete design risk assessment.



Section 4 Design guidance

4.1.12 Project set up:

The concept design stage commences following appointment of a Supplier, and generally starts with a project kick off meeting attended by any relevant stakeholders, Project Manager and the appointed Supplier. The purpose of the meeting is to review the brief as a project team, discuss any issues and agree the details of the project going forwards. The meeting should be recorded in a set of minutes by the Supplier, approved by the Boots Design Manager.

The meetings should confirm or agree changes to the project scope and response to brief.

Suppliers and the Boots Design Manager must discuss the expected extent of concept development required at the briefing stage to ensure sufficient fee is available to collaboratively explore all relevant avenues.

The process for presenting concept proposals to and approvals from the wider Boots Project team will have been agreed at the kick-off meeting. It is the responsibility of the Boots Design Manager to facilitate these meetings and approve the record of decisions made in the form of meetings.

4.1.13 Preliminary Development Design:

The Supplier will develop initial development designs to an outline stage which demonstrate fully the functions, aesthetics, dimensions and general layout of a solution, generally having first been presented with a series of options. At this stage designs will be usually presented in sketch/ visual form such as hand drawings, simple computer models, cardboard mock ups etc.

Boots will expect the Supplier to be prepared to change and adjust the concept design a reasonable number of times in order to get to an approved design solution.

Boots would expect the Supplier to produce a final concept pack at the end of the stage called the 'Design Intent Pack'. It is expected that the concepts would be presented in a series of hand-drawn or 3D computer visuals, sufficient in detail to fully explain the concept proposal to laymen members of the project team. Card mockups may be produced to support the decisions.

4.1.14 Prototype Development Design:

Initial Design Development will be approved in writing by the Boots Design Manager following approval from Project team Stakeholders. Following approval of a Preliminary development design, the unit will be developed in technical detail using 3D CAD drawings to fully explore how the concepts will work in reality. There may be a requirement to update concept visuals if the detailed design impacts the visual appearance materially. This will be the decision of the Boots Design Manager, and the Supplier must allow iterations in their original fee proposal.

The expectation of Boots is that the output of this stage will be a series of 3D CAD Drawings, General Arrangement (GA) CAD drawings, issued as PDF and CAD files, which resolve the design solution to a state at which production details and a working prototype can be developed.

It may be necessary to produce simple models to explain, test, or prove the operation of any complex elements during the detailed design process.

Drawings must be completed and developed to the Boots Drawing Standards, 'Boots Technical drawing & documentation standards for shop fittings and display aids' document. Please refer to Boots Property Central for the latest version.

The process for presenting design proposals to and obtaining approvals from the wider Boots Project team will have been agreed at the kick-off meeting. It is the responsibility of the Boots Design Manager to facilitate these meetings and approve and record any decisions made in the form of meetings, updating the design brief as required.

Section 4 Design guidance

4.1.15 Prototype & trial:

Based upon the approved Prototype drawings an appropriately finished prototype/s to fully show the final production finished units will be produced.

The Supplier must be explicit in the stage 1 brief response about the number and specifications of prototypes expected to be delivered in this phase of the process. In general it is the Boots team expectation that the prototypes will be delivered using the final materials specifications, sizes and shapes. If this is not possible, a detailed explanation must be issued by the contractor in the earlier phases, with proposals given on how appropriate prototyping approval can be achieved;

All development / production tooling commissioned during the design development or prototyping process will be the property of Boots, once payment is received, and this will include all bolsters in the case of injection moulding, unless specifically agreed with the Supplier. Details of the location of tooling must be provided at this stage so that the Boots asset register can be updated.

Prototype Testing:

If prototypes are developed for installation into the Boots 3D Room, then the installation into the 3D Room will need to be managed in adherence with the installation procedures as advised by the Boots Design Manager, as the Boots site in Nottingham is managed by a third party supplier.

If the prototype design is to be tested in store, the testing will run in one or more stores for a fixed period of time, all of which will be arranged by the Boots Design Manager, working with the Group Property Services Team. No equipment will be installed into a Boots Store without Electrical Engineering approval and the approval of the Health and Safety Team.

It is expected that the Supplier will undertake a thorough risk assessment before any design project is trialled in store, and when a project is completed in line with Boots risk assessment guidance. Risk assessments should be presented at prototype approval stages.

If agreed as being within scope of the project, it will normally be the responsibility of the Supplier to install the prototypes within the stores.

The Supplier must provide a schedule of works and appropriate risk assessments to Boots Group Property Services prior to undertaking any work in store to ensure that they are working within Boots recognised procedures.

If the project trial is to be removed from the store after the trial period, any equipment removed from the store during the trial period will be stored by the Supplier and refitted into the store at the end of the trial period by the Supplier. The removed equipment will be protected during removal and storage and will be returned to store in the same condition that it was removed.

It may be necessary to update the prototype in store once the trial has proceeded. This work will not be included within the initial fee proposal, but must be priced and approved by the Boots Design Manager prior to any work being undertaken.

If the unit requires assembly, that assembly should be timed to allow costs of installation to be calculated. Assembly level required should match proposed production units.

Prototype packaging should be as close as production as possible and consider the agreed performance criteria of packaging.

Where rapid prototyping is used to represent injection moulded or other difficult to fabricate parts, resolution must be suitable to evaluate part design. Moving parts and mechanisms should be representative of final design materials and function. Hinges, snap-fits, and slides should be of sufficient quality to test their function and of a material with similar qualities to final production material.

All lighting and electrics should match production. Where substitutions are made for timing reasons, they shall be clearly noted on the prototype. An appropriate production replacement should be made available as soon as possible.

All graphic elements should have an appropriate image supplied by the supplier.

Section 4 Design guidance

4.1.16 Trial learnings:

At the end of the agreed testing period, the Boots Design Manager will facilitate a project team review meeting with all project team stakeholders and contractors. The findings of the test period will be fed into discussions which will determine whether any adjustment to the design is required.

Time to undergo a proper design review will be included in the project timeline.

Following review of prototypes, it may be necessary to adjust final production details. If so this will be confirmed in writing by the boots Boots Design Manager, with exact details of the adjustments. The Supplier will then make the changes and issue a full production pack to Boots.

If rework of the design is required following testing, this work will not be included in the initial response to brief and will be subject to a separate brief and response to brief treated as a revision to the initial brief.

On receipt of a final pack of drawings and specification, the Boots Design Manager will confirm to the Supplier in writing that this stage is complete and the drawings are approved.

The final pack of drawings and specifications will be used as the basis for a tender for the manufacture of the design for in-store installation.

Drawings must be completed and developed to the Boots Drawing Standards, 'Boots Technical drawing & documentation standards for shop fittings and display aids' document. Refer to Boots Property Central for the latest version.

If the testing is deemed to be a success against the pre-determined success criteria, then the development process will continue to the next stage.

Section 4 Design guidance

4.1.17 Tender pack:

Working to the trial learnings provided from the Boots Design Manager, stage 7 is the Project Closure process which Boots follow to close down a project when it is complete. It is called the initiative release process.

The Supplier must upload the design onto the relevant section of the project folder in www.bootspropertycentral.co.uk to allow Boots to use the full technical specification going forwards.

The supplier must adhere to the 'Boots Technical drawing & documentation standards' section of this document.

The following information is required to be uploaded to the ICON system:

- Complete set of manufacturing drawings in electronic format for uploading onto the Boots specification website 'Boots Property Central'.
- A drawing register for large projects where multiple technical drawings will be issued.
- JPEG images.
- Risk Assessment forms, to the Boots specification, and any associated testing documentation.
- All Information relating to electrical circuitry as specified in the document 'Electricity and Illumination within Shopfittings, Display Aids and Signage', refer to Boots Property Central for info.
- An Operation and Maintenance manual, with any replaceable parts identified and marked with the appropriate Boots shop fitting or display aid code.
- Assembly and disassembly instructions.
- Simplified CAD cells for layout planning in electronic format.

The folder structure for the files uploaded onto ICON will be:

- 01 – Drawings PDF.
- 02 – Drawings DXF.
- 03 – JPEG Thumbnails.
- 04 – Risk Assessment.
- 05 – O&M Manual.
- 06 – Assembly Disassembly Manual.
- 07 – CAD Cell.
- 08 – Electrical Content. IE:
 - Electrical Schematic.
 - Component Technical Specifications With CE Certificates.
 - PAT Test Results.
- 09 - Original 3D CAD Engineering.

Section 4 Design guidance

4.2 Change control process:

A reasonable number of changes should be accounted for within the Response to Brief.

These should be discussed with the Boots Brand Design Team prior to the Response to Brief being received by Boots, as each Brief will be variable in structure and complexity.

Any 'response to brief' which has not taken into account the need for extra time to create changes will be viewed unfavourably by the Boots Project Team.

Any extra-ordinary changes to the Design Brief or during the Concept or Development stage may be discussed with the Boots 3D Design Team in order for the Supplier to cover unforeseen changes.

Figure 1 demonstrates guidance to the Supplier for 'reasonable' and 'unreasonable' changes to the Design within the concept and development stages.

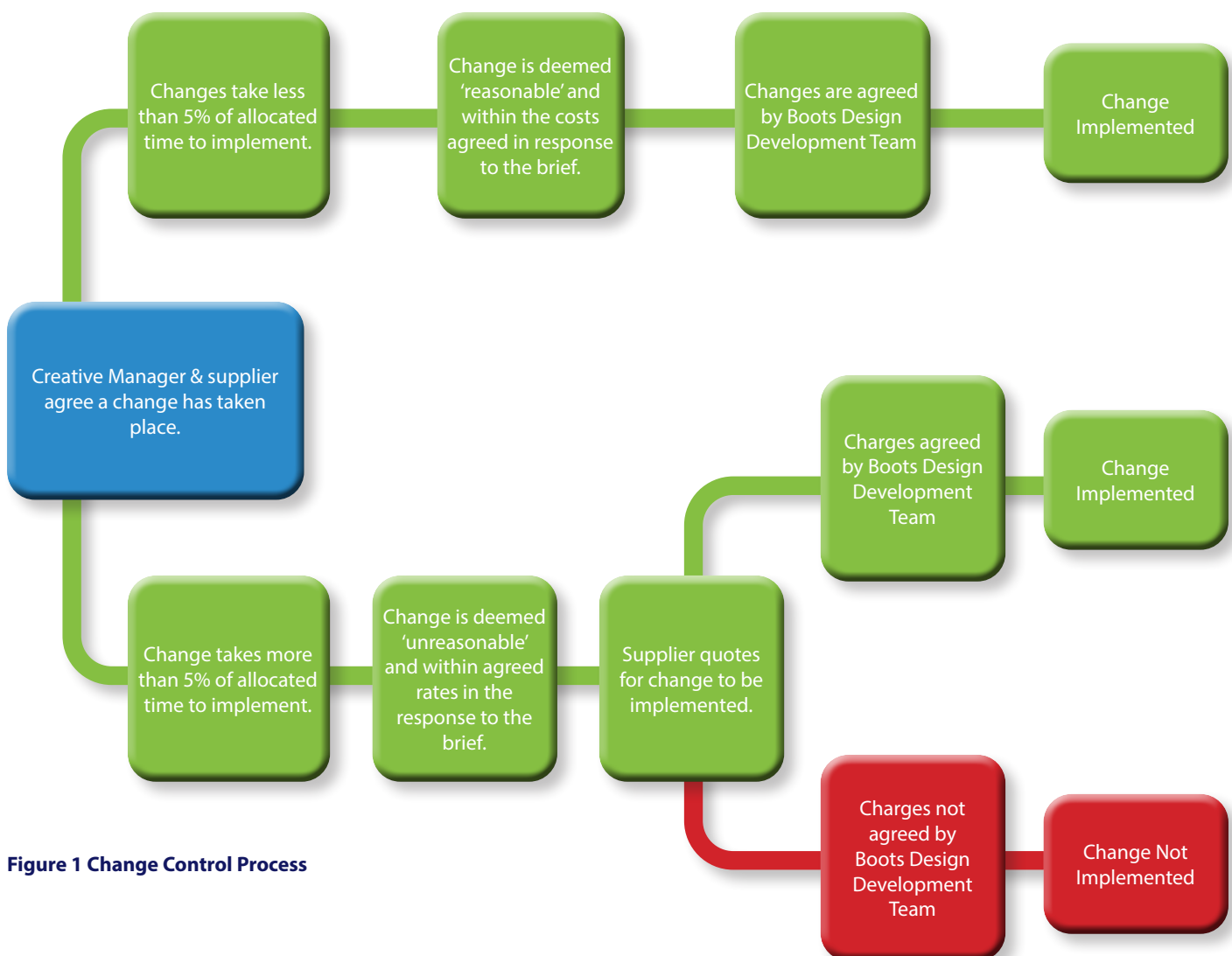


Figure 1 Change Control Process

Section 4 Design guidance

4.3 Design drawings and instructions:

Boots require to own the IP on all furniture designed specifically for them. The ability to re-create these designs consistently is an important element within Boots store development. In order to achieve this, it is crucial that we have complete drawing packs and full specification data, in digital form, for all our fixtures so that we can make swift and accurate design, development and procurement choices.

Supporting documentation is also essential to meet escalating Health & Safety requirements and to protect customers, our business, and our suppliers.

Additionally possessing a complete set of technical drawings and specifications, digitally, allows us to promptly provide all suppliers with identical information resulting in equal tender conditions.

Being able to instantly access complete and up-to-date information, via the Boots Property Central website, will help us all operate our businesses in a professional, cost effective and competitive manner.

4.3.1 Technical drawing:

The following is a list of the minimum information required to complete technical drawing sheets. All drawing dimensions, notation & information must be legible when printed on A4 paper size (**see Appendix A - Drawing Pack Examples**).

All sheets must have:

- Title block to be fully populated including:
 - Shop fitting & display aid code.
 - Supplier name.
 - Drawing number.
 - Drawing description.
 - Materials & finishes (i.e. surface finish, coating, colour reference & gloss level).
 - Project name.
 - Drawing creation date.
 - Drawing revision (in ascending alphabetic order, initial drawing issue must be revision 'A').
 - Drawn by.
 - Weight.
 - Scale.
 - Sheet number & total number of sheets in the drawing pack.
 - Assembly and component codes in line with Boots coding requirements.
- Accurate tolerances that are applicable to the parts and materials / scale used. For example:
Joinery tolerances +/- 1mm per 1000mm.
Metalwork tolerances - Linear: +/- 0.5mm per 1000mm, Angular: +/- 1.5°.

Section 4 Design guidance

Continued...

Third party and 'off the shelf' items must be clearly identified and marked with the appropriate supplier details, including:

- Supplier name.
- Item description.
- Reference number.
- Direct URL link to the items web page.

First sheet :

- Perspective or isometric view of assembled unit.
- No dimensional information required.
- Revision history stating:
 - Current drawing revision.
 - Sheet numbers affected.
 - Revision descriptions.
 - Revision dates.
- Drawing numbers of related parts or assemblies.

General assembly & sub assembly sheets:

- Plan, front & side elevations of the assembly in third angle projection. Sectional elevations should be included wherever relevant to aid understanding. Rear & underside elevations are optional, but should be included when useful. All elevations must be drawn to the same scale.
- Isometric view of the assembly.
- Overall dimensions of the assembly.
- Dimensions of assembled counterparts & any other references viewed as important or useful.
- Fixing details (e.g. Fasteners required, welding details etc)
- Performance criteria (e.g. maximum shelf loading).
- Assembly & manufacturing instructions
- Drawing numbers for related parts or assemblies that are detailed in separate files/folders.

Exploded assembly sheets:

- Exploded isometric view of the full assembly.
- BOM (Bill Of Materials) table numerically identifying each part. Including:
 - Numerical identifier.
 - Part number (if available).
 - Part description.
 - Drawing number (if available).
 - Quantity.

Section 4 Design guidance

Continued...

Individual part drawing sheets:

- Plan, front & side elevations of the assembly in third angle projection. Sectional elevations should be included wherever relevant to aid understanding. Rear & underside elevations are optional, but should be included when useful. All elevations must be drawn to the same scale.
- Fully dimensioned, including:
 - Angles.
 - Thicknesses.
 - Radii.
 - Hole specifications.
 - Critical dimensions (with tolerances applied).
- Drawings of folded metal parts must include an additional flat pattern elevation, with overall dimensions, drawn at the same scale as the main elevations.
- Fixing details (e.g. Fasteners required, welding details etc).
- Multiple show materials, which should always located on the last page, can be collated onto one sheet. Each identified by name or part number with any shop fitting & display aid codes also displayed.

Tolerances:

Where any specific manufacture tolerance affects the fit or build of any equipment, this should be clearly noted and “called-out” on relevant production drawings and specifications. In addition designers should note the 760mm bay width Boots use as standard and ensure that their equipment tolerance takes into account modular “creep” over multiple bays. And ensure that their equipment is properly sized and tested to ensure compliance across all three Boots modular system types.

Electrical specification sheets *

***Refer to latest issue of TBGE1 - Shopfittings, Signage and Displays –Electrical guidance’ available on the Boots Property Central website, for current electrical specifications to be adhered to.**

After uploading to the Boots IR upload portal, all drawings must not be altered or updated without approval.

4.3.2 3D Original Manufacturing CAD Information:

All design cad used to generate manufacture information must be issued to Boots upon final completion and approval. CAD should be in original, production ready manufacture format, with all edit points left usable. CAD should include (in original folder structures):

- All component parts.
- All sub-assemblies.
- All main unit assemblies.
- Any specific or generic part information required to utilise the CAD for production.

It should be noted that all production 3D CAD and data set information will be validated as accurate and production ready under Boots “blue-seal” manufacture control systems. Should it be found that CAD supplied is not accurate or suitable for production, Boots will require the originating supplier to either correct to their satisfaction or pay the costs of completing the CAD to acceptable standards.

Section 4 Design guidance

Continued...

4.3.3* Initiative release process (IR) - *Applies to Boots owned designs only

Why upload?

Throughout the development of new in-store initiatives we will require the associated Initiative Release documentation to share with the Development Design Team and at certain points of the project with a wider selection of key Boots stakeholders. Previous we have used Boots Property Central (ICON) to upload all documentation for approval, however this process has been complicated for some of our suppliers and caused delays to the project timeline. Therefore, please use this method to upload files for Boots to share internally. Once we have all approvals, the data will be transferred to ICON. This system is not a replacement for Boots Property Central (ICON) as this is where the data will sit for BAU.

When to upload?

There will be key stages of the project where we require specific information. These may differ depending on the specific project, but generally this would be during the initial development stages, pre & post prototyping and stages. Uploading allows us to track the development of projects and ensure that our data is current.

What to upload?

Typical documents to upload are:

- PDF Drawings (Including GAs, elevations, sections, details, component lists, specifications etc)
- DXF Drawings (Including GAs, elevations, sections, details, component lists, specifications etc)
- 3D CAD model (.step)
- H&S Test Certificates (Ensure that all tests are carried out in accordance with the standards set out in section 5 of this document)
- O&M Manuals (Covers all maintenance and cleaning procedures, identifies replaceable parts, installation and assembly guidance)
- Sustainability Assessment (Product life cycle, environmental, economic and social impact, renewable materials & recycled content)
- Electrical Schematics (TBGE1, CE Certificates, PAT Tests, schematic drawings, photos, component libraries)
- ESC Certificate (Electrical power consumption, sustainability score, carbon assessment, total life environmental impact)*

*Currently only a requirement for electrically powered elements

How to upload?

We have set up strict file naming principles that ALL digital files must be named. It is very important that all files are clearly named. Boots item codes should be referenced at all times, rather than supplier codes and project name included also. If Boots codes have not been provided, please contact the Design & Development manager responsible for the project. Please note that the underscore symbol should be used to separate the file name.

File naming format: Boots code_Boots Project Item Description_Document Description.File type

Example - PL0210_Zoom Core Gift 1 Shelving Unit C_GA.pdf

Document type code guidance

- GA = General arrangement drawing (PDF or DXF)
- SCH = Electrical schematic drawing (Word or PDF)
- 3D = 3D CAD model (STEP or similar)
- O&M = Operation and maintenance documentation (Word or PDF)
- IM = Installation documentation (Word or PDF)
- RA = Risk assessment (Excel or PDF)
- TC = H&S test documentation (Word or PDF)
- CE = CE Electrical certificate (Word or PDF)
- PAT = Electrical PAT test (Word or PDF)
- PH = Photo (JPG or similar)
- KOP = Kit of parts (Electrical components)

Section 4 Design guidance

Continued...

Where to upload?

Access to the IR upload portal is via a web browser, making it easy for suppliers to share information. In order to gain access, you MUST ensure that the email address of the person uploading the data is registered on the Boots IR Upload Sharepoint site. If not, you will need to request access rights to the upload folder. Once approved and you will receive a notification that you have access to the upload folder below.

Initiative Release Upload Zone - External Uploads

Please note - Access to this folder is restricted.

Note - Suppliers must have their email web domain registered in order to submit documentation via this method.

Currently ONLY files can be uploaded to the upload folder. We are working on a solution to allow folders and sub folders and will update this document when we have a solution in place. Please do not upload ZIP folders!

Note: When a file is uploaded, it is automatically copied to another secure area on our SharePoint site, however this can take a few moments to process. You will not have access to this transfer folder and the original document that you upload will not be visible to you or any other supplier once processed.. A confirmation/error notification email will be issued automatically per file uploaded. In order to minimise email volume in your inbox our recommendation would be to set up a mailbox rule to divert all emails from BootsInitiativeReleaseUpload@boots.co.uk to a separate folder.

4.3.4 Electrical compliance & the TBGE1 Standard - (WBA1E has been superseded by TBGE1)

Overview of TBGE1

All electrical fixtures and fittings supplied to The Boots Group (TBG) must be designed, manufactured, and verified in strict accordance with the TBGE1 Technical Specification. This document serves as the primary driver for compliance with UK and Ireland electrical legislation (including BS 7671 and the Electrical Equipment Safety Regulations). Beyond statutory safety, TBGE1 mandates high-efficiency component selection and standardised wiring configurations and components to ensure estate-wide consistency and reduced operational energy consumption.

UKCA Legal Requirement

The Manufacturer holds the sole legal responsibility for the safety of the product. Compliance with TBGE1 provides the technical foundation required for the manufacturer to legally apply the UKCA Mark (or CE mark for the Republic of Ireland) and issue a formal Declaration of Conformity (DoC). TBG does not accept self-certification at face value; the UKCA mark must be supported by the technical evidence defined within the TBGE1 framework.

Electrical approval process

The submission process follows four mandatory stage gates:

- 1. NAPIT Training & Competency:** Manufacturers must complete a NAPIT electrical compliance course to ensure alignment with UK and BGEE standards. NAPIT offers courses for Electrical Supervisors and Designers. (Note: Manufacturers without an internal qualified electrician may appoint an independent third-party electrician who has passed the NAPIT Boots Electrical Supervisors course). Course and contact details are provided in the appendix.
- 2. Mandatory Technical Filing:** Manufacturers must compile a comprehensive UKCA Technical File for every new design, including:
 - General Assembly drawings (showing isolation and accessibility).
 - Wiring diagrams (loadings, fuse/wire ratings, and components).
 - A NAPIT Boots Electrical Test Certificate (signed by the qualified Electrical Supervisor).
 - Instructions, safety information, and risk assessments.
 - Declarations of Conformity from component suppliers (if applicable).
 - UKCA Declaration of Conformity (signed by a company director) covering all relevant elements.
- 3. Technical Submission & Desktop Review:** Manufacturers must submit their complete Technical File to BGEE (certification@boots.co.uk) for a free Desktop Review prior to manufacturing. If compliant, BGEE will validate the qualified supervisor's sign-off by countersigning the NAPIT Electrical Test Certificate. This countersigned file becomes the official record of the approved unit.

Section 4 Design guidance

- 4. BGEE Audit & In-store Verification:** During production, BGEE will conduct ongoing testing and inspections of physical in-store fixtures to verify they match the countersigned **Technical File** exactly, ensuring consistent quality and operational safety. This process offers increased speed-to-market and professional autonomy. Removing the external sign-off bottleneck and utilizing a free Desktop Review reduces lead times. Furthermore, the mandatory NAPIT training equips manufacturing teams with accredited expertise in retail-specific electrical safety.

This process offers increased speed-to-market and professional autonomy. Removing the external sign-off bottleneck and utilizing a free Desktop Review reduces lead times. Furthermore, the mandatory NAPIT training equips manufacturing teams with accredited expertise in retail-specific electrical safety.

Legally, this aligns with the Electrical Equipment (Safety) Regulations 2016 and Electromagnetic Compatibility Regulations 2016. As the entity placing goods on the GB market, the manufacturer holds full legal design and safety liability for UKCA compliance, including latent defects. BGEE acts as an auditing body—providing independent validation of the accredited supervisor's assessment—to ensure brand standard verification, but this countersignature does not transfer legal liability. Boots will retain a copy of the countersigned Technical File for insurance, safety, and due diligence.

Manufacturer Submission Checklist:

- Certification: Current NAPIT Compliance Course completion (internal or independent).
- Technical Pack: Detailed electrical drawings, wiring schematics, and risk assessments.
- UKCA Declaration: Valid Declaration of Conformity for the specific fixture.
- NAPIT Test Certificate: Completed by the qualified Supervisor, ready for BGEE countersignature.
- Physical Labeling: Permanent electrical label and Safety Inspection Mark applied to the unit.

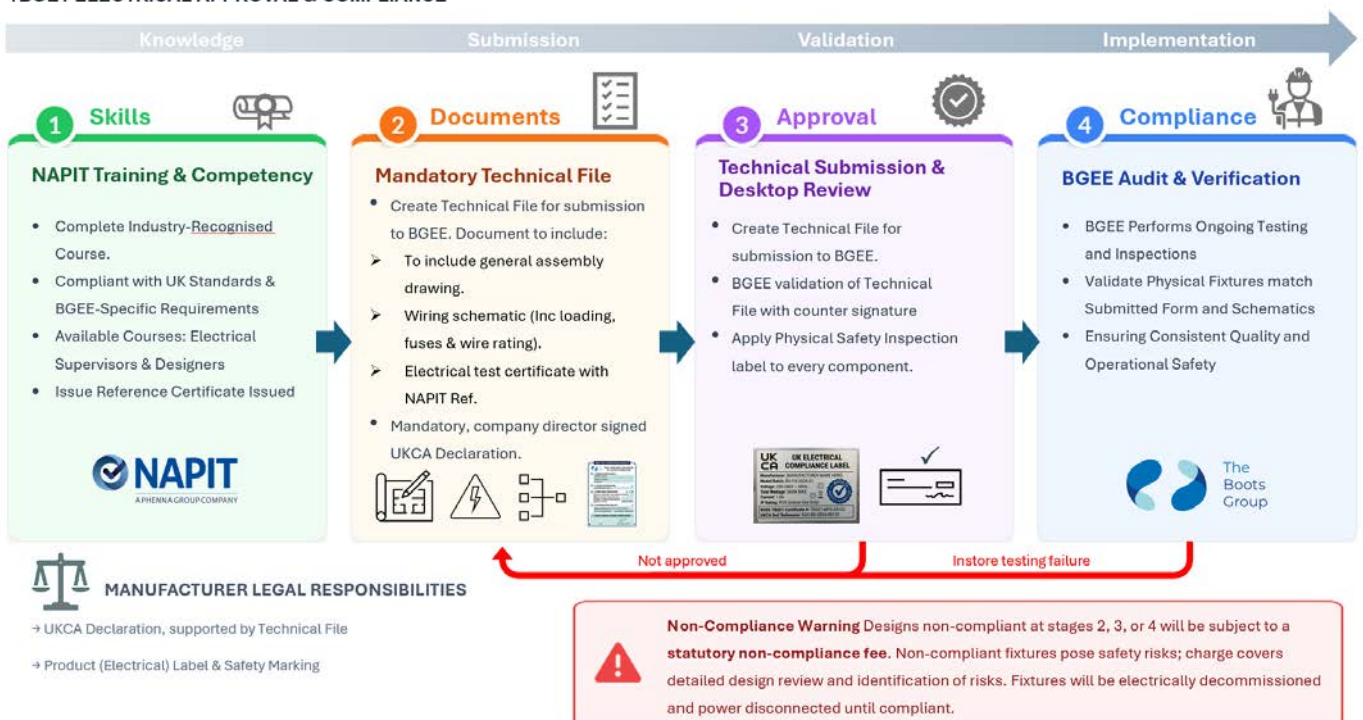
Non-compliance

Failing to comply at stages 2, 3, or 4 incurs a statutory fee. Submissions failing the initial Desktop Review due to missing or incorrect documentation require manufacturer rectification and mandatory BGEE site visits to review the physical unit.

The fee is set at £995, payable to Design Conformity, covering additional design review and safety risk identification costs. Furthermore, physical in-store fixtures deviating from the countersigned Technical File will be immediately isolated, decommissioned, and potentially removed from store at the supplier's expense.

Electrical approval process map

TBGE1 ELECTRICAL APPROVAL & COMPLIANCE



Section 4 Design guidance

Continued...

SCHEMATIC BOM				
ITEM	SUPPLIER CODE	DESCRIPTION	QTY	BOOTS CODES
1	HJ-ELS2001	1m ngn switched fused spur Non-switched 5amp fused spur 1000mm LSF to Female GSI	1	ELS2001
2	HJ-ELS2006	ELS2006 WELAND CONNECTOR 1/5 GSI distribution block	1	ELS2006
3	PROTOTYPE	MOLVENO ROCKER SWITCH	1	PROTOTYPE
4	HJ-ELS1015	Power Plate 75W/24V Input cable 250mm 3x1 0mm ² cable to GSI Output cable 60mm 2x1 0mm ² cable L504	1	ELS1015
5	HJ-ELS2016	1.5m mains connection driver link cable ELS2016	1	ELS2016
6	HJ-ELS1017	3 Way JST Splitter Input 250mm cable end with JST VLP connector Output 2x250 cable end with JST VLP connector	1	ELS1017
7	HJ-ELS1020	10 way distribution block (VH-XH) L504 2x1 0mm ² black cable double insulated Input 250mm JST VLP male connector Output 10pcs JST XHP-2	1	ELS1020
8	HJ-ELS1260	14 Runners cable Input 1770mm L504 2x1 0mm ² double insulated black cable, end with 2P VLP connector Output 4x DC 5.5/2.5mm right angled jack connector socket	1	ELS1260
9	HJ-ELS1261	Under shelf LED strips in AL extrusion and diffuser 695mm Light bar 2W/m, 200lm/w, 128LEDs total, CR80, 4000K. Without endcaps, lens to be full width, no hole on the back of extrusion, White 3A cable, length 200mm, end with JST XH white male connector Size: 695x13.2x11mm Installation way: metal clip & rivet	4	ELS1261
10	PROTOTYPE	2pc Splitter 22AWG L504 double insulated white cable, white connectors Input 200mm DC 5.5/2.5mm (straight type) jack connector Output 100mm XH female connector and 100mm DC 5.5/2.5mm socket panel mount type	4	PROTOTYPE
11	HJ-ELS1262	Corcoca LED strips in AL extrusion and diffuser 720x22x2mm, Light bar 2W/m, 200lm/w, 128LEDs total, CR80, 4000K. Without endcaps, lens to be full width, A hole on the Aluminium profile for cable passing through, White 3A cable, length 200mm, end with JST XH white male connector Installation way: metal clip	3	ELS1262
12	PROTOTYPE	Corcoca LED strips in AL extrusion and diffuser 720x22x2mm Light bar 2W/m, 200lm/w, 128LEDs total, CR80, 4000K. Without endcaps, lens to be full width, A hole on the Aluminium profile for cable passing through, White 3A cable, length 200mm, end with JST XH white male connector Installation way: metal clip	2	PROTOTYPE
13	HJ-ELS1263	16LED Light Panel 542x241mmx4mm 24V 4000K 4w Lightsheet with LED along one side and 500mm lead to DC 5.5/2.5mm (straight type) jack connector, white cable, white connector	1	ELS1263
14	HJ-ELS1264	16LED Light Panel 300mmx147mmx4mm 24V 4000K 4w Lightsheet with LED along one side and 1300mm lead to DC 5.5/2.5mm (straight type) jack connector, white cable, white connector	1	ELS1264
15	PROTOTYPE	3000mm cable JST VLP-DC socket 5.5/2.5 female white cable, white connector a panel mount DC socket. Mounting to a steel plate	1	PROTOTYPE

SUPPLIER HENG-JIE

Boots Part Code

EL - Component
ELS1 - LV subcomponent
ELS2 - ELV subcomponent

Bill of Materials/Components

Item number
Supplier code
Description
Quantity

Supplier

INSTALLATION INSTRUCTIONS		SAFETY INSTRUCTIONS	
INSTALLATION - LED'S MUST BE MECHANICALLY FIXED IN PLACE - CABLES MUST BE RESTRAINED AND TIDY WITH NO ACCESS - THE LED DRIVER MUST NOT BE INVERTED AND MUST HAVE ADEQUATE VENTILATION MAINTENANCE 1. ALL ELECTRICAL PRODUCTS MUST BE TESTED BEFORE AND AFTER INSTALLATION BY QUALIFIED PERSONNEL HOWEVER IT IS THE RESPONSIBILITY OF STORE STAFF TO IDENTIFY ANY DAMAGED OR NON-FUNCTIONING ELECTRICAL COMPONENTS DURING THE LIFE OF THE PRODUCT IN STORE. FAULT IDENTIFICATION WHICH WILL COME UNDER SAFETY IDENTIFICATION OF ELECTRICAL ITEMS, POTENTIAL FAULTS AND ISOLATION LOCATIONS MUST BE CARRIED OUT DURING INDUCTION TRAINING. TYPICAL INDICATORS OF POTENTIAL FAULTS AND FAILURES WILL INCLUDE: - ANY BLACKENING ON ELECTRICAL EQUIPMENT - TRAPPED OR DAMAGED CABLES, CONNECTIONS OR PLUGS - FLICKERING LIGHTS OR LIGHTS NOT WORKING - ANY USE OF DRAWING IS SUBJECT TO DESIGN CONFIRMITY TERMS AND CONDITIONS.		1. ELECTRICAL PRODUCTS SHOULD ONLY BE INSTALLED BY SUITABLY QUALIFIED PERSONNEL 2. THE ELECTRICAL SUPPLY SHOULD BE ISOLATED PRIOR TO ANY WORKS BEING CARED OUT 3. ENSURE THAT NO CABLES ARE TRAPPED OR LAID ACROSS ANY SURFACE EDGES DURING THE INSTALLATION 4. ENSURE THAT THE UNIT IS TESTED FOR SAFETY AFTER THE INSTALLATION	

Boots UK Nottingham NG2 3AA

PROJECT TITLE: FRAGRANCE CABINET

DO NOT SCALE - IF IN DOUBT ASK

UNLESS OTHERWISE STATED ALL DIMENSIONS ARE IN MILLIMETRES (mm) DIMENSION TOLERANCES UNLESS SPECIFIED AS FOLLOWS

SCALE: 1:12

SHEET NO: A3

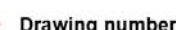
SHEET: 4/6

Safety Instructions

Installation Instructions



General Assembly Drawing



Bill of Materials

Section 4 Design guidance

Continued...
Electrical Certification

Electrical Test Certificate

Test Certificate



Tests Conducted By: Design Conformity Ltd, Western House, Western Street, Hockley, Nottingham, NG1 3AJ

Site: MAXFIELD JARVIS
Location: L812 BLD

Asset: 102279 Description: CHANNEL4AG FRAGA Testcode: 11Q300801 G181B11111
Make: Model: Serial No:

Test	Result	Limit	Status
Visual		NA&A	Pass
Earth Current	100	mA < 5	Information
Earth	0.1	Ohm	Pass
Earth	0.06	Ohm	Pass
Earth	0.07	Ohm	Pass
Earth	0.05	Ohm	Pass
Insulation	199.89	VOhm	Pass
Leakage	<0.1	mA	Pass
Load	0.18	kVA	Information
Polarity			Pass

Comments: For full ac & dc load tests see dc102279 test record sheet

Overall Status: Pass Date Of Test: 23/03/2028

Test Instrument: 18B-0133 Tested By: MSN

Signed By: 

- Earth Continuity
- Insulation resistance
- Differential Leakage
- Load Test
- Polarity

- Name and number of product
- Relative directives
- Tests
- Company directors' signature

UKCA Certificate



GB DECLARATION OF CONFORMITY

Company Name
Company Address

Declares that under our sole responsibility the product

Name and product number

is/are in conformity with the provisions of the following UK regulations, including all amendments:

UK SI 2016 No. 1101 The Electrical Equipment (Safety) Regulations 2016
UK SI 2016 No. 1091 Electro Magnetic Compatibility Regulations 2016

UK SI 2019 No. 539 The Ecodesign for Energy-Related Products and Energy Information (Amendment) (EU Exit) Regulations 2019
UK SI 2012 No. 3032 Restriction of the use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012

The following designated standards have been applied

BS EN 05080-1:2015 Luminaires. General requirements and tests
BS EN 05080-2-4:1998 Particular requirements. Portable general-purpose luminaires
BS EN 55015:2013+A1:2015 Limits and methods of measurement of radio disturbance characteristics of electrical lighting and similar equipment
BS EN 61000-3-11:2001 Electromagnetic compatibility (EMC). Limits. Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems
BS EN 61000-3-2:2014 Electromagnetic compatibility (EMC). Limits. Limits for harmonic current emissions
BS EN 61000-3-3:2013 Electromagnetic compatibility (EMC). Limits. Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems
BS EN 61547:2009 Equipment for general lighting purposes. EMC immunity requirements

EXAMPLE STANDARDS SHOWN - CORRECT PRODUCT STANDARDS TO BE ADDED

The Technical construction File is maintained at the above address

Authorised signatory:

Name: Add Name Note: the person signing the DCD takes full legal responsibility for the product and any prosecutions
Position: Add Position
Date: Add Date issued

The TBGE1 is included in full in the appendix

Section 4 Design guidance

4.3.3 Life cycle assessment (Sustainability)

To align the design and development of new fixtures and fittings with the wider Boots Group ESG policies, we have introduced a mandatory requirement for a sustainability report indicating the environmental impact of all new equipment in our stores. This covers Boots owned fixtures and third party fixtures. Temporary and eventing fixtures are managed via the Boots In-store Marketing team.

Boots have partnered with POPAI, who are a not for profit organisation, driving sustainability standards within the retail industry. To measure the environmental impact of our in-store fixtures, we require all fixture designers and manufacturers to use the Sustain® environmental design calculator.



[POPAI Sustain® Environmental Design Calculator](#)

Sustain® is an indicator tool that works to Life Cycle Assessment (LCA) principles and gives an independent analysis of the environmental performance of displays and point of sale. It provides a vital framework for reporting, measuring factors including CO2e, water, recycled content and recyclability of display and packaging.

It is recommended and endorsed by leading brands and retailers and provides a reporting framework and recommendations for improvement.

The calculator tool is used by designers, manufacturers and suppliers to input individual component data of a total fixture to calculate the total environment impact. A user licence is required and pricing information is available from POPAI. Full training is also available.

All design submissions will require a final Sustain® Environmental result to accompany drawings and other documentation.

Sustain® process map



The Boots Group reserves the right to continually monitor and stipulate minimum standards to reduce environmental impact through better design, reporting and future policy compliance.

Section 4 Design guidance

4.4.1 Operations and maintenance manual (O&M) - See Appendix B - Operation Manual Example:

An operations and maintenance manual is required for all fixtures that will be used in a store environment.

The document is a manual giving an overview of the units day to day use, in-store assemble, maintenance requirements and disassemble and detail any spares available to stores.

The manual should include:

1. Introduction:

- Title page, including: Image of the unit, title including Boots code and description and document contents list.

• Introduction section, including:

Description of the units operational requirements and day to day use. A list of the units components, tools required for assembly and additional equipment if required (E.g. graphics etc).

- Unit overview section, including: an image of the assembled unit, with components labelled plus overall dimensions, weights etc where appropriate.
- Contact information of the manufacturer.

2. Health and safety:

- Important safety point section, including: highlighting all relevant risks in use, assemble, disassemble etc.
- Weight limits if applicable.

3. Operations:

- In-store assembly section, including: a full set of illustrated assemble, adjustment. (Must be tool free).
- End of life section to identify different material recycling streams and disassemble instructions.

4. Cleaning and maintenance regime.

- Maintenance and cleaning regime section, including: a list of the regime required for cleaning and maintenance of the unit and any specialist methods or materials that are required.

4.4.2 CAD blocks

To aid store space planning CAD cells or blocks are used to represent store fixtures. CAD cells are used by Boots format and layout teams, as well as architects and other contractors. Therefore the accuracy of shape, form and size is critical.

The Boots D&D team will provide suppliers a Boots fixture code. This code should be annotated on each CAD cell listed below:

CAD cells should be produced as a single drawing file with the following views saved in the model space:

- Plan view
- Side elevation and section through
- Front elevation and section through

Where feasible, please include associated components such as headers and frames. Guidance is available from the D&D team.

Drawing presentation:

- Please ensure drawing is a hidden line drawing representation of the shopfitting and not a wire frame drawing. We do not require the manufacturing details for this, as the cell is for setting out rather than construction. When there are several shopfittings within one plan we are having problems with file sizes and complexity, often one shopfitting can contain tens of thousands of elements when exported to dwg.

Section 4 Design guidance - cad cells

Continued...

- Do not obscure the cell with colour blocks, as these show through in CAD.
- The 2D representation should not be shaded, block filled, textured, hatched or patterned, or obscured in any way.

Drawing standards;

- The plan should be produced at a scale of 1:1 in the metric drawing system.
- This plan shall be a simplified version of any technical plans that are produced and supplied for sign off.
- Remove all unnecessary detail from any 3D CAD models used to generate the view. E.g. Mechanical fixings or intricate details.
- This will be used to insert into the Boots store plans.

Drawing elements:

- Location of seating/stools/chairs shall always be shown.
- The location if any till shall be clearly indicated
- Any overhangs from the furniture footprint must clearly visible
- Vision priority should be indicated if appropriate.
- The proposal must include clearly labelled services locations for power, data, drainage,
- Door swings and drawer openings must be indicated, with suitable use of dashed lines.

Annotation:

- A unique drawing number should be on every plan, this must include the Boots fixture code.
- The drawing should be clearly labelled as to the type of fitting and code.
- Version control is necessary and should be updated for all revisions.

CAD Detail:

- The drawing should be on one z plane.
- Where possible the proposal should be drawn on 1 level.
- A PDF and DWG must be supplied, for each version or revision.



Section 4 Design guidance - electronic documents and CAD data

4.4.3 Electronic drawing files:

When the design and engineering drawings have been updated and finalised, a complete set of electronic files should be submitted to Boots for archiving and quality control.

A complete set of design and engineering drawings will be submitted to Boots in the following formats:

- General arrangements, exploded views and drawings in PDF format.
- Supporting general arrangements, exploded views and drawings in DWG format.
- Risk assessments in PDF format.
- Operations manual in PDF format.
- Final, production ready 3D CAD files in either Solidworks or Inventor format.

All documentation and data should be the latest version and complete. All Solidworks files should include parts and assemblies in their original, linked file structure and be editable and production ready.

If a supplier or originator does not use the Solidworks platform, please seek approval for progress from your Boots contact prior to commencing any Boots equipment design.

Suppliers should be aware that Boots require an improved level of drawing information and data. It is a requirement that all information issued to Boots contains production ready information and data that allows immediate manufacture from the information provided at the completion of the design, prototype and testing stage. This is particularly applicable to the format and quality of the Solidworks 3D CAD information used to design and specify Boots equipment.

All Solidworks information must be issued complete and in its original structure and file and folder format, preserving all linked files and dependencies. All components, fixings and standard components must also be included within the models and data-sets.

All 3D CAD will be inspected and pre-flight checked as suitable for production prior to Boots "blue-seal" for manufacture process. This will validate all supplied CAD and should deficiencies be found it is the suppliers responsibility to immediately correct these or fund the costs of Boots specialists correcting information to an approved production standard.

Section 4 Design guidance - testing and certification

4.5 design considerations

Furniture Testing (see section 5.0):

- All equipment must be tested before being used in store to Boots BTC test standards (and also be compliant with any relevant British Standards).
- Testing is to determine the strength, durability and stability of furniture.
- Units must be tested as delivered and constructed in accordance with assembly instructions provided.
- Where drawers are tested, these should be completed with full drawer design load in addition to the weights specified in the BTC test standards.
- All testing should be completed with and without loads, extensions etc must be positioned to mimic worst case scenario when in use, this should be noted in the report.
- All test are to be carried out in the order they appear in the standards.
- A competent person must perform the testing, they should use appropriate test equipment safely (BS) complying to all appropriate standard requirements.
- On an individual unit, all safety tests to be carried out on a single unit, strength and durability may be carried out on a second unit.
- Should the supplier not have a qualified person within their organisation it is suggested that the vendor should arrange for testing to be carried out by specialists. Identifying the cost of this within cost structures.
- All Boots equipment should be tested and certified to Boots BTC standards (in addition to any relevant British Standards) and all tests should be properly documented, recorded, photographed and with supporting video footage of the tests.
- All documentation and recording information should be held on file securely by the supplier in addition to being issued to Boots.

It is the suppliers responsibility to adequately test all Boots equipment to the standards detailed above, in addition to any best practice and statutory requirements. All reports and documentation should be issued to Boots at the completion of the design stage.

If for any reason testing is deemed not to be required, this must be approved by your Boots contact and the Boots project team.

Section 4 Design guidance - testing

4.5 Design considerations:

- When engaged by Boots to complete design works we have particular parameters which need to be adhered to, specific Boots design requirements, compliance with relevant standards and environmental requirements.
- Boots require to own the IP on any furniture designed specifically for them.
- Boots Customers and colleagues rely on shop fittings, displays and signage to go about their day to day business in store. The furniture they use must be designed for a life cycle which prevents a risk of injury or physical damage.
- Furniture must comply to relevant standards and to the specific requirements of Boots stores. Consideration must be given to the far ranging age of our customers, from toddlers to well beyond retirement age and any predictable use and miss use this implies.
- When designing for Boots consideration must be made to the fact that we are a large retailer and use a significant number of displays so the use of scalability, modularity and common components can have a significant effect on not only manufacturing cost but complexity to the business.

4.5.1 Boots design requirements:

Furniture dimensions:

Shelf width standard Boots shelving

- Gondola and perimeter bay width 760mm (for standard GLF modular systems).

Section 4 Design guidance - standards and compliance

4.5.2 Standards Compliance

- It is essential that any fixture introduced into a Boots store is compliant with all appropriate international standards of manufacturing and production:
- BS EN 16122:2012 Non domestic storage furniture – test methods for the determination of strength, durability and stability.
- BS EN 16139:2013 Test Level 2 Furniture. Strength, durability and safety. Requirements for non-domestic seating.
- BS EN 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites.
- Performance rating of any material must meet the 'Medium Hazard' category.
- TBGE1 - Shopfittings, Signage and Displays –Electrical guidance.
- BS7449 Specification for inclusion of glass in the construction of furniture, other than tables or trolleys, including cabinets, shelving systems and wall hung or free standing mirrors.
- BS EN 15372:2023 Furniture — Strength, durability and safety — Requirements for non-domestic tables.
- BS EN 1730:2012 Furniture — Tables — Test methods for the determination of stability, strength and durability.

In order to provide concise test criteria for our retail estate, Boots have quantified the type and method of test required for retail fixtures and provided details of these new BTC (Boots Test Certification) codes later in this document. It should be noted that the relevant BTC tests stipulated by Boots are based on the above standards and it remains the vendors responsibility to ensure compliance to all standards, at the same time as satisfying Boots own test criteria. All tests should be carried out by a qualified person or specialist, documented and recorded with both video footage and photography to prove test validity.

Record Keeping:

- It will be assumed that records recording the design development process undertaken during the design development stage will be retained by the supplier, who will be able to demonstrate a 'due diligence' process has been followed upon request.

4.5.3 Ergonomics:

- An ergonomic assessment of the operation of equipment should be undertaken to ensure that any risk of injury is designed out.
- Assessment of ergonomics must be considered throughout a project, however proof of this compliance must be a major part of the trial stage. Compliance to relevant BS ergonomic standards is required.
- This is particularly important in relation to repetitive movements, including reaching, twisting and bending. Ergonomic assessment of the workspace design is a provable requirement. However it should be considered throughout the life cycle of the furniture.
- Design consideration should take into account both standing and seated positions and specific operator and customer use.
- Consideration should also be taken for how the unit will be installed into the store, ensuring that the manual handling of the units is considered during installation.
- All ergonomic design should conform to BS EN ISO 26800:2011.

Section 4 Design guidance - equality and environmental

4.5.4 Equality Act Compliance:

Compliance with Equalities Act (2010) is a legal requirement. All fixtures must consider accessibility as we have a legal obligation to ensure that the design's we develop can be used by all customers and colleagues, irrespective of their ability; as specified in the Equality Act 2010.

Use of Colour and Contrast:

The use of colour and contrast to identify to persons who have difficulty with visibility related issues, the location and orientation of fixtures and features, has been utilised for many years. A percentage of the population have difficulties in differentiating between colours (colour blindness). Some have difficulties with restricted depth perception and aspects affecting eyes that can obscure objects and features that are clearly visible to most persons.

However, most people can differentiate between contrast and shade. Many designers will use colour to attract and to stimulate persons to investigate further. We are not restricting you in this process in any way, shape or form. What we do require of you is that you assess your design in respect of how it may be discernible from the general surroundings in which it is located. The use of contrasting tones can be equally as exciting in the field of design as that of the use of colour, and designers should reference Boots standard flooring and wall finishes for contrast checking.

We require that persons are able to clearly see the furniture or equipment you intend to display in our stores, this is inclusive of the base where it meets the floor. If possible the whole outline would be visible, however, if key visual reference points are discernible, e.g. base, counter top, header, to enable a person to identify that there is an object present and they can determine where it is, this is acceptable.

The use of polished chrome or clear materials as a base may render the base of the item invisible to some persons with sight related disorders. Any plinth arrangement must be clearly discernible from the floor upon which it stands to allow persons to identify that there is something there that they should be aware of. In both these situations, a contrast between the flooring material and the base or plinth is required.

4.5.5 Environmental Considerations

Boots expect environmental consideration to be part of each retail project. An understanding of the life of the furniture is required from creative through manufacture, use and finally disposal and all needs to be considered to minimise environmental impact.

We expect our suppliers to have robust environmental policies which are adhered to when working with Boots.

Material should be used sparingly but sufficiently to allow the appropriate lifespan and not create maintenance issues. Materials should be selected from sustainable sources and be reusable or recyclable where possible. Waste sent to landfill is the last option in respect of waste management and your process should look to reduce or eliminate waste handled in this way.

Packaging should be considered for recycling or reuse where ever possible.

The Furniture should be designed to be easily disassembled to allow for efficient recycling or reuse.

Shipping needs to be minimised where possible, flat pack must be considered where appropriate.

Section 4 Design guidance - structure and stability

4.6 Structure:

- Boots Customers and colleagues rely on shop fittings, displays and signage to go about their day to day business in store. The furniture they use must be designed for a life cycle which prevents a risk of injury or physical damage.
- The structure and stability of a piece of furniture must be such that it is fit for the purpose it has been designed for, comply to relevant standards and to the specific requirements of Boots stores.
- Consideration must be given to the far ranging age of our customers, from toddlers to well beyond retirement age and any predictable use and misuse this implies.
- Safety requirements are based upon the knowledge that furniture and its parts are likely to cause injury only when they are heavy and fall through a significant distance. This is possible if floor standing units overbalance, wall or screen mounted units become detached, heavy parts become detached from units or the structure collapses.
- All components must be rounded, chamfered or deburred to prevent sharp edges.
- Tubular components need to be capped or closed
- Adjustable and movable part must be designed to prevent accidental operation or injury during use.
- Load bearing sections of the furniture must not come loose unintentionally.
- Vertical roll fronts and sliding doors which can cause injury will not move by them selves from any position 200mm from closed position. However it is preferred that these elements are not used in any Boots design unless specifically requested.
- Moving units or moving parts (drawers for example) shall be at least 100mm from floor to avoid pinch points.
- Extension parts and trays will not separate from the unit when subjected to a 200N force to handle.
- Please note compliance to Boots furniture structural requirements can be achieved through desk top confirmation, using appropriate software to prove stability/centre of gravity, weight loading and strength/durability etc, or physical testing may be required. The method to be used must be agreed early on in the project briefing.

4.6.1 Stability:

For compliance to Boots and BS standards, stability must be considered as below;

- All furniture should be designed so that it is stable, and will not fall or topple over in use. The unit should be able to withstand predictable impact from items such as a shopping trolley, storage cage or powered wheelchair, without falling over or becoming unstable.
- The centre of gravity of free-standing or mid-floor equipment is crucial to the stability of equipment and this should be carefully considered when designing the size and type of bases or the footprint of Boots equipment.
- The unit must have an adequate footprint in relation to its height, ensuring that the centre of gravity of the unit is low and well within the overall foot print of the unit. Feet need to be positioned as close to the perimeter of the unit to ensure that the unit is as stable as possible. If necessary, consider adding additional weight to the base of the unit to increase its resistance to toppling.
- Ensure that the unit can be levelled adequately as many Boots stores have uneven floors, and that any adjustable feet are able to support the overall load of the unit.
- If items are added to the unit, once installed these items must not destabilise the basic unit. They must also be securely fitted to the unit so they do not come away.
- The unit should not rely on 'balance' for its stability. If parts are taken away from the unit during its normal use, such as when merchandising a display, the unit must not become unstable.
- The addition of items that are inherently heavy, e.g. display screens, have the potential to destabilise any free standing item. It is vitally important that any additional loadings added are carefully considered relative to the shape and size of any item. These items must be included in any risk assessment conducted and the mitigating factors clearly identified to ensure that the unit is stable. Top heavy free standing display items will almost certainly be rejected by Boots.
- All free-standing or mid-floor equipment should be topple-tested by applying force to angle units to 15 degrees, at which point they should return to vertical when released. If any equipment falls or does not return to its normal, vertical position it should be considered unsuitable for use and documented accordingly.
- All equipment should be certified using Boots own BTC tests and properly recorded and reported.

Section 4 Design guidance - strength, safety and shear

Strength and durability continued...

Strength and durability requirements are fulfilled when:

- There are no fractures of any member, joint or component.
- There are no loosening of joints intended to be rigid.
- The storage unit fulfills the stability requirements.
- The storage unit fulfills its function after the removal of test loads.
- There is no deflection of shelves that exceed 0.5% of the span of the shelf when tested in accordance with test 6.1.15(table 5).

Information for use:

- Info be available in language of country in which it will be delivered to the end user. It shall contain at least the following details:
- Information regarding the intended use, with reference to test severity where appropriate.
- Assembly instructions where applicable.
- Instructions for the care and maintenance of the furniture where applicable.

4.6.2 Safety Critical Fixings:

- Where a component in a display unit is required to be secured with a specific number of fixings, or is vital to the security or integrity of that unit, then the fixings required must be in place, must be the type as identified in the specification and must be recorded as being fitted.
- During manufacture, the fixing points or locations that may be described as being critical must be identified in a manner that is known to the installation teams. They should be numbered or colour coded so they are brought to the attention of the installation teams.
- The fixings, once installed, must be recorded by the fixing teams (including photographic records) and a copy of that record supplied to Boots.
- Adhesive tape, glues or contact adhesives must not be used for permanent fixings. Appropriate, visible mechanical fixing methods must always be used. This ensures that fixings can be visually checked that they are in place and secure.

4.6.3 Sheer and squeeze:

- Such points exist if the distance between two accessible parts moving relative to each other can be less than 25mm or more than 8mm in any position during movement.
- Shear and squeeze points created during set up of an assembly are acceptable however they must not be created by powered mechanisms, (with the exceptions of doors, flaps and extension items,) or during normal operation. Lids must not drop suddenly, lid support mechanisms must comply before and after testing.

4.6.4 Digital screens:

- Television screens must be securely affixed to any support structure (see 4.6.5 - safety critical fixings).
- Screens should use suitable mounting brackets.
- Brackets must be rated to carry the weight of the screen being mounted.
- Brackets must be fitted in exact accordance with manufacturers instructions and use specified fixings.
- Brackets must have a secure mechanical fixing that fixes screens in the mounted position to avoid accidental removal.
- Safety critical fixings must be clearly identified in instructions - refer to 4.6.2 - safety critical fixings.
- For digital screen specifications please refer to Boots Digital Screen Guidance. IE:

All screens must be CE approved.

Internal facing screens must be no brighter than 400 nits.

All screens must be provided with a 2m HDMI cable.

Only media players compliant with WBA IT security requirements will be allowed.

Section 4 Design guidance - strength, safety and shear

Strength and durability continued...

4.6.5 Castors:

- Castors are prohibited for use on ANY chairs intended for use by customers within Boots retail or consulting areas.
- Castors should be avoided wherever possible on any in-store equipment, and if used should be lockable.
- If castors are required by a specific brief this should be confirmed in writing by Boots at the briefing stage and specifically risk assessed and approved for use in Boots retail areas.
- Castors will only be use BY EXCEPTION in Boots stores.

4.6.6 Durability:

- All hinges used on Boots furniture shall comply to BS 16121:2013 (E).
- Hinges shall be guaranteed to a minimum of 40,000 cycles.
- Refer to BS EN 1935 for hinge type, test cycles and ratings.
- All drawer runners should comply to BS 16121:2013 (E).
- Drawer runners should be rated for a minimum of 40,000 cycles if used in their standard horizontal position.
- If runners are used for bespoke angles (other than conventional horizontal fixings), cycle tests must be carried out to guarantee ratings and use over 40,000 cycles. Tested using normal operating weights.

Section 4 Design guidance - materials

4.7. Materials:

Boots are always looking to ensure we deliver the most enjoyable and safe shopping experience for our customers and the safest possible workplaces for our colleagues.

Hazardous and Toxic Materials - Materials used in the manufacture of fixtures and displays to be utilized in Boots properties are, to the greatest extent possible, to be free of toxic or harmful substances, and volatile organic compounds.

4.7.1 Glass / Plastic Components:

- One aspect of this ambition is to minimise where possible the use of glass especially on easily accessible surfaces where they can be struck by items such as shopping trolleys, push chairs, etc. or where persons can lean against them e.g. graphic end panels.
- We expect that designers consider the use of acrylics as a first option as these are inherently safer and do not shatter if struck.
- All glass components used on any fixture must be toughened, tempered, laminated glass or if this is physically impossible to achieve due to glass dimensions, then the glass should be covered with safety film to prevent any risk of cutting from shards of glass.
- Where glass is used, Boots require that you include in your risk assessment the European / British Standard reference number for the glass used.
- Where toughened glass is used on the fixture the glass edges should be protected wherever possible to prevent risk of shattering from impacts.
- Where plastic components are used the plastic used should not have a sharp edge if the plastic component is broken.
- Glass must comply to EN1250-1:2000, clause 8 or has a breakage according to EN 12600:2002 Type B or C.
- Glass - Fixtures and Displays that utilise glass elements as part of the design, for shelving, doors or enclosures, etc., must use tempered glass or laminated glass. Due to its superior strength and durability, tempered glass is the preferred option.

Performance testing:

- Furniture manufacturers are required to undertake performance testing such as horizontal static load, vertical impact and horizontal fatigue tests when using annealed glass that is not supported over its entire area whether or not it is safety glass which complies with BS6206.
- Where glass is used in tables, counter tops or shelves, different tests are required dependent upon the method of support of the glass. Furniture which contains glass must meet the stability requirements of BS 4875.

BS6206: 1981 Impact performance requirements:

- The tests specified in BS6206 determines the performance of safety materials and comprises of a leather bag weighing 45KG being swung against the material which must withstand the impact or break safely as defined. The material is then marked A, B or C according to the drop height endured. Class A is the highest.
- Edgework, bevelling and drilling.
- Exposed edges of glass shall be edge-worked or bevelled (see table 1 of BS952). When machining or drilling toughened glass, reference should be made to GGF Data sheet 4.4. If possible laminated glass should not be drilled because moisture can cause delamination.

Contact between glass and other materials:

- In the construction of furniture there shall be no metal to glass, stone to glass, or glass to glass contact. These materials shall be separated by at least 0.5mm.

Section 4 Design guidance - glass

Doors and fixed glass retained in rebate or groove:

- Refer to BS EN 14749:2016.
- The edge cover provided by the rebate or groove must be at least 4mm or equal to the glass thickness if this is more than 4mm.

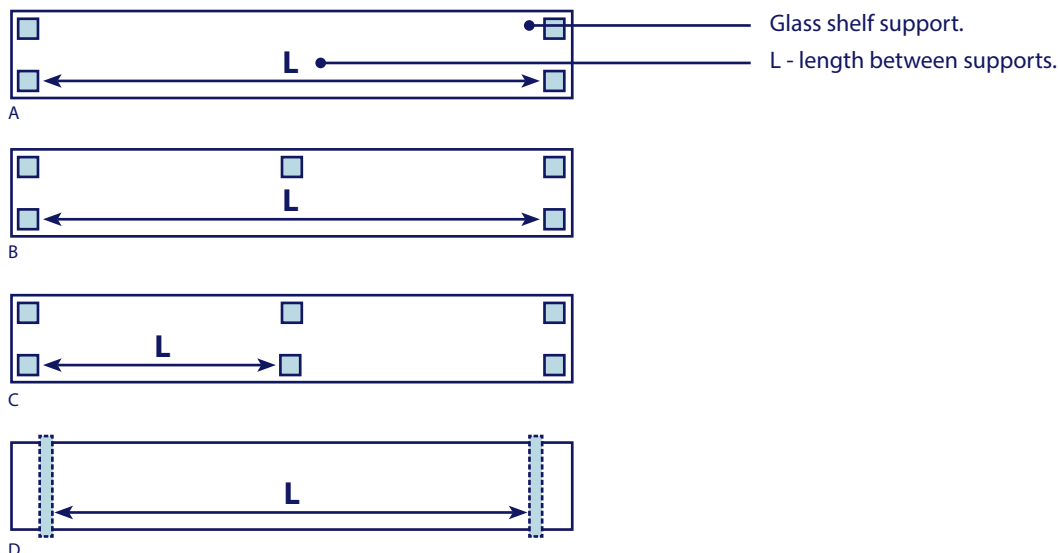
Glass Shelves:

- Refer to BS EN 14749:2016 and BS EN 12600:2002
- The maximum evenly distributed safe load that a shelf can support depends on the glass type, thickness and distance between supports (L), see figure 1.
- When a glass shelf is moved to extreme positions horizontally it should still overlap all supports by at least 4mm. Furniture with doors must comply with this requirement when the doors are open.
- When the furniture or shelving system is tilted to an angle of 30 degrees, plus or minus two degrees, the shelf must still be retained.
- Where doors are fitted, this requirement must be adhered to when the doors are in the open position.
- A shelf must not tip nor the supports fail when a load, equivalent to 50% of the maximum evenly distributed load and covering an area of 75mm x 75mm, is placed at any position on it for at least 10 seconds
- The maximum safe load shall be determined by reference to the choice of material, nominal thickness and length.
- Specific tests applicable BTC004, BTC005, BTC006 and BTC007

Glass Marking:

- All glass should be permanently marked and remain visible when in use.
- The glass shall be marked with the relevant BS.

Figure 1 - glass shelf distance between supports (L)



Section 4 Design guidance - adhesive films

Adhesive backed polymeric films - definitions:

Adhesive backed polymeric film:

- One or more layers of polymeric film laminated together, with an adhesive on one external face. It may also incorporate one or more of the following: Colouring, UV absorbers, UV inhibitors, metal layer(s), metal alloy layer(s), metal oxide layer(s), scratch or abrasion resistant surface, release liner.

Safety films:

- Adhesive backed polymeric film designed so that when applied to a glass pane the final product can be classified in accordance with BS EN 12600 Glass in building – Pendulum test – Impact method test and classification for flat glass. Safety film is also known as ‘Anti-Shatter film’ or ASF since it is intended to retain broken glass fragments and hence reduce shattering of glass over a wide area.

Thickness:

- The total thickness of the safety film is the sum of the thicknesses of its polyester film layers, excluding the thicknesses of adhesive and other coatings.

Cure time:

- The time required for the adhesive backed polymeric film to achieve its expected adhesion level and for visual effects due to water haze/water bubbles/etc. to disappear.
- NOTE: Cure time should not be confused with performance time. Safety film will, with correct installation procedures, achieve good performance before complete cure.

Section 4 Design guidance - adhesive films and substrates

Components and substrates:

Adhesive backed polymeric film can be applied to any glass substrate as described below:

- Basic soda lime silicate glass products in accordance with BS EN 572-1.
- Float glass in accordance with BS EN 572-2.
- Polished wired glass in accordance with BS EN 572-3.
- Drawn sheet glass in accordance with BS EN 572-4.
- Patterned glass in accordance with BS EN 572-5.
- Wired patterned glass in accordance with BS EN 572-6.
- Wired or unwired channel shaped glass in accordance with BS EN 572-7.
- Supplied and final cut sizes in accordance with BS EN 572-8.

Special basic products:

- Borosilicate glasses in accordance with BS EN 1748-1-1.
- Glass ceramics in accordance with BS EN 1748-2-1.
- Basic alkaline earth silicate glass – Float glass in accordance with BS EN 14178-1.
- Mirrors – manufactured from silvered float glass in accordance with BS EN 1036-1.
- Thermally toughened soda lime silicate safety glass in accordance with BS EN 12150-1.
- Heat soaked thermally toughened soda lime silicate safety glass in accordance with BS EN 14179-1.
- Thermally toughened borosilicate safety glass in accordance with BS EN 13024-1.
- Thermally toughened alkaline earth silicate safety glass in accordance with BS EN 14321-1.
- Heat strengthened soda lime silicate glass in accordance with BS EN 1863-1.
- Chemically strengthened soda lime silicate glass in accordance with BS EN 12337-1.
- Laminated glass and laminated safety glass in accordance with BS EN ISO 12543-1.
- Coated glass in accordance with BS EN 1096-1.

Section 4 Design guidance - mirrors

4.7.2 Mirrors:

- Mirrors are to satisfy the requirements described above for glass and plastic components, i.e. acrylic should be the first consideration.
- All mirrors, be they acrylic or glass must be secured in place by mechanical means. Adhesives, adhesive tapes, glues or Velcro products must not be used as a primary method of fixing.
- Glass - Fixtures and Displays that utilise glass elements as part of the design, for shelving, doors or enclosures, etc., must use tempered glass or laminated glass. Due to its superior strength and durability, tempered glass is the preferred option.
- **Mirrors - Fixtures or Displays that incorporate mirror(s) as part of the design, must use acrylic or shatterproof safety film mirror. Mirrors must be mechanically secured to the fixture, such as a channel capturing the mirror, and cannot rely on adhesive alone.**
- **It is Boots standards that no glass or mirror should ever rely singularly on tape, glue or adhesive fixing. Without exception all mirrors and glass components should be secured using mechanical fixings, fittings or retaining frames.**

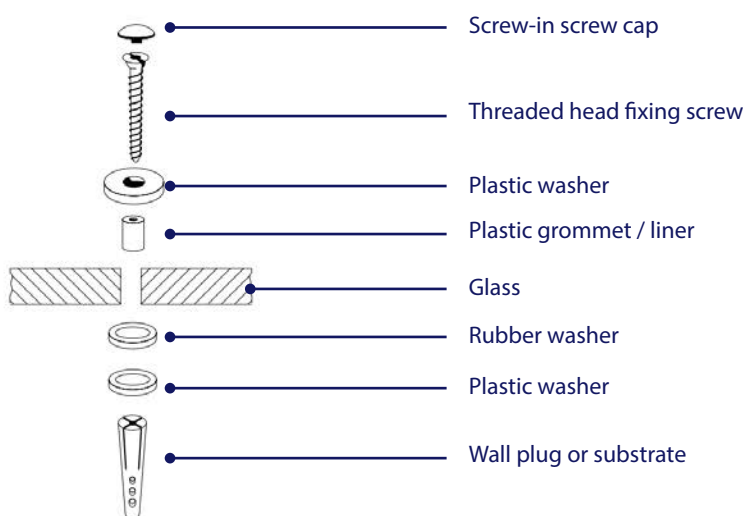
Mirror fixing:

- When fixing mirrors, care must be taken to ensure that the backing and edges cannot be attacked by chemical action from substrates.
- Where practicable an air space of 3 to 5 mm should be provided between the mirror and the background to allow free circulation of air behind the mirror.
- The mirror should be fixed securely but free from stress to avoid distortion and risk of breakage.
- The mirror should not be fixed to a very uneven surface but where there is any unevenness, nylon washers should be used as packing.
- Make sure that the substrate is strong enough to take the weight of the mirror and thick enough to take the full length of the fixing method provided.

Mirrors can be fixed in a number of ways as follows:

- Screw fixing of drilled mirrors.
- Mirrors should be fixed as shown in the diagram. It is essential that there is no glass to metal contact, therefore ensure that a plastic collar is used between the screw and the mirror.
- Nylon or suitable rubber washers must be used to ensure a freely ventilated air space between the mirror and the wall.
- Do not overtighten the screws. This will cause distortion and may lead to breakage of the mirror.

Figure 1 - mirror fixing recommended method:



Section 4 Design guidance - mirror fixings

Clip fixing of undrilled mirrors:

- Manufacturer's instructions must be followed according to the type of clip fixing set supplied. For a typical set of clips the following general recommendations apply. (For small mirrors 4 clips may be supplied.).
- Because clip fixing involves glass to metal contact, edge treatment of the glass is very important. The edges of the mirror must therefore be arressed, ground or polished.
- The mirror should be fixed securely but free from stress to avoid distortion and risk of breakage.
- The mirror should not be fixed to a very uneven surface.
- Make sure that the wall or door substance is strong enough to take the weight of the mirror and thick enough to take the full length of the fixings provided.

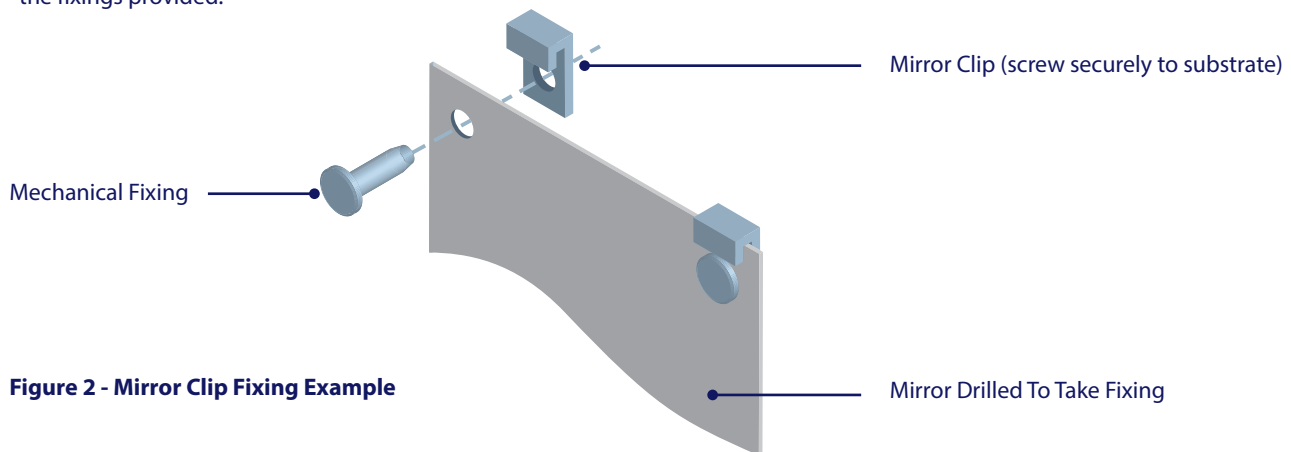


Figure 2 - Mirror Clip Fixing Example

Edge fixing of undrilled mirrors:

- Edge beads can be used either against a flat surface or on battens spaced not more than 375 mm (15 in) apart, faced with a continuous cushion such as a non hardening glazing strip.
- The mirror should be fixed securely but free from stress to avoid distortion and risk of breakage.
- The mirror should not be fixed to a very uneven surface.
- Make sure that the wall or door substrate is strong enough to take the weight of the mirror and thick enough to take the full length of the fixings provided.

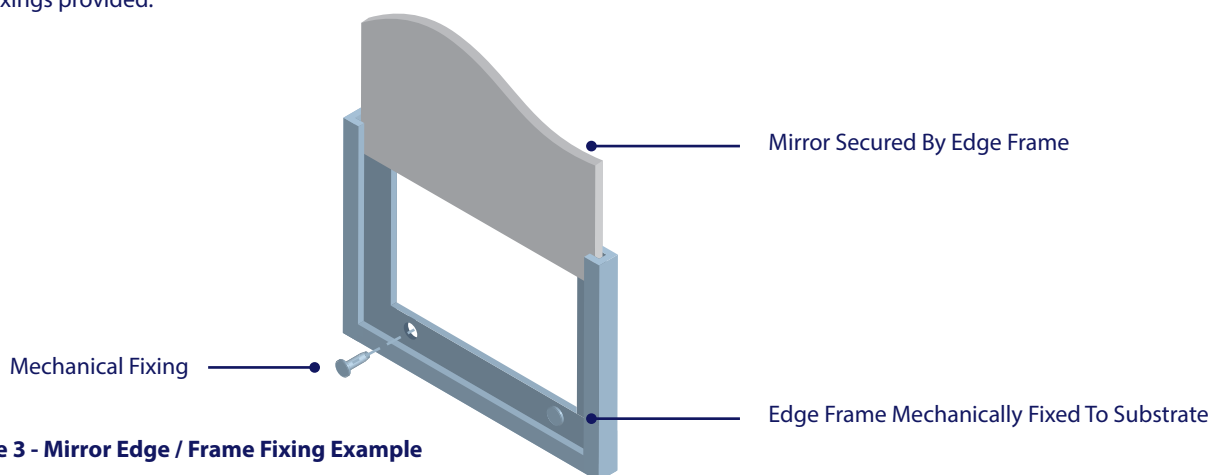


Figure 3 - Mirror Edge / Frame Fixing Example

Secondary fixing methods:

- Where the mirror has to be set flush in a wall or panel, appropriate adhesive may be used. Alternatively self adhesive foam pads can be used as recommended by foam pad manufacturers. The number and size of pads to be used is critical and will depend on the weight of the mirror being fixed. Pad manufacturers specify the area of their pad required in relation to the weight of the mirror to be fixed.
- When using adhesive it is important that the adhesive does not create sealed areas on the back of the mirror. Generally it is advisable to apply in vertical strips.
- Note: At no point is it acceptable for adhesive alone to be used as a fixing method for glass or mirrors.

Section 4 Design guidance - environmental, electrical, sound, stools and chairs

4.7.3 Environmental considerations:

- Boots expect environmental consideration to be part of each retail project. An understanding of the life of the furniture is required from creative through manufacture, use and finally disposal and this needs to be considered to minimise environmental impact.
- We expect our suppliers to have robust environmental policies which are adhered to when working with Boots.
- Material should be used sparingly but sufficiently to allow the appropriate lifespan and not create maintenance issues.
- Materials should be selected from sustainable sources, with recycled content and be reusable or recyclable where possible as landfill needs to be reduced to a minimum.
- Packaging should be considered for recycling or reuse where ever possible.
- The Furniture should be designed to be easily disassembled to allow for efficient recycling or reuse.
- Shipping needs to be minimised if possible, flat pack must be considered where appropriate.

4.8 Electrical and illumination:

- All electrical components should comply with TBGE1 - Shopfittings, Signage and Displays –Electrical guidance’ as specified on the Boots Property Central website (www.bootspropertycentral.co.uk).
- This specification is updated regularly and it is essential that upon commencing the design of a new fixture or fitting the current specification is located and used for the design of that fixture.
- Any fixture that doesn't comply with the electrical specification will not be approved.

4.9 Noise and use of sound:

- Any fixture design should not generate noise in its normal operation.
- Music or instructional videos which use sound should not be used in store, unless this has specifically been agreed as part of the commercial agreement.

4.10 Stools and Chairs:

- Any chairs, stools or tables to be used in Boots stores must comply at minimum to the relevant BS standards of the geographical store location, documentation to confirm compliance must be provided before approval will be given.
- Where stools, chairs or other seating is provided for customers or colleagues to sit upon, Boots require that you include in your risk assessment the European / British Standard reference number for the seating used.
- The furniture must be stable by design and be capable of supporting a minimum weight of 120Kg / 20 stones.
- Your risk assessment must consider the stability of these items in terms of persons climbing on and off them.
- BS EN 16139:2013 Test Level 2 Furniture. Strength, durability and safety. Requirements for non-domestic seating.
- BS 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites.
- Under no circumstances should castors be used on any seating that is for use in the retail area.

5

BTC test certification



The Boots Group

Section 5 BTC test certification

5.0.1 BTC Testing:

In order to provide concise test criteria for our retail estate, Boots have quantified the type and method of tests required for retail fixtures and provided details of these new BTC (Boots Test Certification) codes in the following pages.

The source information for each BTC test has been derived from BS or UL test criteria and all Boots equipment should carry relevant test certification using appropriate or applicable BTC tests.

For the avoidance of doubt: The BTC test criteria is in addition to any statutory or British Standard requirements and it is the vendors responsibility to evaluate (with due diligence) any applicable test criteria they deem necessary.

As an absolute minimum all equipment used in Boots estate should have passed relevant BTC testing, with individual compliance certificates issued for each piece of equipment.

Applicable Test Standards:

- BBS EN 16122:2012 Non domestic storage furniture – test methods for the determination of strength, durability and stability.
- BS EN 16139:2013 Test Level 2 Furniture. Strength, durability and safety. Requirements for non-domestic seating.
- BS EN 7176:2007+A1:2011 Specification for resistance to ignition of upholstered furniture for non-domestic seating by testing composites.
 - Performance rating of any material must meet the 'Medium Hazard' category.
- TBGE1 - Shopfittings, Signage and Displays –Electrical guidance.
- BS7449 Specification for inclusion of glass in the construction of furniture, other than tables or trolleys, including cabinets, shelving systems and wall hung or free standing mirrors.
- BS EN 15372:2023 Furniture — Strength, durability and safety — Requirements for non-domestic tables.
- BS EN 1730:2012 Furniture — Tables — Test methods for the determination of stability, strength and durability.

Note:

Where any drawers or extensions are tested, these should be fully loaded to maximum designed weight in addition to any test weights defined in the BTC test criteria.

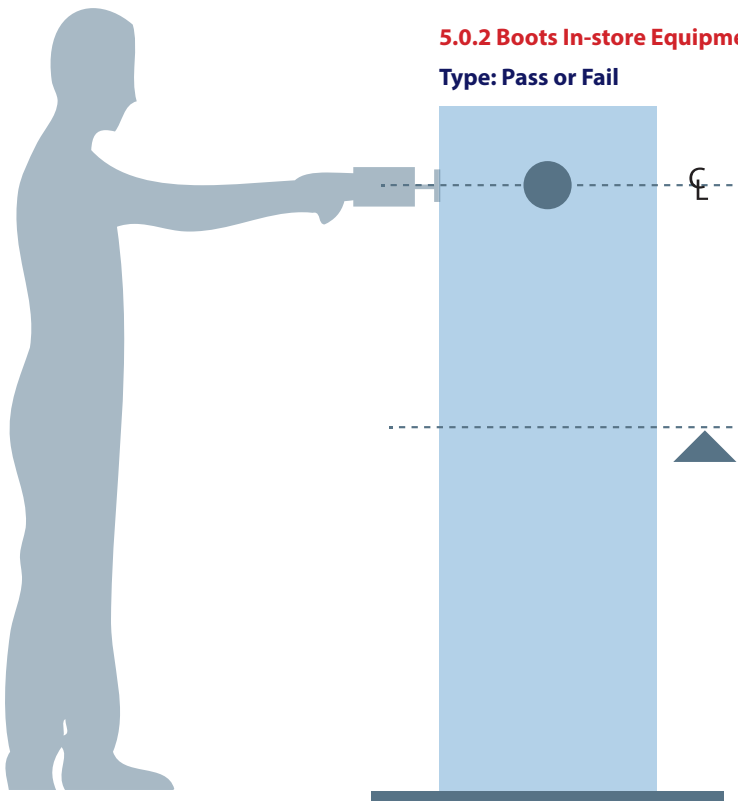
Test Conditions:

- All testing should be carried out in clean, indoor environments that have adequate lighting for quality photographic records and video recording.
- Lighting levels should be a minimum of 1000 lux, 1000mm from floor level. Measured with a lux meter to allow for well lit photography and commonality of visual record keeping. Noting that 1000 lux is comparable to a store environment.
- Flooring should be a rigid, horizontal flat material (not carpet). A smooth contract vinyl type flooring material that is similar to the type used in Boots stores would be preferable.
- Photographic records should be made of each test showing the force meter readings as applied to each BTC code.
- Video recordings of each test should also be made, clearly showing the test conditions and readings.
- Photographic and video records should be coded, dated and held in vendors records as proof of testing and certification. In addition to providing digital copies to Boots for our records.
- It is imperative that photographs and video shows each test and its outcome.
- All video and photographic records must be stored and considered as confidential.
- Non-disclosure agreements may apply and all Boots records are considered confidential.

Section 5 BTC physical stability test certification for items 1067mm or taller

5.0.2 Boots In-store Equipment - Measured Force Topple Test Certification BTC001

Type: Pass or Fail



Location Of Test:

1372mm high OR 100mm from top of unit (whichever is greater).

Size And Type Of Force Delivery:

100mm dia rigid disk.

Applied Force:

180N applied gradually over three second period using a calibrated force meter or unit reaches an angle of 10 degrees, whichever occurs first

Direction Of Test:

Separately to all sides of UNLOADED units.

Floor Type:

Rigid, horizontal and flat surface (not carpet or cushion floor)

+ 1372mm high

Below 1372mm in height, topple test does not apply



PASS:

Unit does not topple or tip over at application of full force, or the unit inclines to an angle of ten degrees without tipping over.



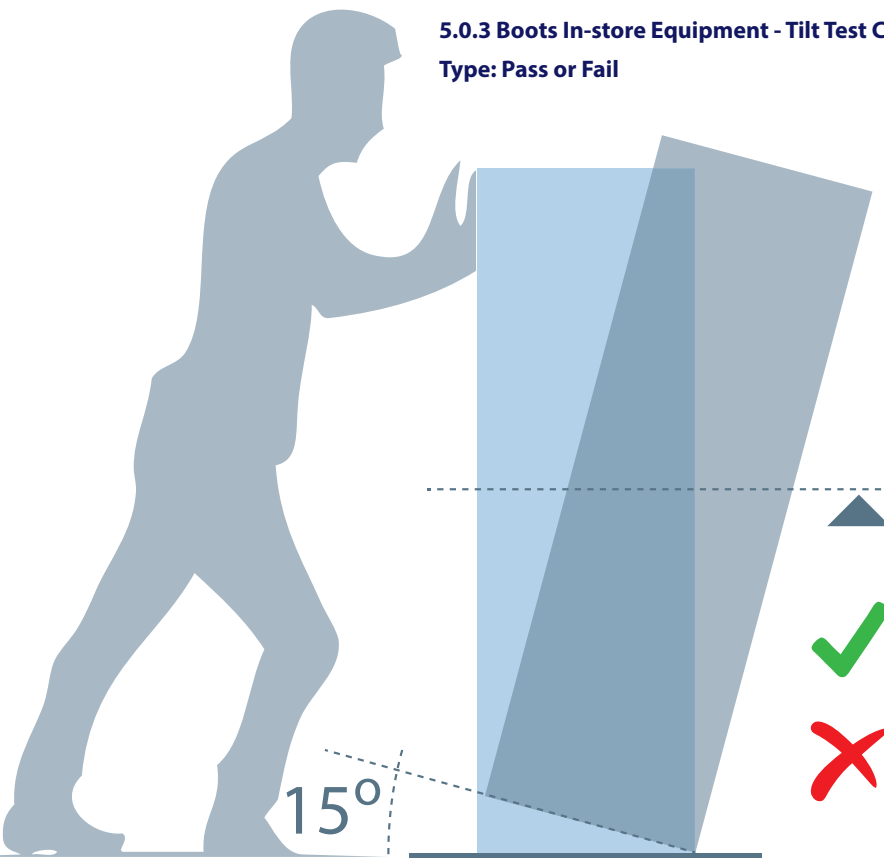
FAIL:

Unit tips over and is classified unstable.



5.0.3 Boots In-store Equipment - Tilt Test Certification BTC002

Type: Pass or Fail



Location:

Top corner of unit.

Type Of Force:

Sufficient to tilt unit to 15 degrees from vertical.

Test:

Release from 15 degrees instantly.

Result:

Unit should immediately return to original vertical position.

Direction Of Test.

Separately from all sides of unit.

+ 650mm high

Below 650mm in height, topple test does not apply



PASS:

Unit returns to normal vertical position when immediately released from a 15 degree angle.



FAIL:

Unit does not return to vertical and tips over. Therefore is classified as unstable.



The Boots Group

Section 5 BTC physical stability test certification for items 1067mm or taller

5.0.4 Boots In-store Equipment - Bump Test BTC003

Type: Pass or Fail

Test Criteria:

Apply test at the highest point of the test unit or 1200mm high (whichever is greater).

Using a forcemeter or calibrated pendulum, apply a "bump" force of 660N once in a single impact to the position described above. Repeat in all directions.



PASS:

The test item does not fall or topple.
The test item returns to its normal stable position after the test is complete.

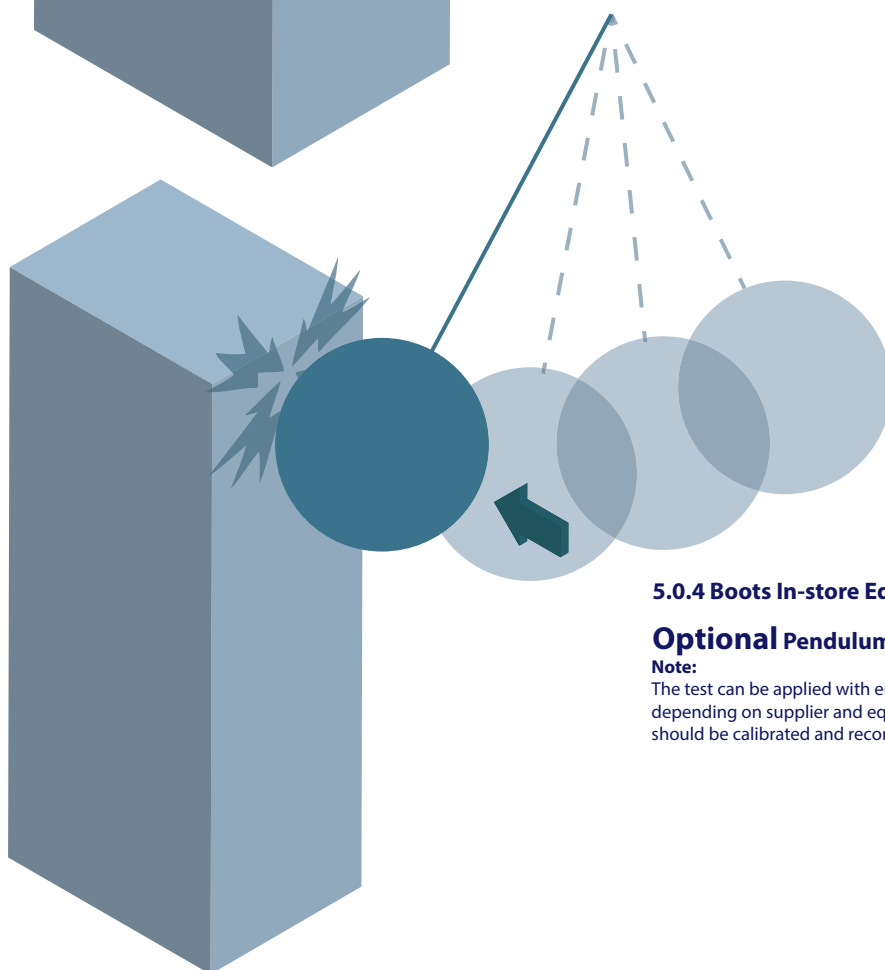
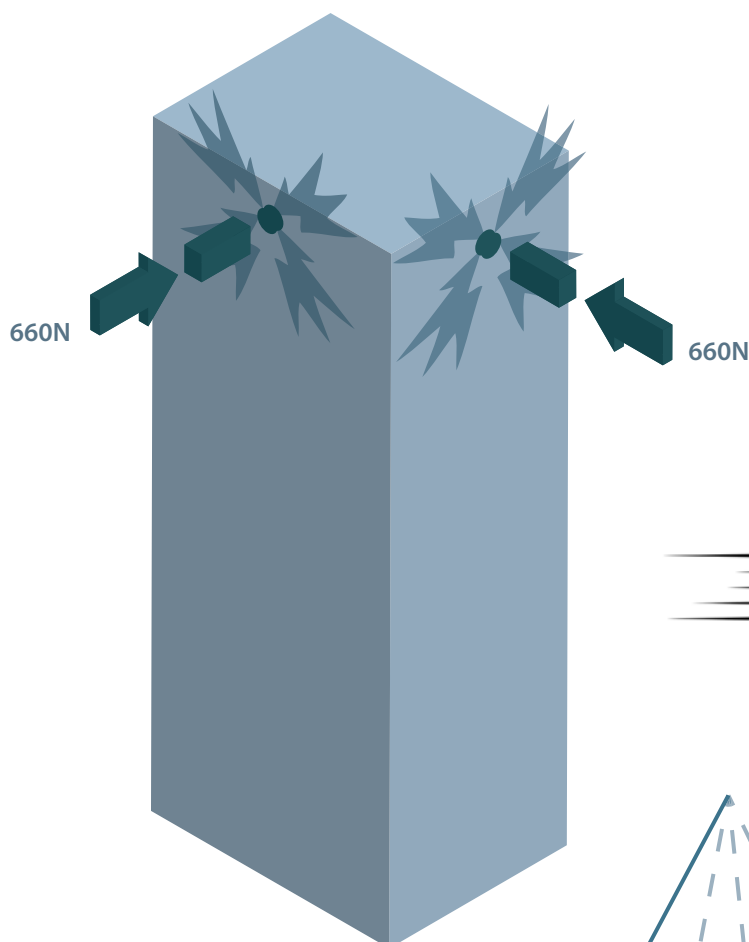


FAIL:

The test item topples or becomes unstable.



Note: 660N represents an equivalent force of a mobility scooter traveling at speed



5.0.4 Boots In-store Equipment - Bump Test BTC003

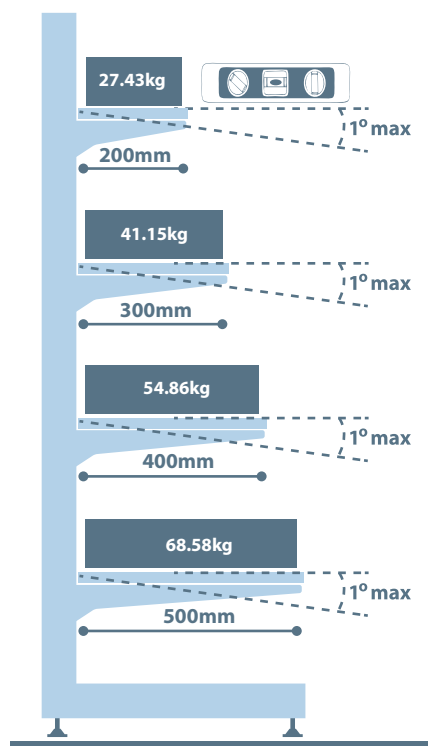
Optional Pendulum Or Force Meter Test

Note:

The test can be applied with either a forcemeter or a weighted pendulum depending on supplier and equipment preference. Either option however should be calibrated and recorded to give an instant impact force of 660N.



Section 5 BTC weight loading certification



5.0.5 Boots In-store Equipment - Static Weight Load Standard 760mm Shelf Angle Deflection Test BTC004 Type: Pass or Fail

Test Criteria:

Using a calibrated level ensure shelves are perfectly horizontal prior to weight loading.
Apply evenly distributed (minimum) weights shown to each shelf.
After one hour check any angle deflection at each bracket location using a calibrated level.



PASS:

Shelf angle does not deflect beyond one degree from horizontal.



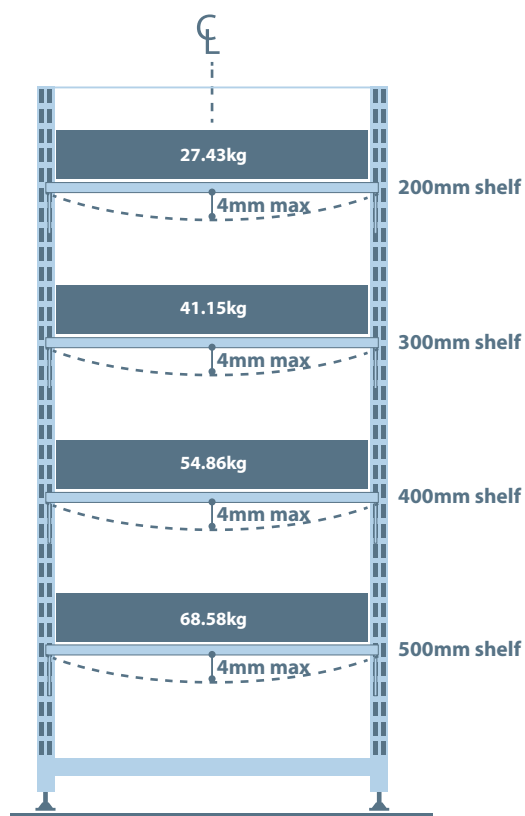
FAIL:

Shelf angle deflects more than one degree from horizontal level.

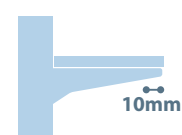
Angle Over Load Test:

Test Criteria:

Using a calibrated level ensure shelves are perfectly horizontal prior to weight loading.
Apply evenly distributed DOUBLE the weights shown to each shelf. IE: 200mm shelf 55kg. 300mm shelf 83kg. 400mm shelf 110kg. 500mm shelf 138kg.
After one hour check and record any angle deflection at each bracket location using a calibrated level.



5.0.6 Boots In-store Equipment - Static Weight Load Standard 760mm Shelf Deflection Test BTC005 Type: Pass or Fail



Shelf Width Deflection Measure Point:

10mm from front edge of shelf at centre line.

Test Criteria:

Using a calibrated level ensure shelves are perfectly horizontal prior to weight loading.
Apply evenly distributed (minimum) weights shown to each shelf.
After one hour check any shelf deflection at the centre of each shelf, 10mm back from the front edge.



PASS:

Shelf has a deflection of LESS than 4mm vertically measured on the shelf CL, 10mm from front edge.



FAIL:

Shelf has a deflection of MORE than 4mm vertically measured on the shelf CL, 10mm from front edge.

Deflection Over Load Test:

Test Criteria:

Using a calibrated level ensure shelves are perfectly horizontal prior to weight loading.
Apply evenly distributed DOUBLE the weights shown to each shelf. IE: 200mm shelf 55kg. 300mm shelf 83kg. 400mm shelf 110kg. 500mm shelf 138kg.
After one hour check and record any shelf deflection at the centre of each shelf, 10mm back from the front edge.

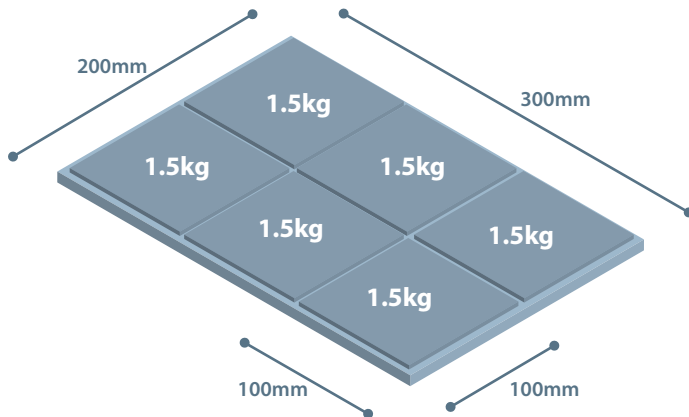


Section 5 BTC weight loading

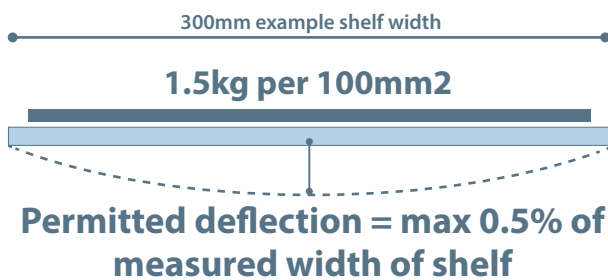
5.0.7 Boots In-store Equipment - Static Weight Load

Custom Size Shelf Deflection Test BTC006

Type: Pass or Fail



EXAMPLE SHELF: 300mm wide x 200mm deep



Test Criteria Based On Illustrated Example:

Calculate mm² area of shelf.
Load shelf with evenly distributed weight loading of 1.5kg per 100mm².
After one hour check any deflection at the centre line of the shelf front, 10mm back from the front edge.

PASS:

Shelf does not deflect beyond (measured) 0.5% of the width of the shelf.

FAIL:

Shelf deflects beyond (measured) 0.5% of the width of the shelf.

Example Calculation (based on illustration).

Shelf width: 300mm
Shelf depth: 200mm
Shelf area: 60,000mm²

Weight Loading:

1.5kg per 100mm x 100mm square (10,000mm²)

Total Weight Loading:

$6 \times 1.5\text{kg}$ ($6 \times 10,000\text{mm}^2 = 60,000\text{mm}^2$) = 9kg.
Permitted Deflection (0.5% width) Example Calculation.
W (shelf width) = 300mm
D (permitted deflection) = 0.5% of W

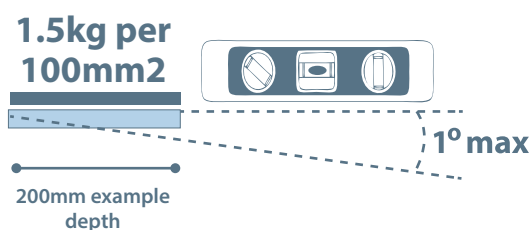
Therefore $D = W \times 0.5\%$ ($300\text{mm} \times 0.005$) = 1.5mm



5.0.8 Boots In-store Equipment - Static Weight Load

Custom Size Shelf Angle Deflection Test BTC007

Type: Pass or Fail



Test Criteria Based On Illustrated Example:

Using a calibrated level ensure shelves are perfectly horizontal prior to weight loading.
Load shelf with evenly distributed weight loading of 1.5kg per 100mm².
After one hour check any angle deflection at each bracket location using a calibrated level.

PASS:

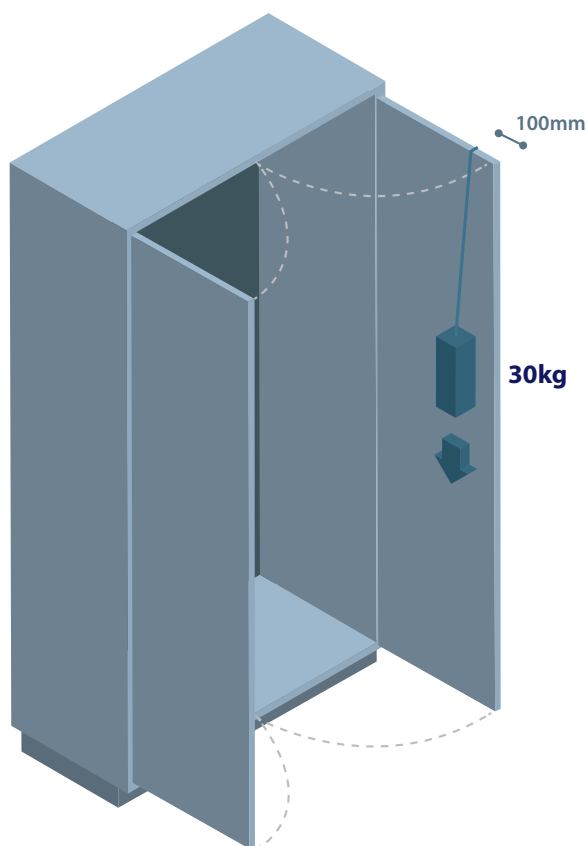
Shelf angle does not deflect beyond one degree from horizontal.

FAIL:

Shelf angle deflects more than one degree from horizontal level.



Section 5 BTC testing of pivoted doors



5.0.9 Boots In-store Equipment - Vertical Load Test Of Pivoted Doors BTC008

Type: Pass or Fail

Test Criteria:

Load the door as shown with the 30kg weight. The weight should be suspended 100 mm from the edge furthest from the hinge.
Open and close the door ten full cycles (back and forth) from a position 45degrees from fully closed to a position 10 degrees from fully opened, up to a maximum of 135 degrees from the fully closed position.
Opening and closing can be done by hand using 3 sec to 5 sec for opening and 3 sec to 5 sec for closing.



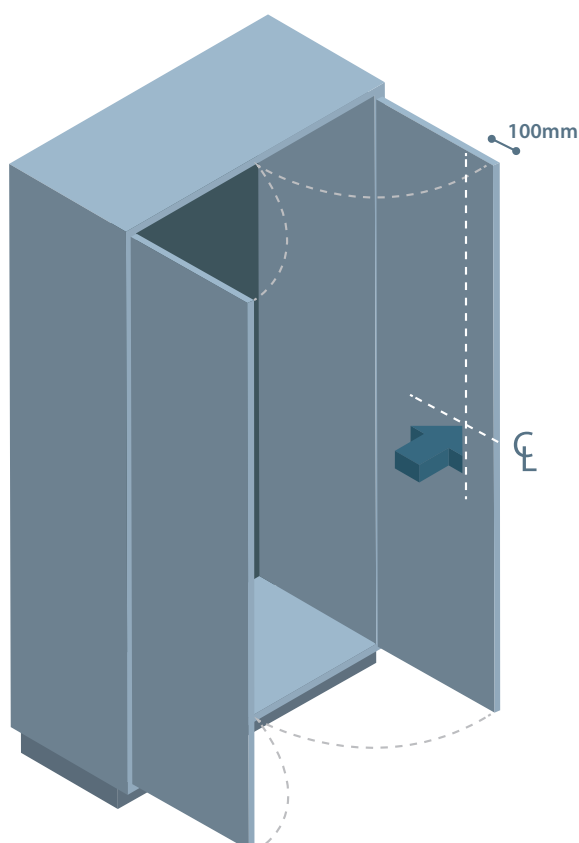
PASS:

The door does not deflect from its original position and all fixings and hinges remain fixed in place with no movement.



FAIL:

The door suffers deflection, is out of alignment, or any fixings or hardware fails or becomes loose.



5.0.10 Boots In-store Equipment - Horizontal Load Test Of Pivoted Doors BTC009

Type: Pass or Fail

Test Criteria:

Apply 60N horizontal static load perpendicular to the plane of the door on its horizontal centre line using a calibrated force meter.
Apply force 100 mm from the edge furthest from the hinge, as shown.
Apply the load ten times.



PASS:

Doors hinges and hardware remain intact and stable, with door remaining in original fitted position with no deflection or adjustment.



FAIL:

Doors, hinges and fixings fail, or doors become misaligned and are not in original fixed positions.



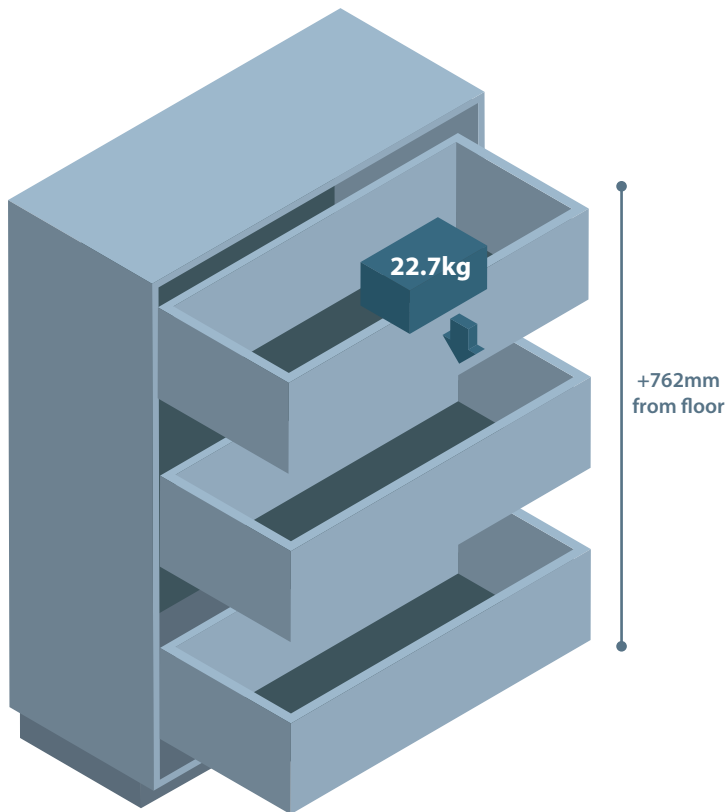
Doors General Information:

Sliding doors should be avoided in all designs.
All doors and drawers must be lockable.
All catches and hardware must comply with BS EN 16122:2012.
Where possible all locks should be keyed alike.



The Boots Group

Section 5 BTC testing of sliding drawers (appurtenance loading)



5.0.11 Boots In-store Equipment - Topple Test Of Drawers +762mm BTC010

Type: Pass or Fail

Test Criteria:

At the height of 762mm (or greater), load the drawer as shown with the 22.7kg weight. The weight should be located on the front edge of the drawer. The drawer should be fully extended to the limit of its runners. Drawers should be fully loaded with maximum designed weight.



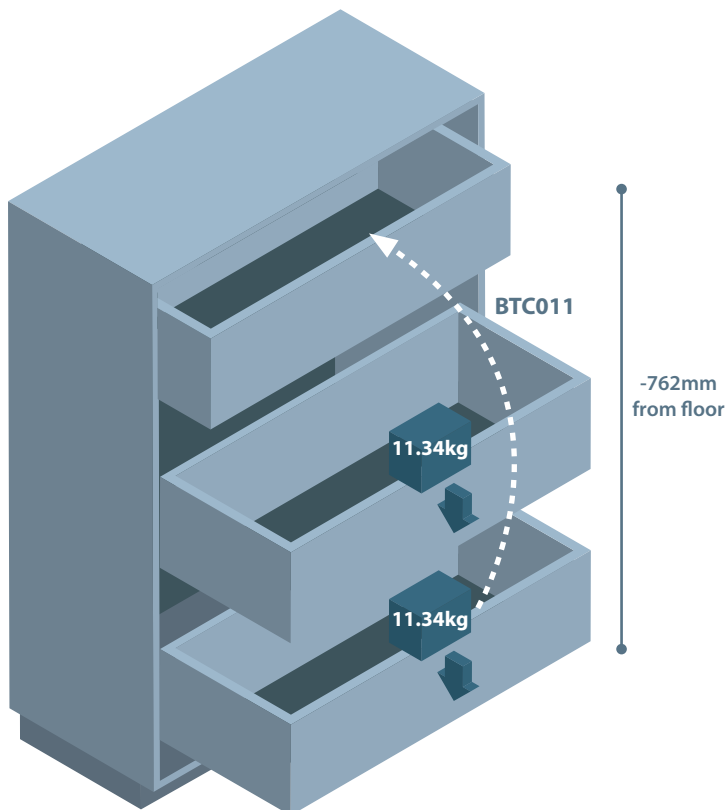
PASS:

With the drawer fully loaded to the above specification the unit does not topple or become unstable.



FAIL:

The unit topples forwards under the specified load and is deemed unstable.



5.0.12 Boots In-store Equipment - Topple Test Of Drawers -762mm BTC011

Type: Pass or Fail

Test Criteria:

Below the height of 762mm, load drawers as shown with 11.34kg of weight per drawer. Weights should be located on the front edge of drawers. Drawers should be fully extended to the limit of their runners. Drawers should be fully loaded with their maximum designed weight.



PASS:

With drawers fully loaded the unit does not topple or become unstable.



FAIL:

The unit topples forwards under the specified load and is deemed unstable.



5.0.13 Boots In-store Equipment - Climb Test Of Drawers BTC012 Type: Pass or Fail

Test Criteria:

Load bottom two drawers as shown with 11.34kg of weight per drawer. Weights should be located on the front edge of drawers. Drawers should be fully extended to the limit of their runners. After ten seconds remove the weight from the lowest drawer and place onto the drawer above the higher weight. Weight ten seconds and repeat with the lower weight to simulate a child climbing the drawers step by step. Drawers should be fully loaded with their maximum designed weight.



PASS:

With drawers fully loaded in steps the unit does not topple or become unstable.

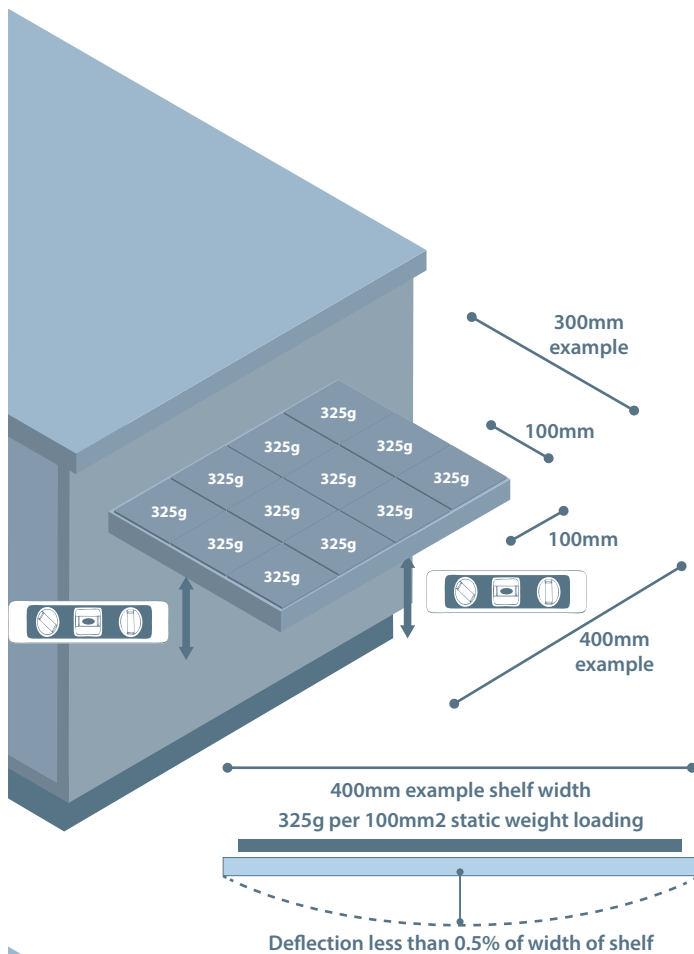


FAIL:

The unit topples forwards under the specified load and is deemed unstable.



Section 5 BTC testing of hinged flaps



5.0.14 Boots In-store Equipment - Hinged Flap Static Weight Test BTC013

Type: Pass or Fail

Test Criteria Based On Illustrated Example:

Calculate mm² area of shelf.

Load shelf with evenly distributed weight loading of 325g per 100mm². After one hour check any deflection at the centre line of the shelf front, 10mm back from the front edge.



PASS:

Shelf does not deflect beyond (measured) 0.5% of the width of the shelf.



FAIL:

Shelf deflects beyond (measured) 0.5% of the width of the shelf.

Example Calculation (based on illustration).

Shelf width: 400mm

Shelf depth: 300mm

Shelf area: 120,000mm²

Weight Loading:

325g per 100mm x 100mm square (10,000mm²)

Total Weight Loading:

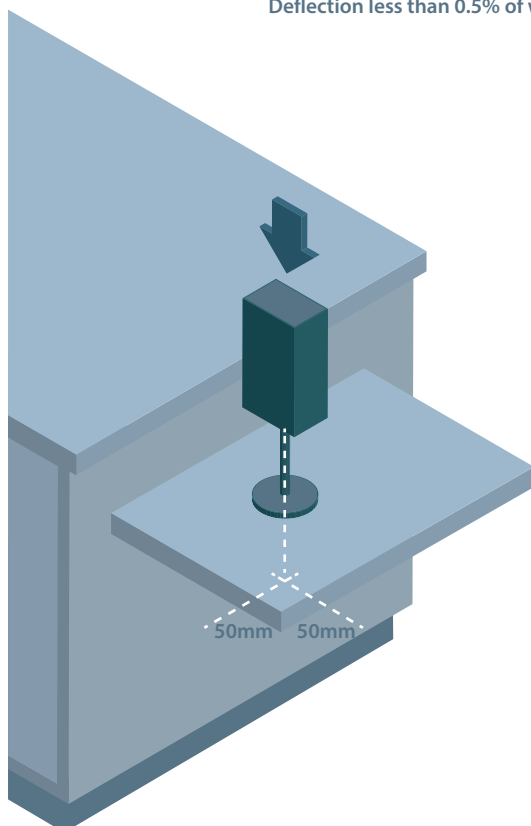
12 x 325g (12 x 10,000mm² = 120,000mm²) = 3.9kg.

Permitted Deflection (0.5% width) Example Calculation.

W (shelf width) = 400mm

D (permitted deflection) = 0.5% of W

Therefore D = Wx0.5% (400mm x 0.005) = 2mm



5.0.15 Boots In-store Equipment - Hinged Flap Point Load Test BTC014

Type: Pass or Fail

Test Criteria:

Position of test to be carried out 50mm from the front corner of the flap as shown in the illustration.

Using a calibrated force meter apply a point loading of 200N vertically downward on the flap for a period of three seconds, then release.

Repeat for ten cycles and record any deflection or deformation.



PASS:

Flap remains in original position with all hardware and fixings firm and undamaged.



FAIL:

The flap suffers deflection, or hinges, fixings and hardware fail, resulting in the flap being loose, unsafe or unusable.



Flaps General Information:

All hinges and fixings should confirm to BS EN 1622:2012 (E).

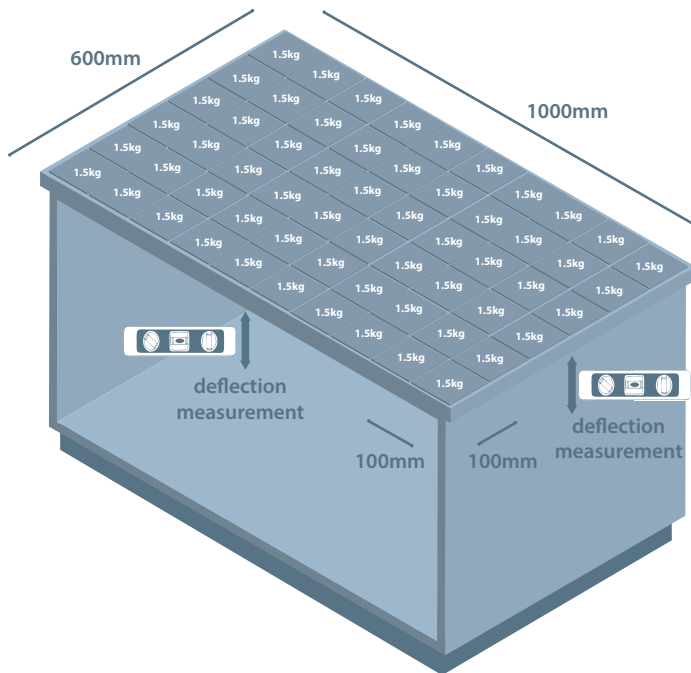
Hinges should have a minimum durability of 10,000 cycles.

Hinge ratings should be included in specifications and operational manuals.

Flap locking and retaining mechanisms should be fully risk assessed and tested prior to Boots approval.

All locking mechanisms should conform to BS EN 1622:2012 (E).

Section 5 BTC static and point loading of horizontal surfaces



5.0.16 Boots In-store Equipment - Horizontal Surface Static Weight Test BTC015

Type: Observation and Recording

Test Criteria Based On Illustrated Example:

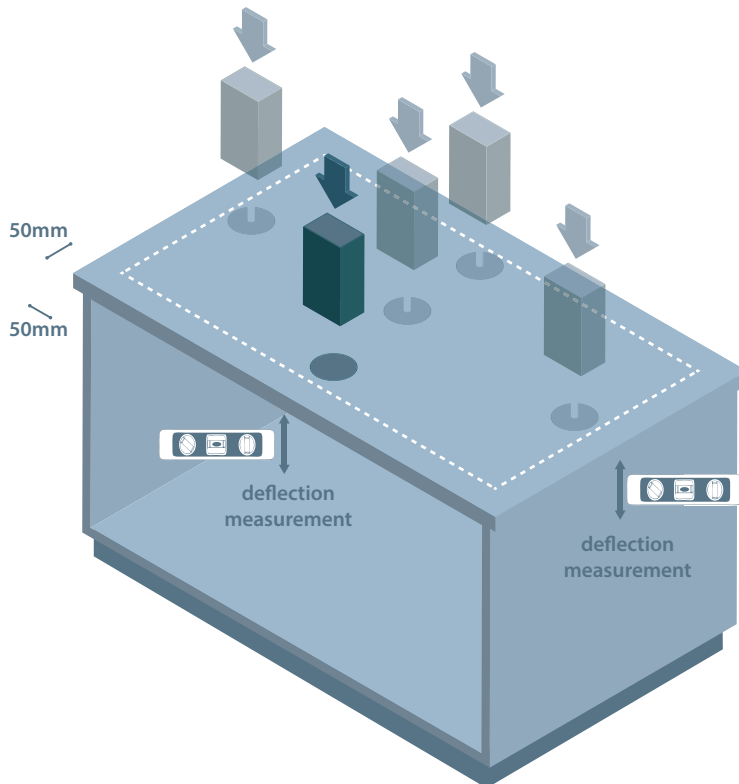
Calculate mm² area of surface to be tested.
Load surface with evenly distributed weight loading of 1.5kg per 100mm².
After one day check any deflection at the centre line of the surface front and sides, 10mm back from the edge and record.



PASS:
No material failure of surface. NOTE: BTC014 is not a pass or fail test.



FAIL:
Surface material cracks, sags or fails therefore is classified unsuitable for use.



5.0.17 Boots In-store Equipment - Horizontal Surface Point Load Test BTC016

Type: Observation and Recording

Test Criteria:

Mark a test area 50mm in from each side of the horizontal surface to be tested. Using a calibrated force meter apply 200N of downward force using a pressure pad formed from a 100mm dia rigid disk, for three seconds. Repeat in each illustrated position for ten cycles. Check any deflection at the centre line of the surface front and sides, 10mm back from the edge and record.



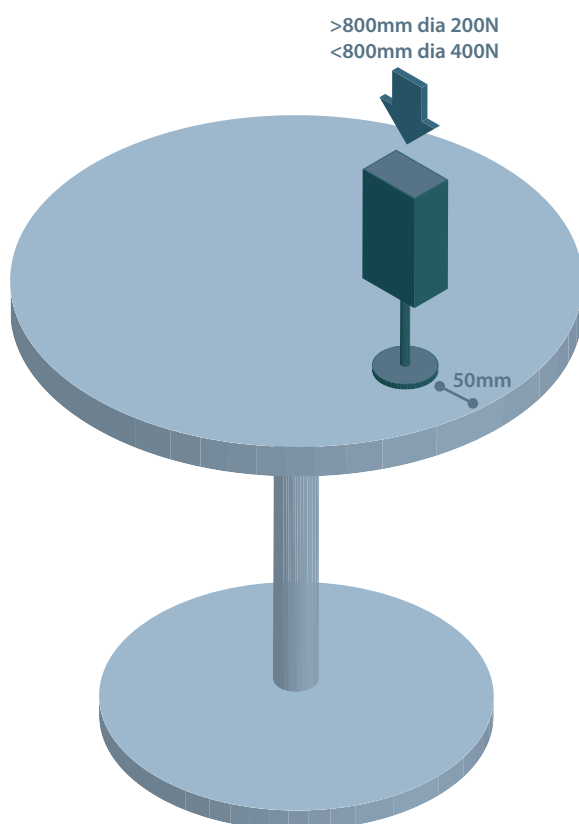
PASS:
No material failure of surface. NOTE: BTC015 is not a pass or fail test.



FAIL:
Surface material cracks, sags or fails therefore is classified unsuitable for use.



Section 5 BTC static and point loading of horizontal surfaces



5.0.18 Boots In-store Equipment - Stability Under Vertical Load (Circular Tables) BTC017

Type: Pass or Fail

Test Criteria:

Using a calibrated force meter (fitted with a smooth, rigid, 100mm dia loading pad) apply the following loads (based on the diameter of the table) not less than 50mm closer to any edge.
Table diameter below 800mm - 200N.
Table diameter above 800mm - 400N.



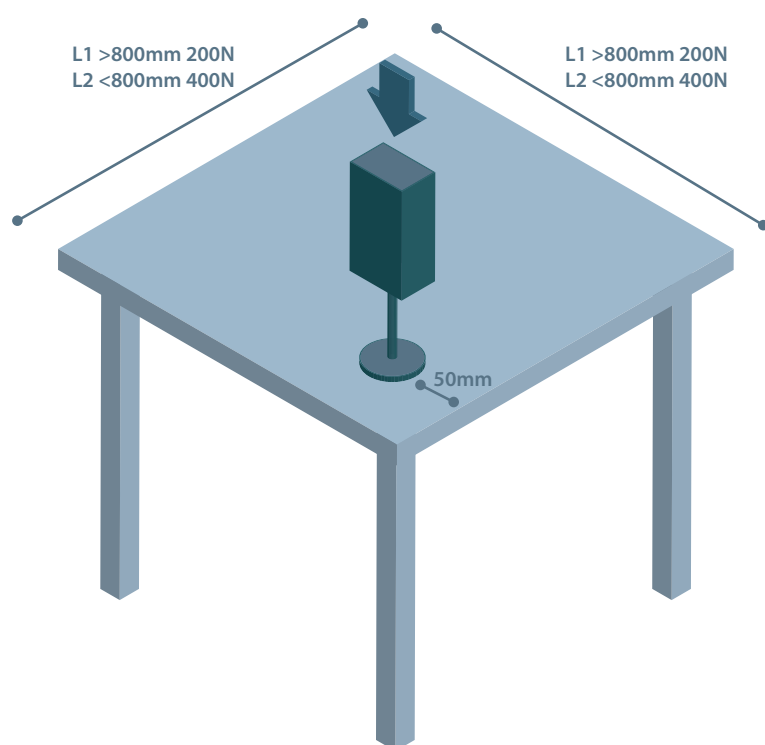
PASS:

Table does not tip or overturn when the specified forces are applied and;
a) there are no fractures of any member, joint or component;
b) there are no loosening of joints intended to be rigid;
c) table fulfils its functions.



FAIL:

Table tips and overturns when forces are applied.
There are any failures of a, b or c listed above.



5.0.19 Boots In-store Equipment - Stability Under Vertical Load (Rectangular Tables) BTC018

Type: Pass or Fail

Test Criteria:

Using a calibrated force meter (fitted with a smooth, rigid, 100mm dia loading pad) apply the following loads (based on the dimensions of the table) not less than 50mm closer to any edge.
Table dimension (L1) below 800mm - 200N.
Table dimension (L2) above 800mm - 400N.



PASS:

Table does not tip or overturn when the specified forces are applied and;
a) there are no fractures of any member, joint or component;
b) there are no loosening of joints intended to be rigid;
c) table fulfils its functions.

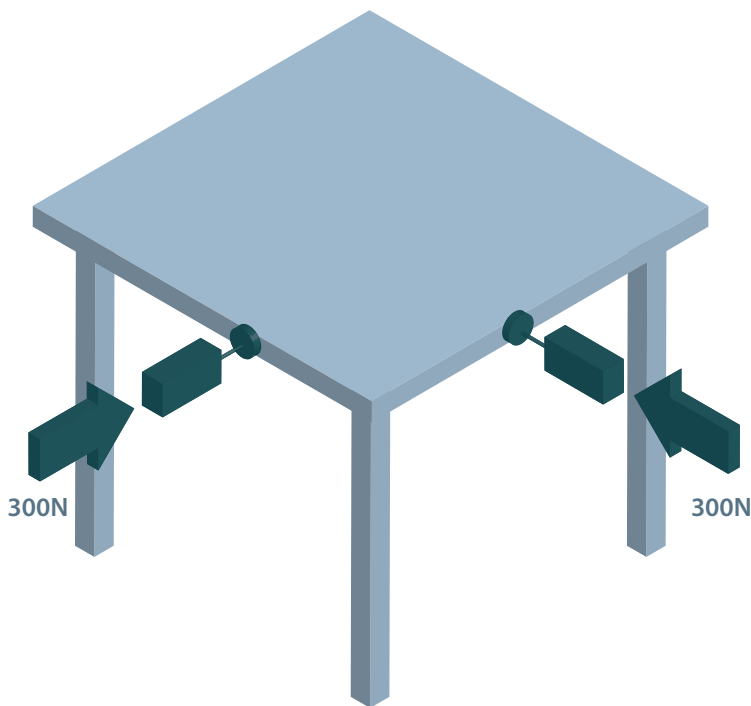


FAIL:

Table tips and overturns when forces are applied.
There are any failures of a, b or c listed above.



Section 5 BTC static load testing



5.0.20 Boots In-store Equipment - Table Stability Test (upper) BTC019

Type: Pass or Fail

Test Criteria:

Using a calibrated force meter (fitted with a smooth, rigid, 100mm dia loading pad) apply a force of 300N to each upper face of the table to confirm rigidity and stability.

Apply force sharply in a smooth motion over a 1 second period. Repeat in all directions.



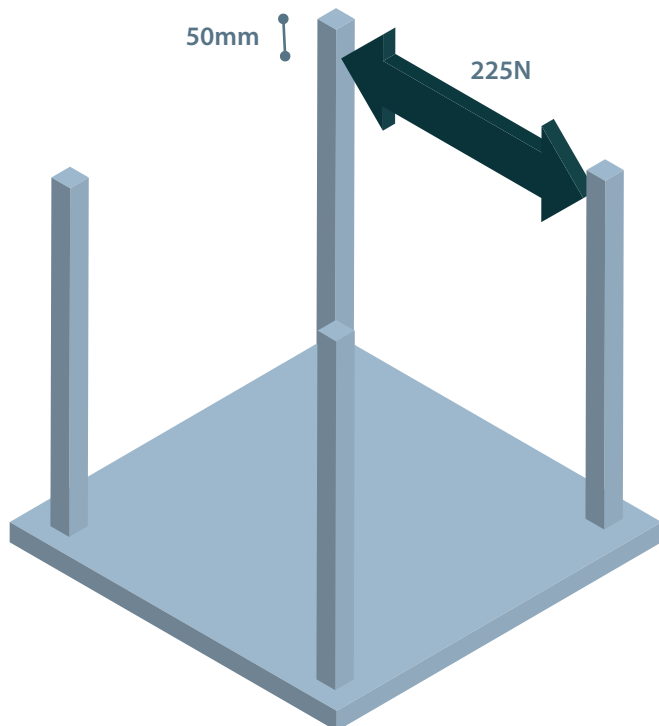
PASS:

Table does not tip or overturn when the specified forces are applied and;
a) there are no fractures of any member, joint or component;
b) there are no loosening of joints intended to be rigid;
c) table fulfils its functions.



FAIL:

Table tips and overturns when forces are applied.
There are any failures of a, b or c listed above.



5.0.21 Boots In-store Equipment - Table Leg Out Test BTC020

Type: Pass or Fail

Test Criteria:

Turn table over and using a calibrated force meter (fitted with a smooth, rigid, 100mm dia loading pad) apply a force of 225N between table legs pushing outwards between both legs forcing them apart, 50mm down from top of leg.

Repeat in all directions.



PASS:

Legs return to original position post testing;
a) there are no fractures of any member, joint or component;
b) there are no loosening of joints intended to be rigid;
c) table fulfils its functions.

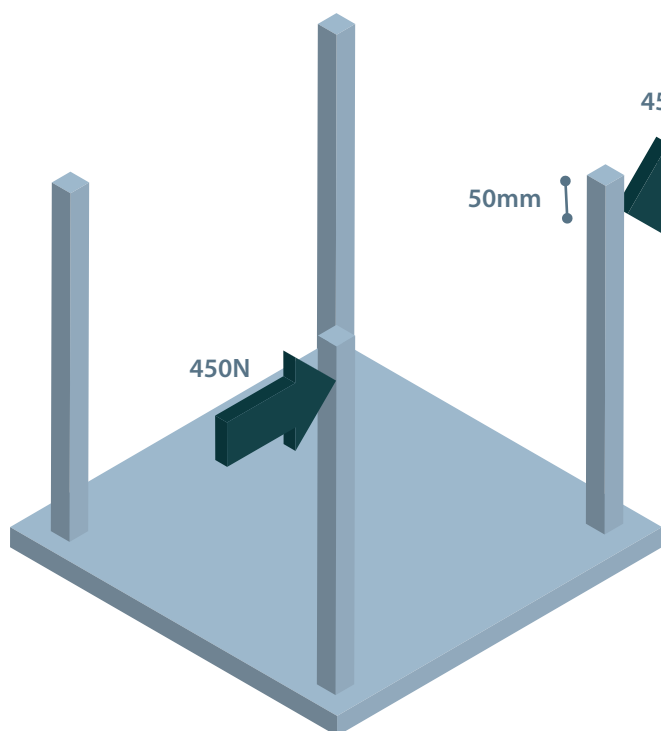


FAIL:

Table legs break or fail during test.
There are any failures of a, b or c listed above.



Section 5 BTC static load testing



5.0.22 Boots In-store Equipment - Table Leg In Test BTC021

Type: Pass or Fail

Test Criteria:

Turn table over and using a calibrated force meter apply a force of 450N outside table legs pushing them inwards towards each other, 50mm down from top of leg. Repeat in all directions.



PASS:

Legs return to original position post testing;
a) there are no fractures of any member, joint or component;
b) there are no loosening of joints intended to be rigid;
c) table fulfils its functions.



FAIL:

Table legs break or fail during test.
There are any failures of a, b or c listed above.



6 Manufacture guidance



Section 6 Manufacture guidance

Part 6 Manufacture:

6.1 Manufacture Process Illustration

The following process illustration shows the normal steps Boots expect suppliers to take in ensuring that all Boots requirements for production of equipment are met. It is critical that suppliers understand that Boots specifications are set and approved. Therefore no change, modification or deviation from approved specifications are permitted. Unless following Boots change control process and with the instruction and approval of your Boots contact.



Section 6 Manufacture guidance

Part 6 Manufacture:

Boots maintain strict control of manufacturing standards and specifications. It is therefore crucial for all suppliers to understand the importance of Boots specifications and ensure that any production materials, substrates, finishes and methods adhere strictly to the specifications provided within the production documentation.

No deviation, modification or change of materials, finishes or processes is permitted without the written approval and instruction of Boots. Any change of specification during manufacture or supply will create a material breach of supply conditions and could result in the cancellation of orders or resupply of goods to the correct specification.

6.2 First off production sample:

- Manufacture the first off in exactly the same way as production items, in the same factory using the same materials otherwise this is another prototype which cannot be signed off.
- Use manufacturing materials only.
- Use only the production CAD data provided. Any suggested modifications or changes must be approved and managed using the Boots change control process.
- Any elements that are not suitable for full prototype manufacture (rapid prototype, tooled items etc) should be produced in materials and methods that are as close to production type as possible. With a dedicated discussion and approvals process for these elements that allows for Boots to be confident final production solutions will be suitable to task and viable.

6.3 Packaging:

- Furniture packaging must be designed to be fit for purpose based upon the design brief taking into account manufacture process, delivery route and method, storage requirements and implementation requirements.
- Packaging materials should take into account Boots CSR and environmental policies and be recyclable, reusable and designed for maximum performance and where possible from single material types to ease recycling.
- Where fragile items are being shipped it is the suppliers responsibility to properly test and certify their packaging designs as suitable to task. And where considered necessary, carry out drop tests and transit tests to prove the efficacy of packaging solutions.
- Sustainable packaging materials should be used where possible. The inclusion of single use plastic should be avoided. Packaging carbon impact should be included in the Sustain® calculation.

6.4 Instruction drawings:

- Instruction drawings are required on all furniture to assist ease of installation, assemble and disassemble they must be designed to specs as laid out in section 4.1.12 and 4.1.13.

6.5 Labels:

- All fixture components supplied to The Boots Group must include a QR code to support traceability, logistics, and maintenance. The QR code must contain plain text only and must follow the standard format defined below to ensure consistent scanning across iPhone, Android, and industrial scanning devices. Suppliers may generate QR codes using any reliable tool, including the web-based generator available at www.qrcode-monkey.com/#text, provided the output is a standard ISO/IEC 18004 compliant QR code, contains no embedded URLs, and uses only unformatted plain text.

<https://www.qrcode-monkey.com/#text>

- The QR code content must be a single continuous line of text using the following structure, with each field separated by a vertical bar ("|") and no line breaks, spaces, or additional formatting:

BOOTS_CODE=<Boots Code>|COMPONENT=<Component Name>|SUPPLIER=<Supplier Name>|DATE=<DD-MM-YYYY>|LOCATION=<Location of Manufacture>

- No additional text, branding, or characters are permitted within the QR code content. The date must strictly follow DD-MM-YYYY format. Suppliers are responsible for ensuring that each QR code is fully scannable and validated using standard smartphone cameras prior to shipment.
- The physical QR code should be directly printed or applied as a sticker in a discreet location and not interfere with any customer facing surfaces. e.g. not on the front face or top surface of a counter, but inside a cupboard or on the underside of a shelf. If the component is too small to locate a QR code, please make the D&D team aware.

ALL QR CODES MUST BE PRINTED AT 20 x 20MM

Example QR Code Content (for reference)

BOOTS_CODE=123456|COMPONENT=SHELF UNIT|SUPPLIER=ABC Manufacturing Co.|DATE=24-04-2026|LOCATION=China



Section 6 Manufacture and quality assurance

Part 6 Manufacture:

6.6 QC Process Illustration:

The following process illustration shows suggested steps for achievement of consistent, acceptable quality standards. It is the vendors responsibility to ensure that Boots equipment has a supporting and robust quality management process, where all equipment is of consistent manufacture type and finish and free from defects. It is also the suppliers responsibility to ensure that protective packaging materials are of sufficient design and quality to protect Boots equipment up to point of final delivery.



Boots retail equipment standards are consistent and we require our suppliers to provide adequate quality control, recording and delivery services for all of our projects. Where required we expect specific QC models to be adopted to support individual order types and it is the suppliers responsibility to provide QC process documents upon request.



Section 6 Manufacture and quality assurance

Part 6 Manufacture:

6.7 Quality control process:

- The Supplier will only manufacture and assemble Goods from Customer approved facilities and locations. The Supplier must request in writing to the Customer for a change in manufacturing facility and location. A requested change by the Supplier for a change in manufacturing facilities and locations must prove beneficial to the Customer in Quality, Cost and Delivery. The Customer must approve any such change in writing (such approval will not be unreasonably withheld).
- The Supplier is required to have and maintain a quality system program that will ensure that the design, manufacture, assembly and inspection of the Goods is consistent with ISO 9001 (or able to demonstrate) regulatory requirements and will consistently deliver Goods to the specification requirements. The Supplier will inform the Customer, in writing, of any quality system changes that may affect the Goods or of any deviation from agreed process standards and / or process work instructions. All changes will need to be approved by the Customer if it pertains to the Customer Goods, prior to implementing the change.
- The Supplier is expected to document, deploy, and maintain, as required by ISO 9001 (or able to demonstrate), all Quality Policies, Quality Plans, Procedures, and Work Instructions. These documents are expected to be readily available and understood by all personnel involved with manufacture of the Goods. The Customer reserves the right to obtain for review and approval of current Process Control / Quality Plans and Work Instructions used in manufacture of the Goods upon written request to the Supplier.
- The Supplier shall ensure that all documents relating to the Customer, including drawings, specifications, contracts, policies, procedures, and work instructions are under version/revision control and that only the current documents are available to personnel with responsibility for Goods quality (as per ISO 9001 requirements or able to demonstrate). A system by the Supplier shall be maintained for the effective control of documentation.
- The Supplier shall establish and maintain procedures for identification, collection, indexing, filing, storage, maintenance, and disposition of quality records. This shall include incoming material quality, such as proof of conformance (through incoming quality inspections or other means) to the established specifications, through final outgoing quality inspections. This is inclusive of all regulatory requirements such as RoHS requirements, and shall be made available to the Customer upon demand. Records shall be kept in accordance with the Supplier's own retention procedures but must, as a minimum meet the minimum retention periods.

6.8 Quality Assurance:

- The supplier is expected to ensure product quality by use of various tools and independent sampling of the various products' attributes (such as out of box audits and ongoing reliability testing).
- The testing should be carried out on samples taken according to a statistically derived sampling plan and taking into account the production rate. Packaged out of box audit to assure all in box items are included, assure cosmetics conform to specifications, & assure an acceptable customer out of box experience is also required. Units should be sampled across build population in order to accurately assess the build. All failures during product quality plan testing are priority and take precedence over other investigation.

6.9 Quality Metrics:

- The supplier is expected to use quality metrics as a basis to monitor the quality of its products and processes.
- Manufacturing Yields.
- Rework Yields.
- Test/Inspection Station Yields.
- Mean Time Between Failure data (MTBF).
- Out of Box Audits (AQL).
- Ongoing Reliability Testing/or Extended Life Testing as agreed upon.
- Quality Assurance Tests (QAT).

7

Implementation



Section 7 Implementation

7.1 Housekeeping:

- It is important that any unit introduced into store is easy to clean and any waste products are easily disposed of.
- Any fixture that produces waste, such as product testing for example, must include a place for disposal for that item in a waste bin or other hygiene unit.
- Adequate facilities for wipes and tissues should be provided if testing is to be undertaken.
- All inaccessible nooks and crannies which cannot be easily cleaned should be designed out of the fixture design.
- If the fixture design requires specialist cleaning instructions or cleaning products, these should be provided with the equipment along with instructions on how to replenish if required.

Operational Maintenance and Hygiene:

- Fixtures and Displays of a permanent nature shall be easy to clean, and any waste products they may generate are to be easily disposed of. The design of any fixture that produces waste, such as product testing for example, must include provisions for the disposal of that item, whether in an associated waste bin or an integrated hygiene receptacle. Where applicable, storage for wipes and tissues should be provided where product testing is to be undertaken.
- Inaccessible nooks and crannies, which cannot be easily cleaned, should be avoided.

7.2 Installation:

- Consideration must be made when preparing the fixture for installation of the limitations of the Boots retail environment.
- Space in store is limited, and it is essential that adequate space in store is allocated for the installation to progress safely and without risk. Access to the shop floor in Boots stores can be challenging and it is essential that you understand the limitations that you are likely to experience and prepare accordingly.
- Any unit should be able to go through a standard doorway etc. Access to first and second floors should be taken into account, ensuring lifts or stairs can be navigated.

7.3 Delivery:

- Fixture and Display designs should consider Knock-Down-Flat (KDF) construction whenever practical, for ease of shipping and handling. Whether fully assembled, or delivered KDF or in multiple components, Fixtures must fit through a standard doorway (36" x 80") when being delivered to Stores.
- A detailed installation instruction sheet must be supplied with each Fixture or Display. At a minimum, such instruction sheets must include the following;
- Detailed list of each part of component in the box, along with quantities. - Illustration of each component in the box. - Individual part or reference number for each part or component. - Detail list of tools required for assembly. - Detailed (step by step) assembly instructions
- If a Fixture or Display will be used in combination with an existing fixture in the store, the installation instruction sheets must clearly state what components will be removed (if any). For example, if a shelf level display will be installed on existing store shelving, part of the instruction sheet may direct the installer to remove the ticket strip (or leave it in place). - In cases where existing fixtures, or portions of fixtures are to be removed as part of installation, instructions must also include directions for the safe and proper disposal of the removed items.

8

Appendices



Appendix 1 BTC Compliance Certificate v6

ISE First Off Compliance Certificate 2.0 - To be completed by supplier or brand for all Third Party Equipment

Supplier:

Date:

Designed By:

Address:

Project Title:

Contact email:

Product Code:

Tel:

Safety Considerations:

Sharp Edges And Corners Checked	☐
Gaps And Trapping Checked	☐
Child Assessed For Specific Risks	☐
Risk Checked For Slips, Trips And Falls	☐
Stacking/ Storage Risk Check	☐
Health Assessment Checked	☐
Mechanisms Compliance	☐
Thermal & Fire Compliance	☐
Water And Liquids Compliance	☐
Pressure Compliance	☐
Chairs And Stools Compliance	☐

Boots Test Certification:

	Pass	Fail	N/A
BTC001 - Measured Force Topple Test	☐	☐	☐
BTC002 - Tilt Test	☐	☐	☐
BTC003 - Bump Test	☐	☐	☐
BTC004 - Static Weight Std 760mm Shelf	☐	☐	☐
BTC005 - Static Weight Angle Deflection	☐	☐	☐
BTC006 - Static Weight Custom Shelf	☐	☐	☐
BTC007 - Static Weight Angle Deflection	☐	☐	☐
BTC008 - Vertical Load Test - Doors	☐	☐	☐
BTC009 - Horizontal Load Test - Doors	☐	☐	☐
BTC010 - Topple Test Drawers +762mm	☐	☐	☐
BTC011 - Topple Test Drawers -762mm	☐	☐	☐
BTC012 - Climb Test Drawers	☐	☐	☐
BTC013 - Hinged Flap Weight Test	☐	☐	☐
BTC014 - Hinged Flap Point Load Test	☐	☐	☐
BTC015 - Surface Static Load Test	☐	☐	☐
BTC016 - Surface Point Load Test	☐	☐	☐
BTC017 - Stability Under Vertical Load	☐	☐	☐
BTC018 - Stability Under Vertical Load	☐	☐	☐
BTC019 - Table Stability Test (Upper)	☐	☐	☐
BTC020 - Table Leg Out Test	☐	☐	☐
BTC021 - Table Leg In Test	☐	☐	☐

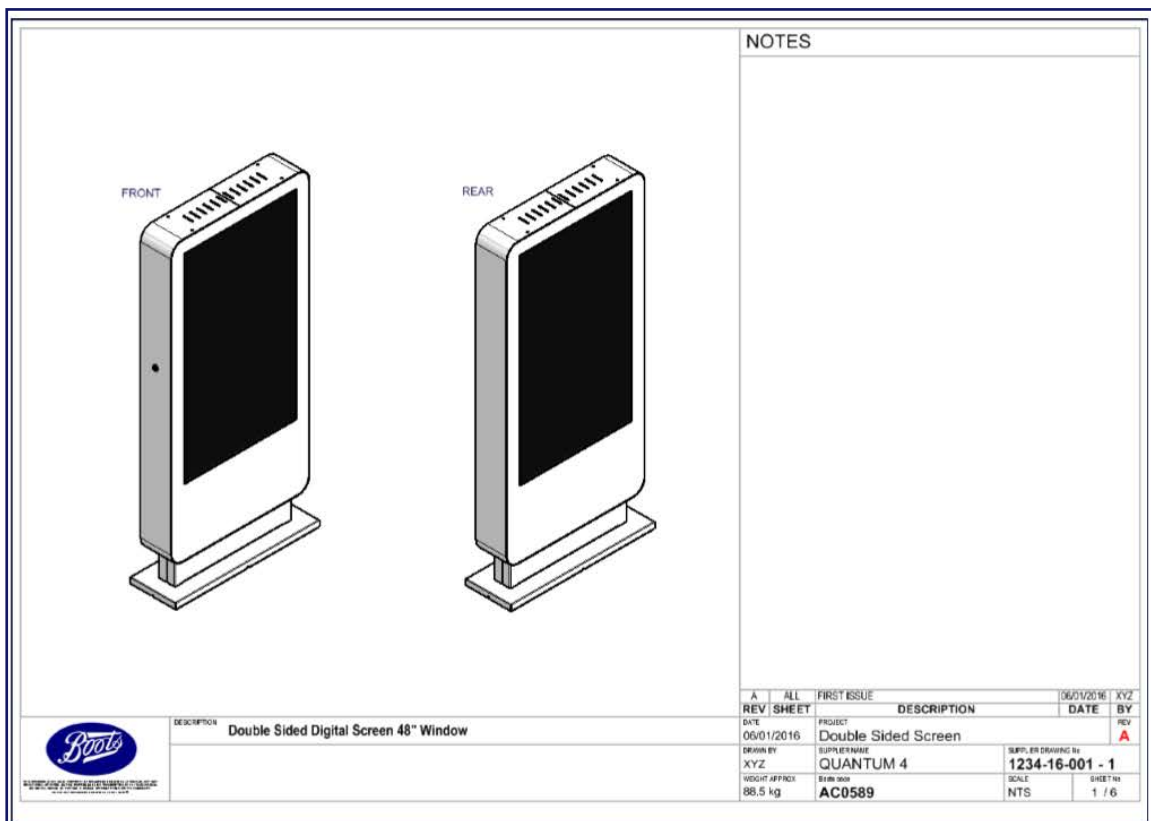
Glass & Mirrors:

Glass	Toughened	Laminated	Safety Film	N/A
	☐	☐	☐	☐
Boots Glass Guidance Compliance?		Yes	No	
		☐	☐	
Mirrors	Toughened	Laminated	Safety Film	N/A
	☐	☐	☐	☐
Mirror Fixings	Frame/Clip	Screw	Other	N/A
	☐	☐	☐	☐

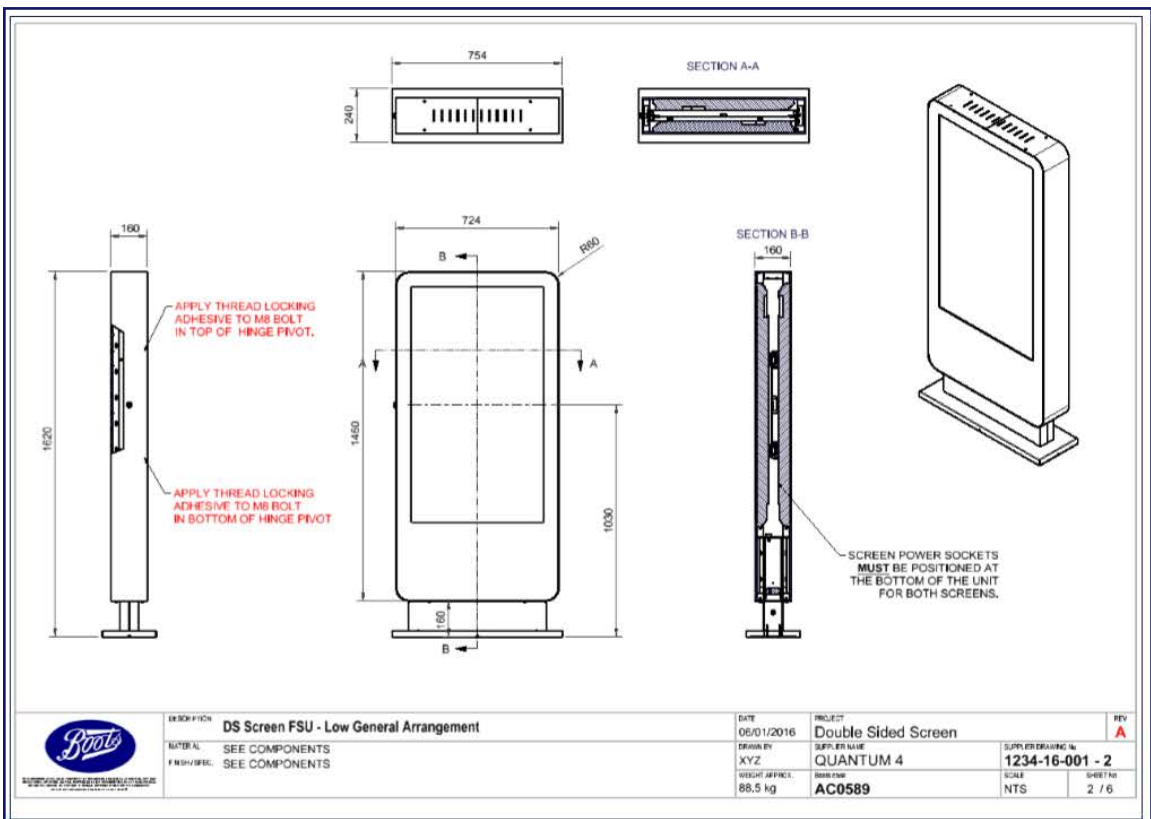
General:

Tool free assembly for store colleagues?	Yes	No	N/A
	☐	☐	☐
Safety critical fixings required? (If Yes, provide details in O&M manual)	Yes	No	N/A
	☐	☐	☐
Manufacture engineering and CAD issued to Boots specifications?	Yes	No	N/A
	☐	☐	☐
Design risk assessments complete?	Yes	No	
	☐	☐	
H&S testing report complete?	Yes	No	
	☐	☐	
O&M manual complete?	Yes	No	
	☐	☐	
Installation/assembly manual complete?	Yes	No	
	☐	☐	
Boots prototype approval granted?	Yes	No	
	☐	☐	
Sustainability assessment complete?	Yes	No	N/A
	☐	☐	☐
Full transfer of design IP to Boots? (Boots initiatives only)	Yes	No	N/A
	☐	☐	☐

Appendix A - Design guidance - Drawing pack examples

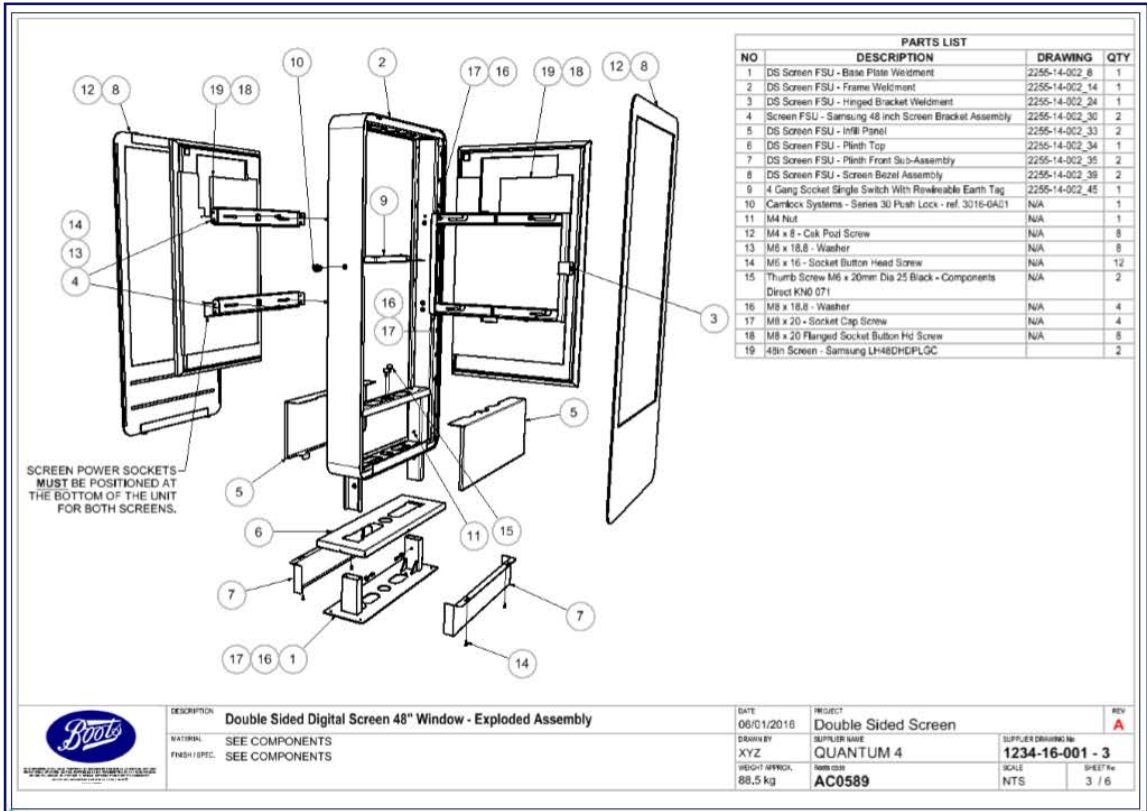


4.4.1 Example front page

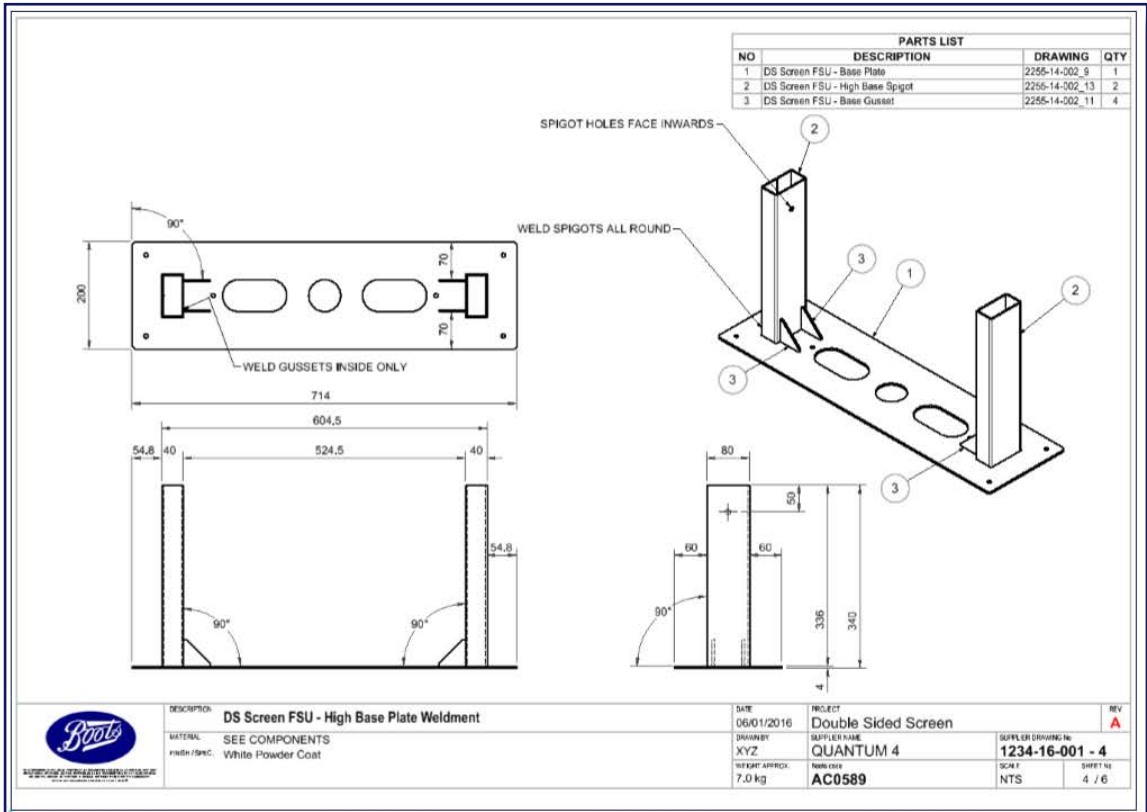


4.4.2 Example main page

Appendix A - Design guidance - Drawing pack examples

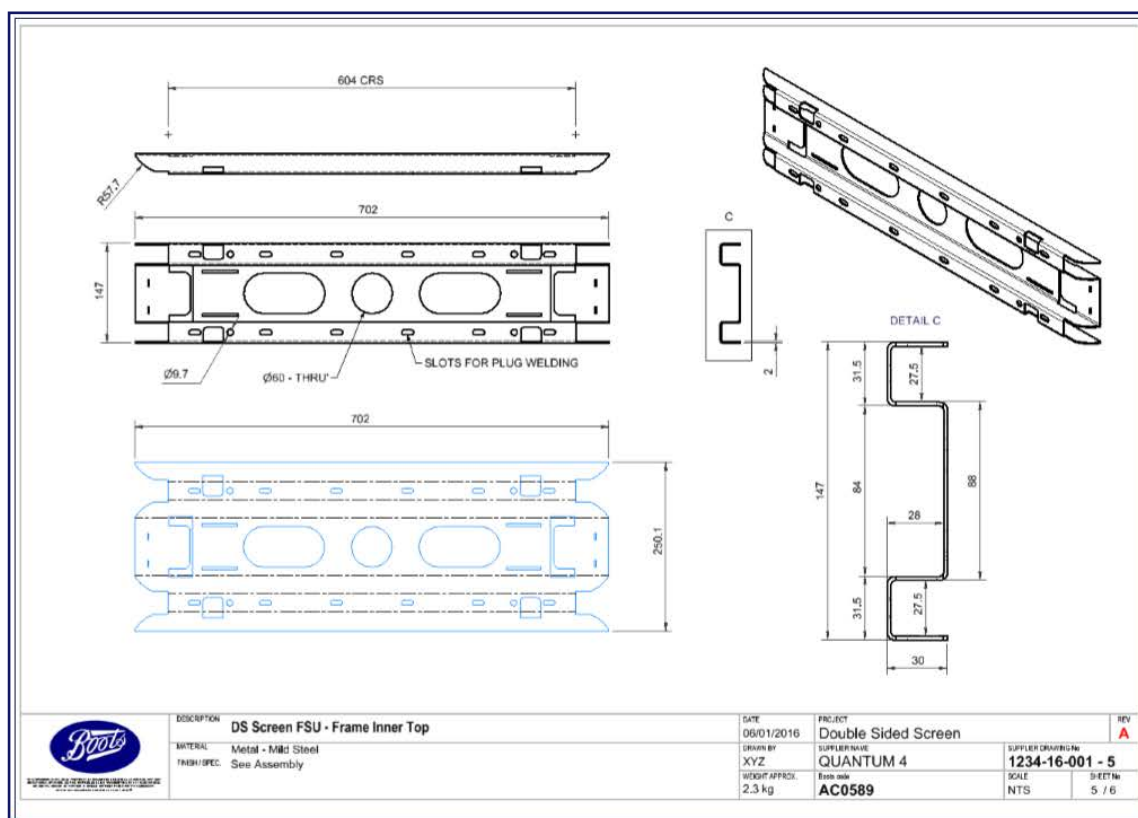


4.4.3 Example exploded view



4.4.4 Example sub-assembly

Appendix A - Design guidance - Drawing pack examples



4.4.5 Example component drawing

The above page examples are for reference only and as a guide as to the level of information Boots require. As a minimum standard Boots expectation is that all information, dimensions, data, material specifications and finishes will be included within the drawing packs provided. Should this prove not to be the case it shall be considered a failure of the supplier to provide agreed details and a failure of their contractual obligations.

Should it be found that manufacture or specifications cannot be used for immediate manufacture Boots reserve the right to contra-charge the originating supplier for rectification costs to bring drawings and CAD up to acceptable standards.

Appendix B - Design guidance - O&M manual examples

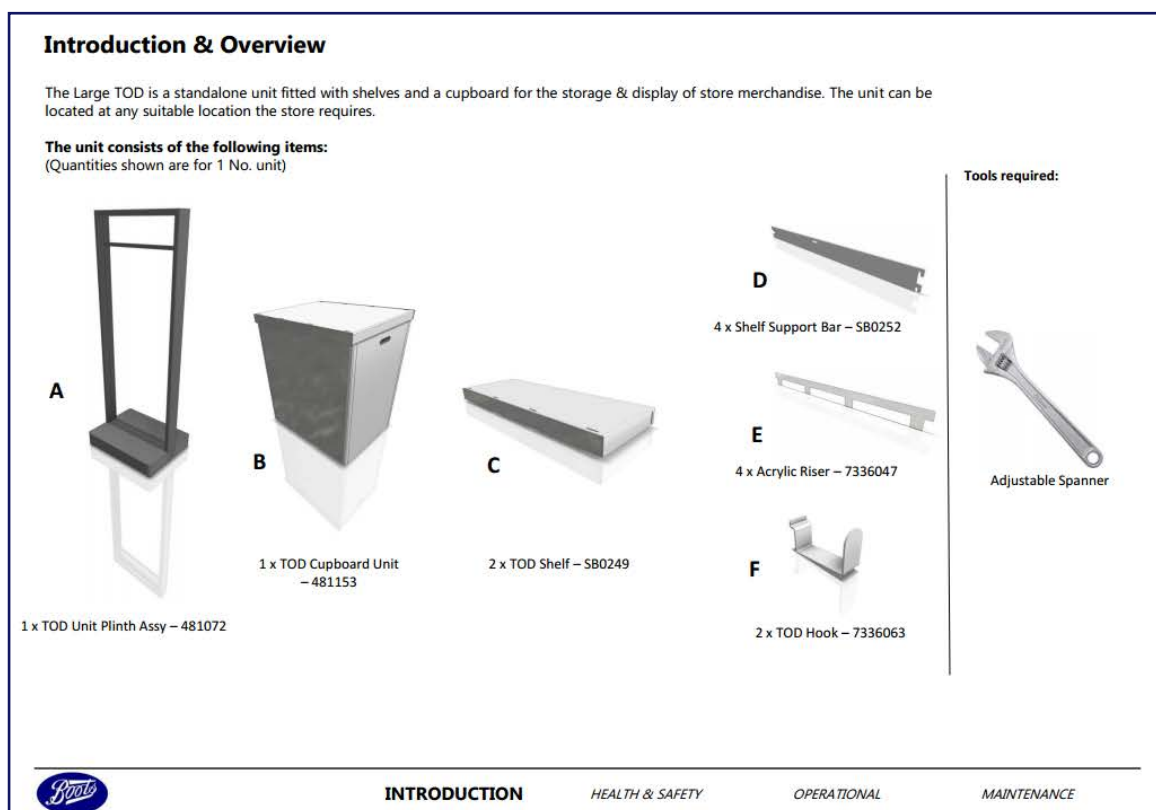
The following pages (Appendix B) illustrate the type, style and standard of operation and maintenance (O&M) manuals that should be provided for all Boots equipment.

It is the vendors responsibility to properly provide manuals to Boots approved samples and where necessary update and maintain the integrity of the data should modifications or changes be made.

Appendix B - Design guidance - O&M manual examples



4.4.7 Example front page



Appendix B - Design guidance - O&M manual examples

4.4.8 Example introduction page

Introduction & Overview

Additional items that may be used with this unit if required:
(Quantities shown are for 1 No. unit)

1 x Fragrance TOD 500 Header Graphic Panel – 7335989

1 x Fragrance TOD 500 Half Height Graphic Panel – 7335997

1 x Fragrance TOD 500 Full Height Graphic Panel – 7336004

2 x Fragrance TOD 500 Cupboard Graphic Panel – 7336012

2 x Fragrance TOD 500 Shelf/Cupboard Self Adhesive EPOS Strip – 7336055

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4.4.9 Example overview page

Unit Overview

Shelf Support Bar

TOD Shelf

Unit Plinth Assy

TOD Hook

Acrylic Riser

Acrylic Riser

Cupboard Unit

GENERAL INFORMATION
NET WEIGHT: 61.5kg (approx)
MEASURES L x D x H: 636 x 501 x 1660mm

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Appendix B - Design guidance - O&M manual examples

4.4.10 Example unit overview page

Important Safety Points

- Assembly requires a **MINIMUM of 2 persons!**
- The unit should be checked weekly for stability & positioning and adjusted accordingly. This should only be carried out by a suitably qualified person.
- All shelves can be moved or adjusted by store staff.
- Larger, heavier items should be checked post installation by the store manager, **prior to the store going live.**
- Any damaged or broken items should be **removed from the shop floor immediately** and replaced accordingly
- **DO NOT** climb, stand, sit or lean on the unit.
- Only use shelves for their specific use.
- Do not place shelves where they could pose a **risk at head height.**
- **Equally & evenly load** shelving areas.
- Ensure the unit is sighted level, with all levelling **feet in contact with floor surface at all times**
- Unit shelves should not exceed the external footprint of the main unit.
- Unit should not protrude into or block **emergency walkways or exits**
- Weekly checks should be made to ensure all feet are properly adjusted and in contact with the floor.

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4.4.11 Example important safety points page. **Always reference the turning off of any electrics before any cleaning or moving of items or components. This reference is to be bold and underlined.**

Unit Assembly

1. Position unit plinth assy (A) into its intended location. Then making sure it is in the correct orientation, lift & position the TOD cupboard unit (B) to sit on top of the plinth, lift using the side handles only and taking care to avoid any damage to either component.
2. Position shelf support bar (D) between plinth assy frame (A) and hook on one side at a time. Make sure all hooks are fully located into slots! Repeat this process for all required support bars.

Minimum Persons required:

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4.4.12 Example assembly page. **If there is a critical fixing type or component, reference MUST be called out clearly in the operations manual, explaining in detail the type, use and application.**

Appendix B - Design guidance - O&M manual examples

4.4.13 Example cleaning instructions page

Cleaning

- Ensure unit is stable before commencing any cleaning
- Excess water should be soaked up using a dry, lint free cloth whilst the unit is in an electrically isolated position. Best results can be obtained using a microfiber cloth
- Care should be taken to keep excessive water away from all steel components
- Unitary should only be cleaned using a damp lint free cloth using a non abrasive soap based cleaner
- Do not clean any of the unit with polish or solvent based cleaner
- Do not attempt to use commercially available cleaning products as they may cause bleaching or staining
- Any items that are damaged or un fit for purpose should be removed immediately and replaced accordingly.
- The above can be carried out by a suitably qualified person
- **DO NOT use any abrasive pads on lacquered surfaces or any alkaline cleaning agents such as potassium hydroxide, sodium carbonate or caustic soda, or any powerful acidic products, heavily abrasive scouring agents or lacquer dissolving cleaning agents.**

*INTRODUCTION**HEALTH & SAFETY**OPERATIONAL***MAINTENANCE**



Health and Safety Policy

Caring for people is what we do

Boots is a brand built on our values of care and trust. Care is at the heart of our mission to help people get better, live well and feel good for life. How well we do that is a key measure of our overall business success.

Leadership is key to a strong and continually improving safety culture, keeping the safety of people at the heart of our business. This Policy sets the tone of that leadership and is supported by safety systems which are regularly reviewed to ensure they remain relevant and effective.

Caring for people is part of the DNA of Boots, and we value the health and safety of every person that visits, works, or otherwise maybe affected by our operation. Working together, we will continually seek to reduce harm and share ideas for improvement on ways to manage safety risks. We'll also conduct governance reviews and audits to identify where we're strong and where we have opportunities to improve.

We recognise that our leaders and team members are our greatest asset, helping us deliver on the trust and responsibility placed on us. We will ensure they have the environment, equipment, skills, knowledge, and procedures to invoke this Policy and, in return, expect engagement and collaboration on our overarching ambition of care.

Anthony Hemmerdinger

Senior Vice President and Managing Director, Boots UK & ROI

November 2024



Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)



The Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

Definitions

Boots Certification

Certificate issued by Boots Property to confirm compliance to technical and safety requirements as required by Boots UK.

Boots Electrical Certificate

Electrical certificate issued by appointed electrical specialists and forms part of the supporting documentation needed for Boots Certification. Required for electrified Display aids, Shopfitting's, Signage and Temporary displays.

Boots Property Central (BPC)

Online software portal used by Boots Property teams to store and manage data for both internal and external use. www.bootspropertycentral.co.uk

Digital Equipment

Screens, media players, routers, totems and associated digital equipment used in display aids, shopfitting's, signage and temporary displays.

Display Aids

Temporary, semi-permanent and permanent inserts for shopfitting's and temporary displays, both illuminated and unilluminated.

Electrical Supervisor

Qualified Supervisor or equivalent approved by Boots Engineering, Gold Card Electrotechnical Certification Scheme (ECS) carrier. This person can be either directly employed or an approved subcontractor, but who is working on behalf of the manufacturing company.

NAPIT

Government-approved and UKAS-accredited membership scheme operating in the building services and fabric sector. Organisers of the Boots Electrical Design and Certification, and Electrical Design and Awareness courses.

Shop fittings

Semi-permanent and permanent fixtures used for the display of products, both illuminated and unilluminated.

Signage

Semi-permanent and permanent signs used for in-store communication, both illuminated and unilluminated.

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

Temporary displays

Temporary fixtures used for the display of products, both illuminated and unilluminated, typically for events and short promotions.

Type Approval

Certification of equipment to ensure it meets technical and safety requirements. Issued before installation it represents all subsequent digital equipment, display aids, shopfitting's, signage and temporary displays.

UKCA

UK Conformity Assessed, the conformity marking used for goods sold in Great Britain (England, Wales, and Scotland). Demonstrates that products comply with UK product legislation.

1.0 Introduction

The primary objectives of The Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1) is as follows:

- 1.1 Update of previous related documents Boots TBGE1 Guide requirements and expectations.
- 1.2 Alignment to Health and Safety at Work Act and UK legislation.
- 1.3 Support of Boots Certification including standardisation of the sourcing, design and usage of electrical components within digital equipment, display aids, shopfitting's, signage and temporary displays.
- 1.5 Enforcement of energy performance requirements to reduce operational carbon within Boots stores.

2.0 Application for Boots Electrical Certification

The below certification process for digital equipment, display aids, shopfitting's, signage and temporary displays will be mandatory for manufacturers and agents seeking to produce retail furniture with electrical componentry within it for use in Boots stores

- 2.1 Oversight of the design, factory testing and submission for Boots Electrical certification are the responsibility of the appointed Electrical Supervisor.
- 2.2 Electrical Supervisor's from Boots suppliers must complete NAPIT's Boots Electrical Design and Certification course.
- 2.3 Boots suppliers will need to ensure at least one designer working on Boots projects has completed NAPIT's Electrical Design and Awareness course.
- 2.4 It's recommended that key production staff such as QC managers and Line Leaders undertake NAPIT's Electrical Awareness and Basic Electrical Testing Course.

Stage 1: Compliance

2.5 Electrical wiring diagrams for the proposed designs must be submitted with the general assembly drawings by the manufacturers Electrical Supervisor for review by Boots for initial design approval. Designs will be reviewed for compliance with this policy in conjunction with the creative brief. Confirmation of compliance does not constitute approval for installation.

Stage 2: Certification

- 2.6 An Electrical Supervisor shall provide the following document for review before testing:
 - a. General Assembly drawings showing power supply, isolation point and access point for maintenance. See Appendix.

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

b. Wiring diagram showing all components, OEM suppliers name, cable and fuse ratings, driver loads and LED efficiency. See Appendix.

c. Factory electrical test certificate showing load, insulation, leakage, polarity and results See Appendix.

d. A UKCA certificate for the equipment in the manufacturers name. See Appendix.

2.7 An example of the production digital equipment, display aid, shopfitting, signage or temporary displays must be made available to Boots or an appointed agent to carry out inspection and testing for Type Approval prior to installation.

Stage 3: Verification

2.8 Boots reserve the right to carry out re-inspection and testing of digital equipment, display aid, shopfitting, signage or temporary displays that have been Type Approved, either during manufacturing, distribution or while in-store. Failure to meet Type Approval could result in the digital equipment, display aid's, shopfitting's, signage or temporary display's not being installed, distributed or electrified in-store. Any or all costs associated will be borne by the manufacturer or agent.

Early engagement with Boots Properties is recommended to minimise delays and help ensure Boots Certification is timely and efficient.

3.0 General Requirements

The latest version of this policy is available on Boots Property Central within the Creative Team Development/Design policy document.

3.1 All digital equipment, display aid's, shopfitting's, signage or temporary display's, both permanent and temporary must comply with this policy which requires a Boots Electrical Certificate. Only Boots Certified equipment will be connected to Boots power.

3.2 Manufacturers of electrified display aid's, shopfitting's, signage or temporary display's must employ suitable qualified technical staff and have appropriate insurance to comply with meeting country regulatory requirements. All design responsibility shall be with the manufacturer or agent including compliance with UKCA in the UK and IECEE CB Certification potential for export outside of Europe. Boots suppliers must have completed the due diligent requirements for both themselves and Tier 1 suppliers.

3.3 Electrical design must be carried out by a person with both relevant electrical and design qualifications. Boots suppliers are required to ensure at least one electrical designer working on Boots projects has attended NAPIT's Electrical Design and Awareness course, and this course is recommended to all manufactures and agents.

3.4 The digital equipment, display aid's, shopfitting's, signage or temporary display's needs to arrive on site pre-wired and certified ready for connection into the appropriate socket or power supply. Where this is not possible due to unit size or design, a specific procedure must be agreed in writing by Boots Engineering which will cover for onsite assembly and testing requirements.

3.5 Installation works connecting to Boots electrical infrastructure must be undertaken by suitable qualified persons. Records of post installation inspection and testing must be maintained by the installation company and made available to Boots Property team upon request.

3.6 Suppliers of Boots equipment must ensure wiring diagrams including the appropriate EL and or DV part codes. All parts codes are listed on Boots Property Central with drawings. Deviations from this must be made in writing to Boots Engineering prior to design. Allowance of 2 weeks assessment time must be considered, and last-minute requests may not be approved.

3.7 Total energy consumption of the shopfitting, display aid, signage or digital equipment must be within the targets set out in Creative Brief – Technical Requirements (see Appendix), but no greater than:

Wall units maximum lighting load of 120W per m2 (C energy rating or above) and maximum 50W per m2 for screens not brighter than 700 nits.

Mid-floor units maximum lighting load of 80W per m2 (C energy rating or above) and maximum 50W per m2 for screens not brighter than 400 nits

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

4.0 General Electrical Requirements

- 4.1 All digital equipment, display aid's, shopfitting's, signage or temporary display's must be either UKCA or CE certified.
- 4.2 All EPOS circuits and outlets on the sales floor shall be powered 24hours, 7 days a week, 365 days a year. This circuit shall be referred to as the Essential Circuit and shall be separated and dedicated. Power outlets on this circuit shall be a red Type G socket outlet.
- 4.3 In UK stores all other power shall be derived from Trading Circuit, or non-essential circuits, and a local point of isolation incorporating overload protection shall be provided at the shop fitting or beside the sign or digital equipment. This shall be the designated start/end point of respective responsibilities between the shopfitting manufacturer and the installing electrician. This isolation point shall be positioned in a location that is accessible by staff and a minimum of 100mm from the floor level.
- 4.4 All appliances such as hairdryers and hair styling products circuits shall be powered on a dedicated trading circuit referred to as the Hot Tools Circuit. Only one socket per powered appliance is allowed; only Boots approved switching units are allowed. The socket will be controlled by a dedicated key switch managed and controlled by store staff.
- 4.5 The switched fused isolator can be any colour to blend in with the unit, but it shall not be locked or concealed in an inaccessible location. The position of this isolator shall be agreed as part of the technical review and the position shall only become fixed once this review has been undertaken and signed off. The isolation should be identifiable to untrained staff within 30 seconds, in line with industry guidelines.
- 4.6 Where units incorporate lighting and power on the same non-essential circuit as mentioned above, additional lower rated overload protection shall be provided for the lighting equipment circuit. Reliance to clear any faults on lighting equipment shall not be down to the local 13Amp fuse on the non-essential supply .
- 4.7 Ventilation grills, compartments and enclosures shall be identified in the General Assembly drawings and within the operating manual to ensure they are not obstructed by display materials, merchandise, or any combustible materials during use.
- 4.8 No exposed connections will be allowed within the shopfitting, display aid or sign.
- 4.9 All components with the digital equipment, display aid's, shopfitting's, signage or temporary display's must be securely fixed.

5.0 Control Gear

- 5.1 Over and above warranty requirements all Boots owned equipment shall be capable of being operated 24 hours a day all year around without the risk of catastrophic failure.
- 5.2 Control gear which operates on the following frequencies (+ or – 2 kHz) must never be used: 14.5, 19.5, 29 & 58 kHz.
- 5.3 The driver design load shall be maintained at >0.90 power factor or above.
- 5.4 The power factor and THD of the driver or power supply shall be provided for the whole system at design stage and be included in calculation information.
- 5.5 All ELV fuse fusing characteristics (including any on board spares) must be FF type, Very Fast Acting fuses.
- 5.6 ELV fusing design must consider both the driver operating characteristics and the ELV circuit maximum current carrying capacity.
- 5.7 Control gear must be installed as per manufacturer's recommendations.
- 5.8 All Boots owned ELV components shall be 24V DC only.
- 5.9 Control gear must have a maximum inrush current of <30A.
- 5.10 In Boots equipment, only drivers listed in Appendix are approved for use. Applications for alternative control gear must be made to Boots Engineering and adhere to the test parameters of DC002.

6.0 Cables

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

6.1 All cables shall be compliant with current legislation both local and regional and a minimum of low smoke and fume (LSF). In Boots owned equipment, all cables shall be BASEC approved and double insulated. Whether LV, ELV or data.

6.2 In Boots owned equipment, connectors shall be fitted to reduce site works and manufacturers must use electrical sub-components.

6.3 All cables must be adequately secured by mechanical fixings in accordance with the manufacturer's recommendations. (Metal staples are not to be used for this). Excessive cable looping shall not be allowed, and cables shall be cut to length. External thermal influences on cable must be considered.

6.4 ELV cable current rating shall be greater than the ELV fuse rating protecting the circuit.

6.5 In UK stores segregation of LV and ELV cables must be maintained in accordance with BS 7671, and where multi-discipline installations occur (e.g. power and audio/data) there must be co-ordination between specialists to ensure cables are kept segregated.

6.6 All cables must be rated at a minimum of 5A for ELV and a minimum of 16A for LV

6.7 All cables must be sized to comply with Volt drop regulations in BS7671

7.0 Connections and Components

7.1 Connections and components must be installed assembled and used as per manufacturer's recommendations.

7.2 Components such as strain relief bushes and P clips should be used in accordance with BS7671 in order to protect cables and connections.

7.2 In Boots owned equipment, manufacturers must use electrical subcomponents.

7.3 All connectors must be rated at a minimum of 3A for ELV circuits and 16A for LV circuits

8.0 LEDs

8.1 All LED's must be a minimum of grade C efficiency. In Boots owned equipment, grade B efficiency and no greater than 2W per meter in linear application.

8.2 LEDs used in lightboxes and headers should be no greater than 50W m2.

8.3 In Boots equipment, only Boots approved LEDs are approved for use. Applications for alternative control gear must be made to Boots Engineering and adhere to the test parameters of DC001.

8.4 Light engines mounted on fire propagating material (anything other than metal) must be mounted on aluminium a minimum 1.6mm thick.

8.5 All light engines must be adequately fixed. All light engines mounted on the vertically or inverted (upside-down) plane must be mechanically fixed using fire rated fixings.

8.6 Over and above warranty requirements all light engines shall be capable of being operated 24hours a day all year around without the risk of catastrophic failure.

8.7 LED solutions are to be tested in-situ to ensure that adequate, stable, thermal dissipation occurs, and that PCB's mounted close to product do not cause product degradation. No continuous temperature rises allowed.

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

9.0 CHARGING POINTS

9.1 Only Boots approved phone chargers are approved for use within Boots store and can be found Appendix.

9.2 USB charging points are not approved in Boots stores

10. Batteries

10.1 Rechargeable batteries are not permitted in Boots stores. Only disposable alkaline batteries are approved for use.

10.2 The maximum allowed wattage is 2.5W at 12VDC giving a maximum discharge current of around 208mA.

10.3 The maximum number of cells (either AA, C, or D type) will be restricted to a maximum of 8, this giving a nominal voltage of 12VDC. The recommended battery type will be Varta industrial pro range of AA, C or D size batteries.

10.4 Batteries must be fitted in an approved battery holder such as the RS 185-4680 (D cell) this must have an FF type 20mm fuse fitted to the positive terminal, rated no higher than 1A.

11. General Digital Requirements

11.1 Compliance with TBG security requirements is mandatory. A copy of the security requirements can be found on Boots Property Central and any digital installation must be approved by Boots IT department.

11.2 All digital content deployed in store must be approved by the Boots or in compliance with Boots Partnering Wi-Fi content directive.

11.3 In Boots equipment, only Boots digital equipment listed is approved for use.

11.4 Media players will be provided by a Boots appointed agent to ensure TBG security compliance and only Boots media players may be connected to the Boots Media Network.

11.5 Boots media players are connected to the Boots Media Network via a LAN connection.

11.6 Approval to use the Boots Partnering Wi-Fi must be sought by the Boots IT/Creative/Property Departments.

12.0 Screens

12.1 All screens must fulfil the following performance requirements:

- a. All screens must be HD compatible format
- b. No brighter than 400 nits (cdm2) for use within displays
- c. No brighter than 700 nits (cdm2) for use within store
- d. A minimum of 2500 nits when used in parallel to external facing windows

12.2 Screens must be supplied with 2m HDMI cable for connection to a Boots media player.

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

13.0 Tablets

13.1 Charging for tablets must be done within the back office with no access to chargers within displays unless the tablets are fixed in place with no access to the power cable.

13.2 Boots owned tablets must include Wi-Fi compatibility.

14.0 Labelling

14.1 All digital equipment, display aid's, shopfitting's, signage or temporary display's must carry labelling with the following information including: Manufacturer name and contact details, Product code, Batch number and UKCA logo.

14.2 Boots owned shop fitting, display aid or signage must include the Boots code on the label.

15. Environment

15.1 All manufacturers and agents shall ensure that while undertaking design, manufacturing, supply of and disposal of equipment for Boots stores, Boots waste handling policies and Boots engineering energy efficiency policies are followed and adhered to.

16.0 Conflicts

16.1 Conflicts must be directed to certifiaction@boots.co.uk

17.0 Terms

17.1 Manufacturers must comply with the Boots Standard Terms & Conditions (issued on request) and the following Boots policies:

- a. Click on title for link to 'Code of Conduct for Ethical Trading'
- b. Click on title for link to 'Environmental Policy'
- c. Biodiversity Policy (issued on request)
- d. Wood, Paper and Pulp Sourcing Policy (issued on request)

18.0 Application of electrical design for Boots Certification

18.1 The health and Safety of our staff and customers is of paramount importance and takes precedent over any functional or operational need.

18.2 Consideration shall be given to the environment within which a shopfitting will be located, especially how the unit will be powered. The life of the unit and how it will be packaged and delivered.

18.3 Flexibility and speed of installation are important considerations and shall be incorporated within the design of the unit. Wherever possible complete systems made up and tested in the factory shall be brought to site and wherever possible assembly on site shall be kept to a minimum.

18.4 Replaced equipment shall be disposed of in accordance with all current in country directives/policies and in the context of Boots policies. The actual need to illuminate a shopfitting shall always be considered before undertaking this task.

Appendix D - Boots Group Electrical Policy for Digital, Display aids, Shopfitting's, Signage and Temporary displays V1 (TBGE1)

18.5 Equipment maintenance shall be easily undertaken, with little or no de-stocking required.

18.6 Many of the products in Boots stores are sensitive to heat and Ultraviolet light. Clarity must be sought by the manufacturer from Boots before design is undertaken. Special measures may be required in these instances.

18.7 The maximum permissible temperature on the displayed product (i.e., merchandise) caused by illumination shall not exceed 25°C.

18.8 Designs shall allow for constant ventilation to ensure the control gear does not exceed the component manufacturer's recommended operating temperature (Ta).

18.9 All power supplies shall be mounted in suitably ventilated locations, to ensure that compliance with the manufacturer's stated ambient operating temperature is achieved.

Appendix

Manufacturers Responsibilities

Wiring diagram

Factory Test Certificate

UKCA

Electrical Supervisors will have access to BPC

Approved components

Approved parts

Environment Policy Statement

The Boots Group regards the care of the environment as an integral part of running a responsible and successful business. We therefore seek to conduct our business activities in a way that reduces as much as is reasonably possible our direct and indirect impacts on the environment, while at the same time promoting practices that help to protect the environment and support sustainable development. The Boots Group is committed to reporting annually on environmental, social and governance issues and progress.

The Boots Group is committed to ensuring that due consideration is given to the impact our operations have on the environment at all levels. The Boots Group has in place a Boots Group Governance and Oversight Policy with a dedicated section for Environmental that includes expectations for every location, function, department and team members related to the management of environmental matters and to drive ongoing improvements even when it exceeds local legislative requirements¹. In addition, the Company maintains environment-related objectives that are tracked and monitored and cascaded to each business to align with local objectives and initiatives. Each business is responsible for continuously demonstrating its management of environmental matters. Monthly reports are provided to relevant Executive Committee or any of its Delegated Committees offering insights into environmental incidents across businesses.

In carrying out our business activities, The Boots Group endeavours to:

- engage in due diligence processes, including environmental, as it relates to mergers, acquisitions divestiture activities, and new business initiatives;
- responsibly source the materials, water and energy that we use – we hold suppliers and vendors for our owned brand products and our goods not for resale to our own environmental standards;
- promote and encourage re-use and recycling where practical, and ensure responsible management of any unavoidable waste materials;
- minimise impacts from emissions or wastes that cannot be avoided;
- provide our people access to environmental management systems, information and resources to manage and support our objectives;
- be transparent and open with our customers and other stakeholders;
- regularly consult with internal and external stakeholders to seek feedback on our performance in areas where we have environmental and social impact; regularly complete internal and external environmental audits;
- seek to continually improve the environmental performance of our operations;
- ensuring compliance with relevant environmental laws and regulations;
- seek to educate and empower our customers on key environmental issues such as waste and carbon emissions reduction by introducing new innovations and solutions such as recycling schemes and online healthcare services.

¹ Given the footprint of our operations, local legislative requirements will vary by market

